

Recent Advances in Pre-trained Language Models

姜成翰

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防疫訊息

更新日期：111年3月30日

本校防疫訊息－依據CDC 3月28日公告提醒本校教職員工生落實各項防疫措施

全校教職員工生大家好：

依據中央流行疫情指揮中心 3月28日發布的公告，請全校教職員工生在校內活動時，除用餐、室內外運動、以及指揮中心規定的例外情形之外，其他活動均應全程配戴口罩。講課、以及室內外拍攝個人/團體照，雖屬於指揮中心規定得免戴口罩的活動，但本校建議如無法與他人保持社交距離時，仍應該配戴口罩。其他防疫措施提醒事項如下：

大家好：

疫情升溫，保健中心也接獲同學反應：有許多學生在上課時並未配戴口罩，可能造成防疫漏洞。尤其曾發生配戴口罩的同學被迫與不戴口罩的同學分在同一組討論，令配戴口罩的同學感到很大的壓力。

保健中心在此籲請本校各單位再度對內加強宣導：

學生上課時應一律配戴口罩，也請各單位提醒教師上課時應提醒學生全程配戴口罩，尊重自己也尊重他人，不應該讓遵守規定配戴口罩的同學承受他人不配戴口罩的防疫壓力，謝謝大家～

Prerequisites

- [【機器學習2021】自注意力機制 \(Self-attention\) \(上\)](#)



- [【機器學習2021】自注意力機制 \(Self-attention\) \(下\)](#)



Highly Related Topics

- 【機器學習2021】[BERT簡介](#)



- 【機器學習2021】[GPT的野望](#)



Highly Related Topics

- 【深度學習於人類語言處理 2020】[BERT and its family](#)



Outline

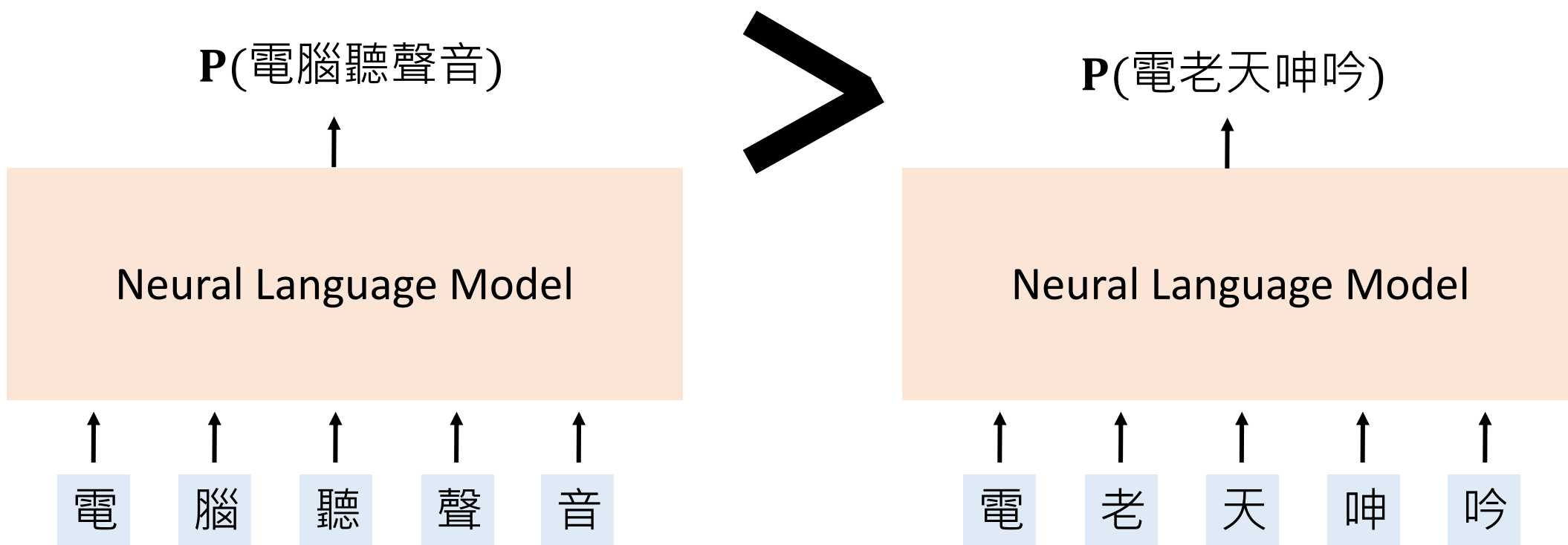
- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
- Closing Remarks

Outline

- Background knowledge
 - Pre-trained Language Models
- The Problems of PLMs
- The Solutions of Those Problems
- Closing Remarks

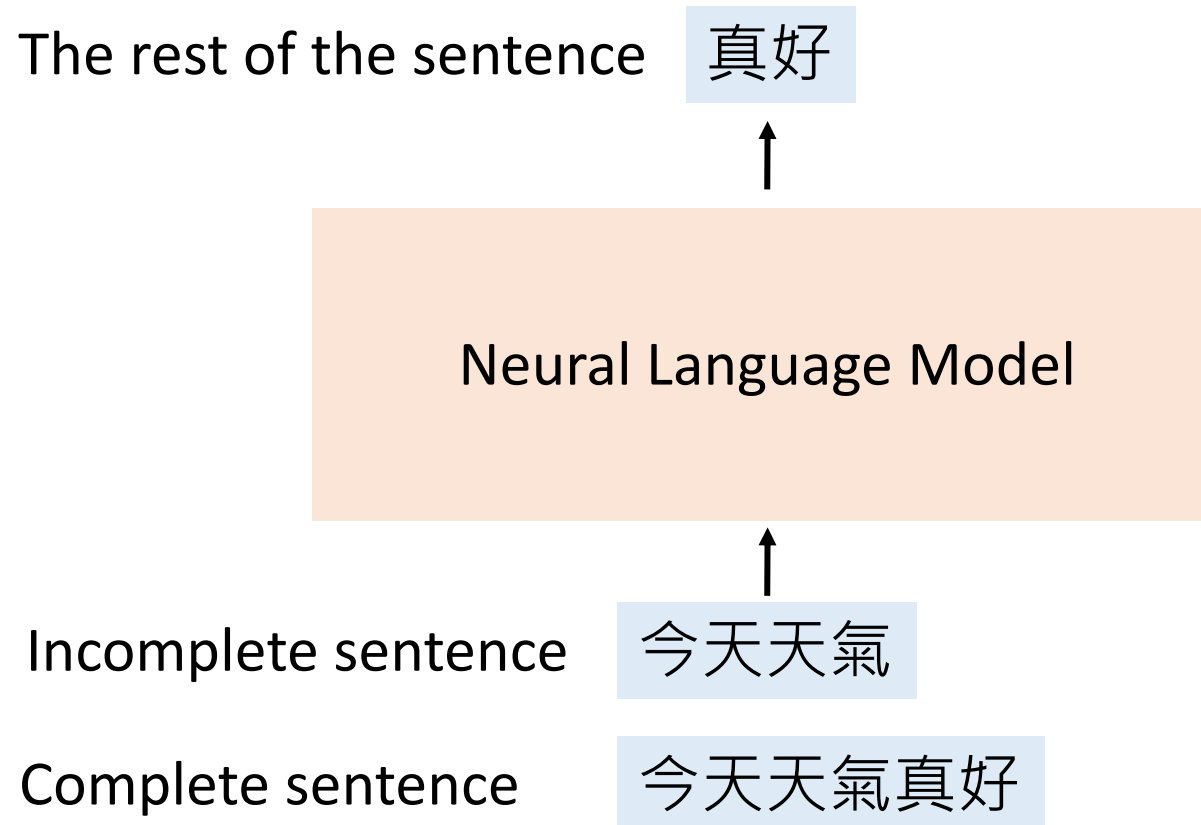
Background: Pre-trained Language Models

- Neural Language Models: A neural network that defines the probability over sequences of words



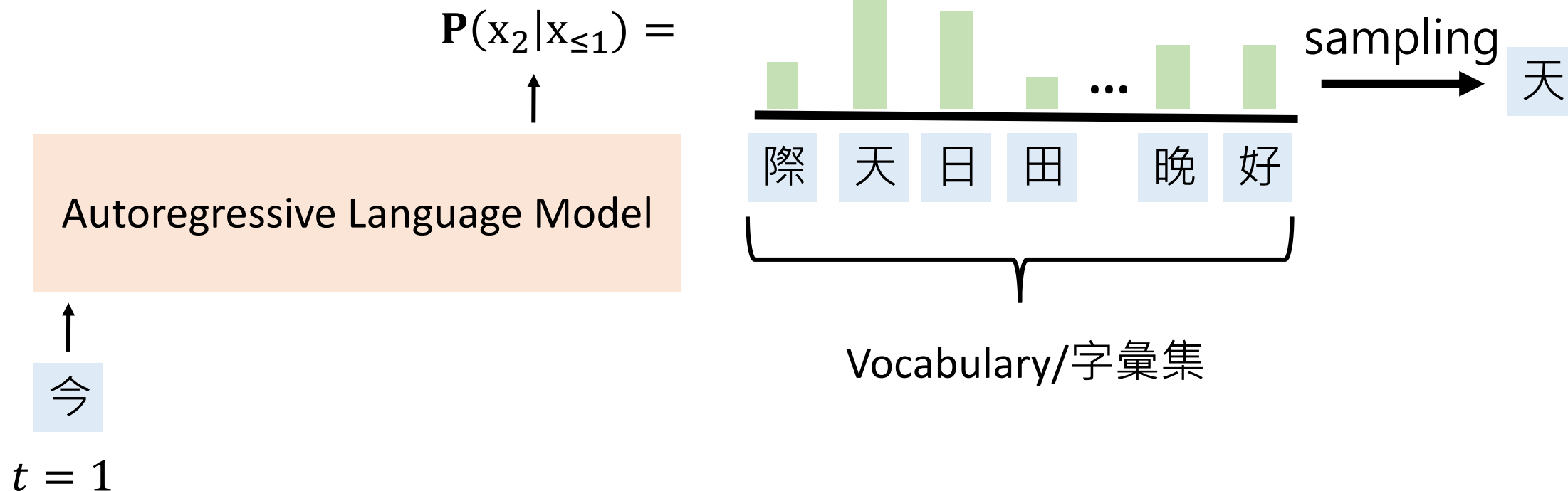
Background: Pre-trained Language Models

- How are these language models trained?
 - Given an incomplete sentence, predict the rest of the sentence



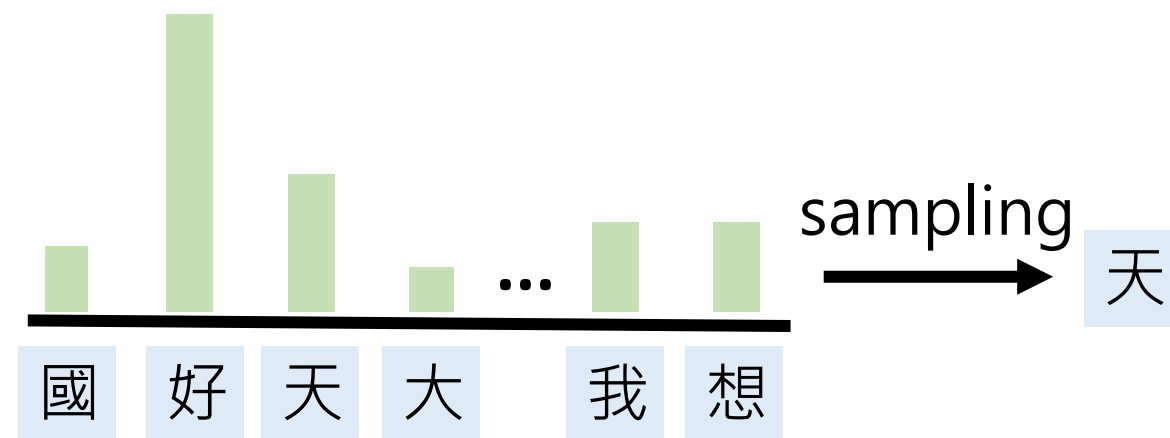
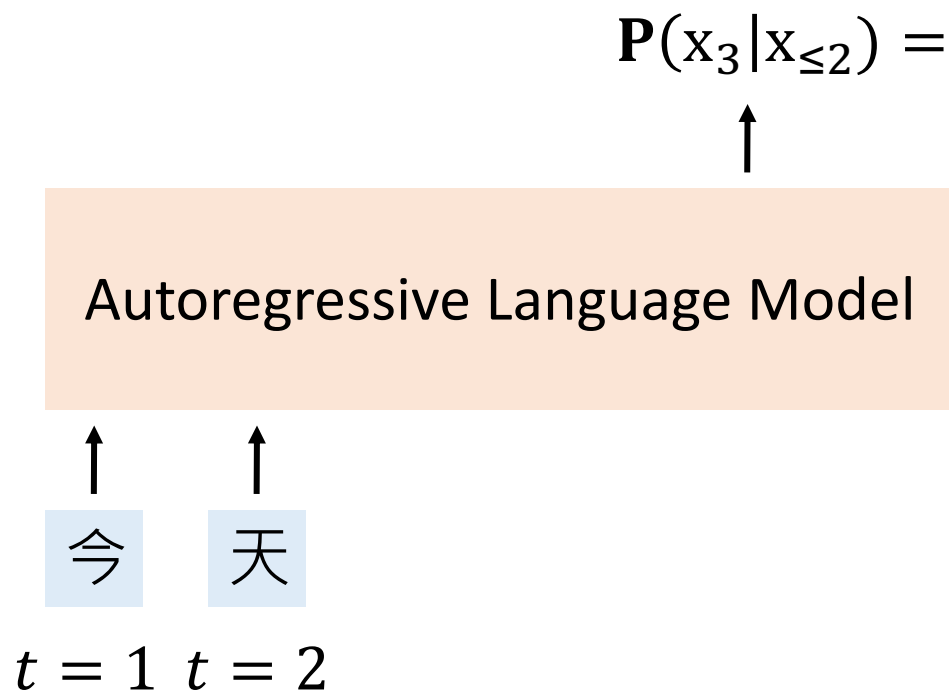
Background: Pre-trained Language Models

- Autoregressive Language Models (ALMs): Complete the sentence given its prefix
 - Sentence completion



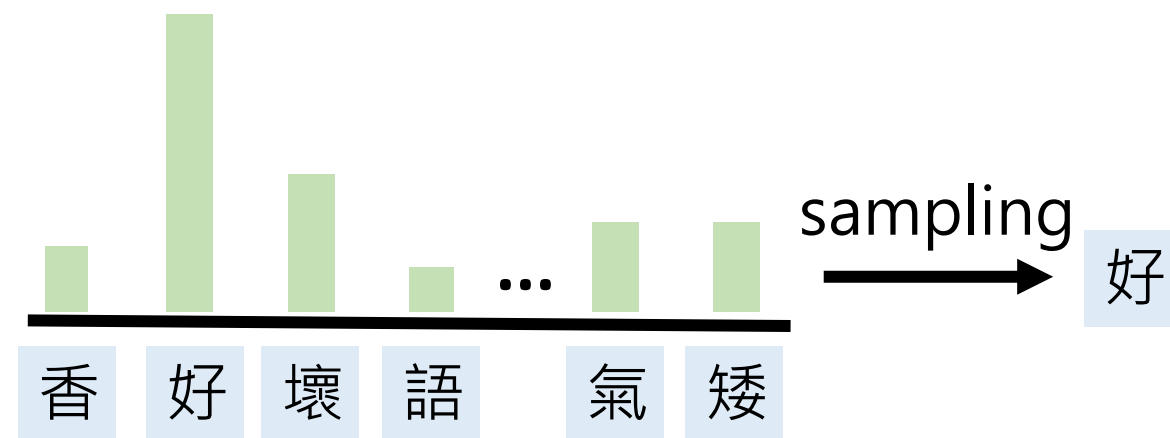
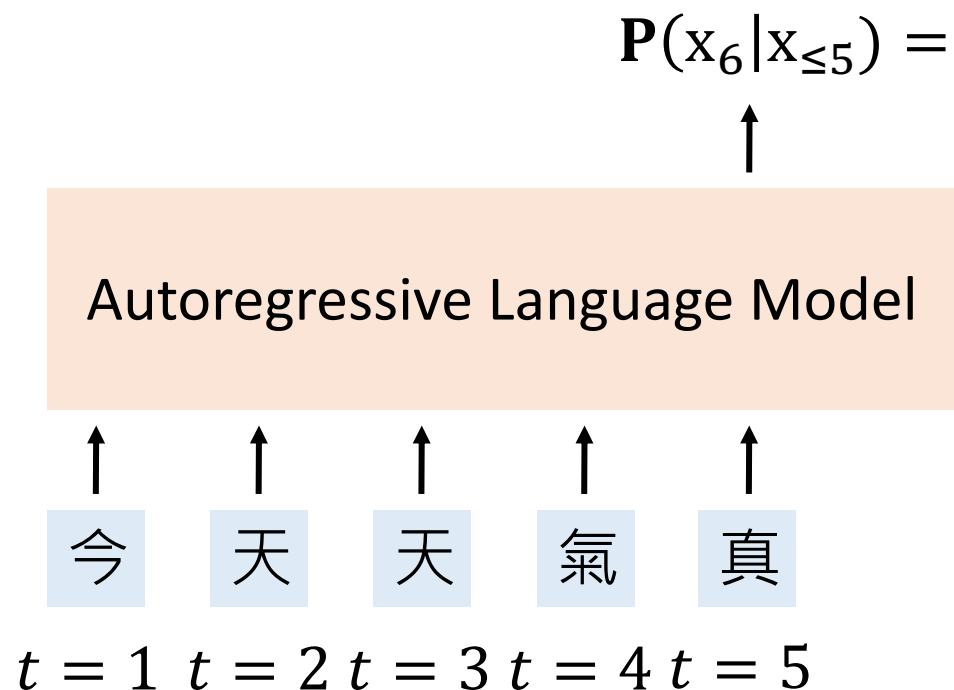
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- Autoregressive Language Models (ALMs): Complete the sentence given its prefix
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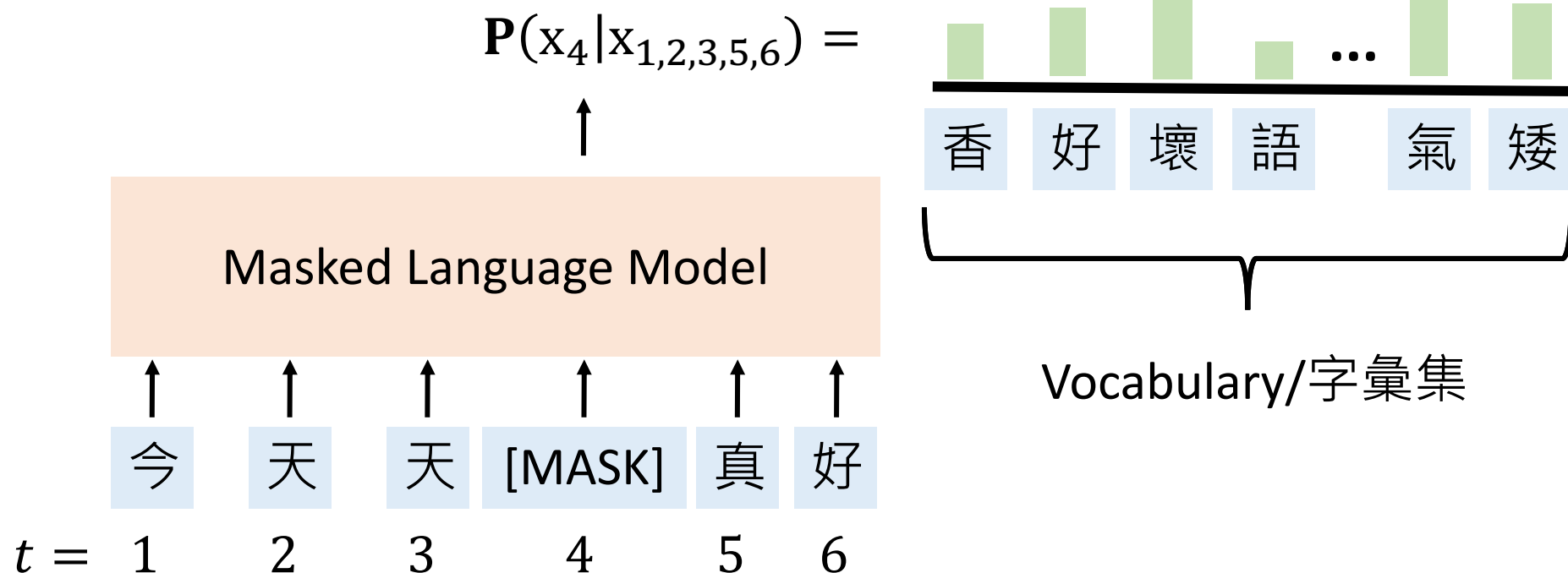
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- Autoregressive Language Models (ALMs): Complete the sentence given its prefix
 - Sentence completion



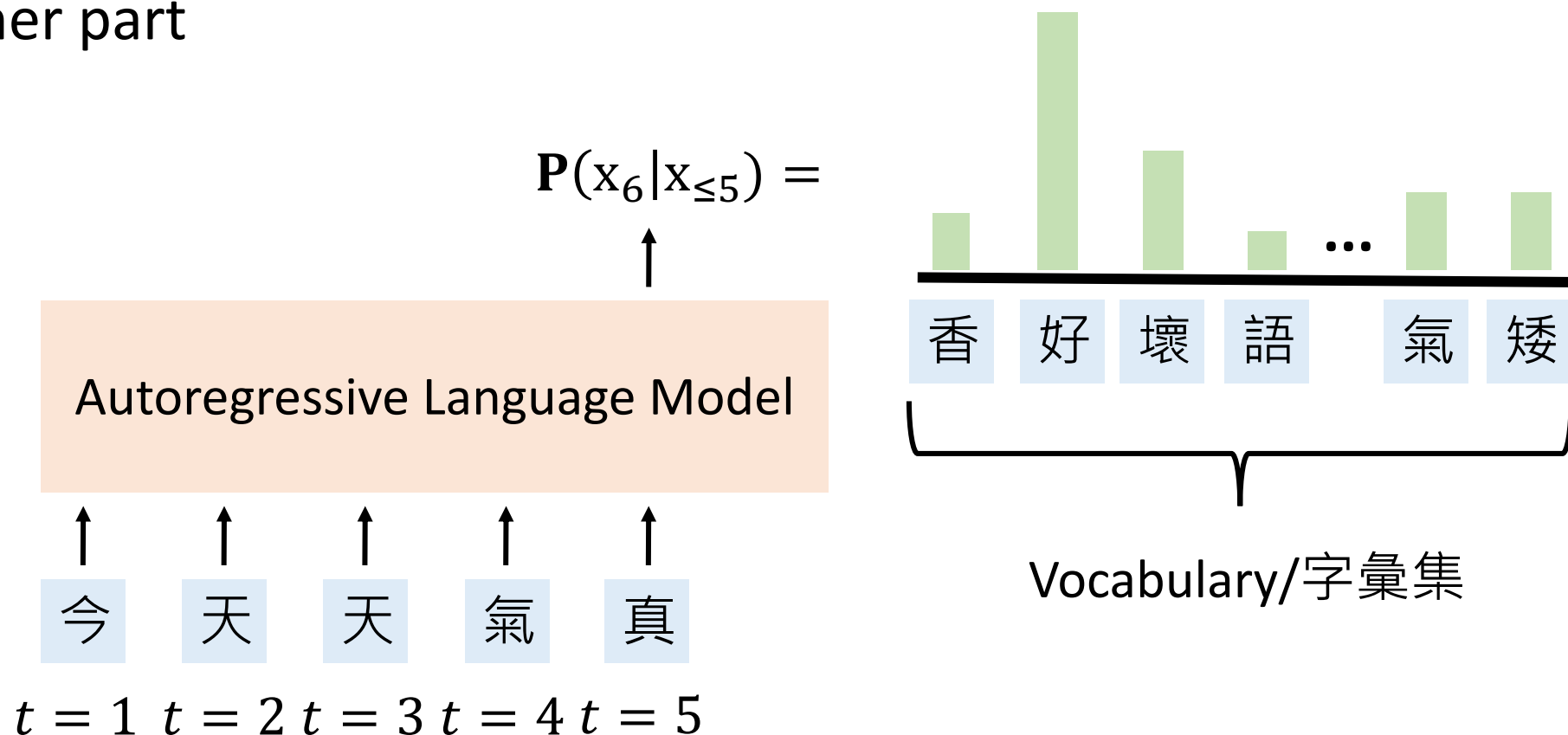
Background: Pre-trained Language Models

- Masked Language Models (MLMs): Use the unmasked words to predict the masked word
 - Cloze



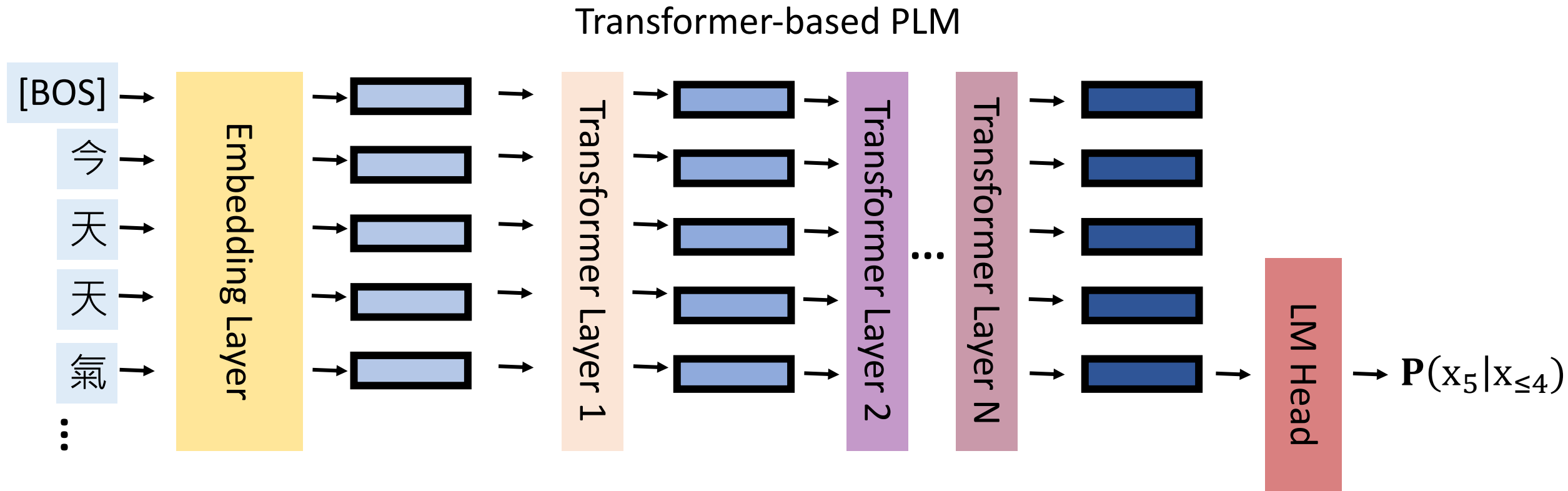
Background: Pre-trained Language Models

- Training a language model is self-supervised learning
- Self-supervised learning :Predicting any part of the input from any other part



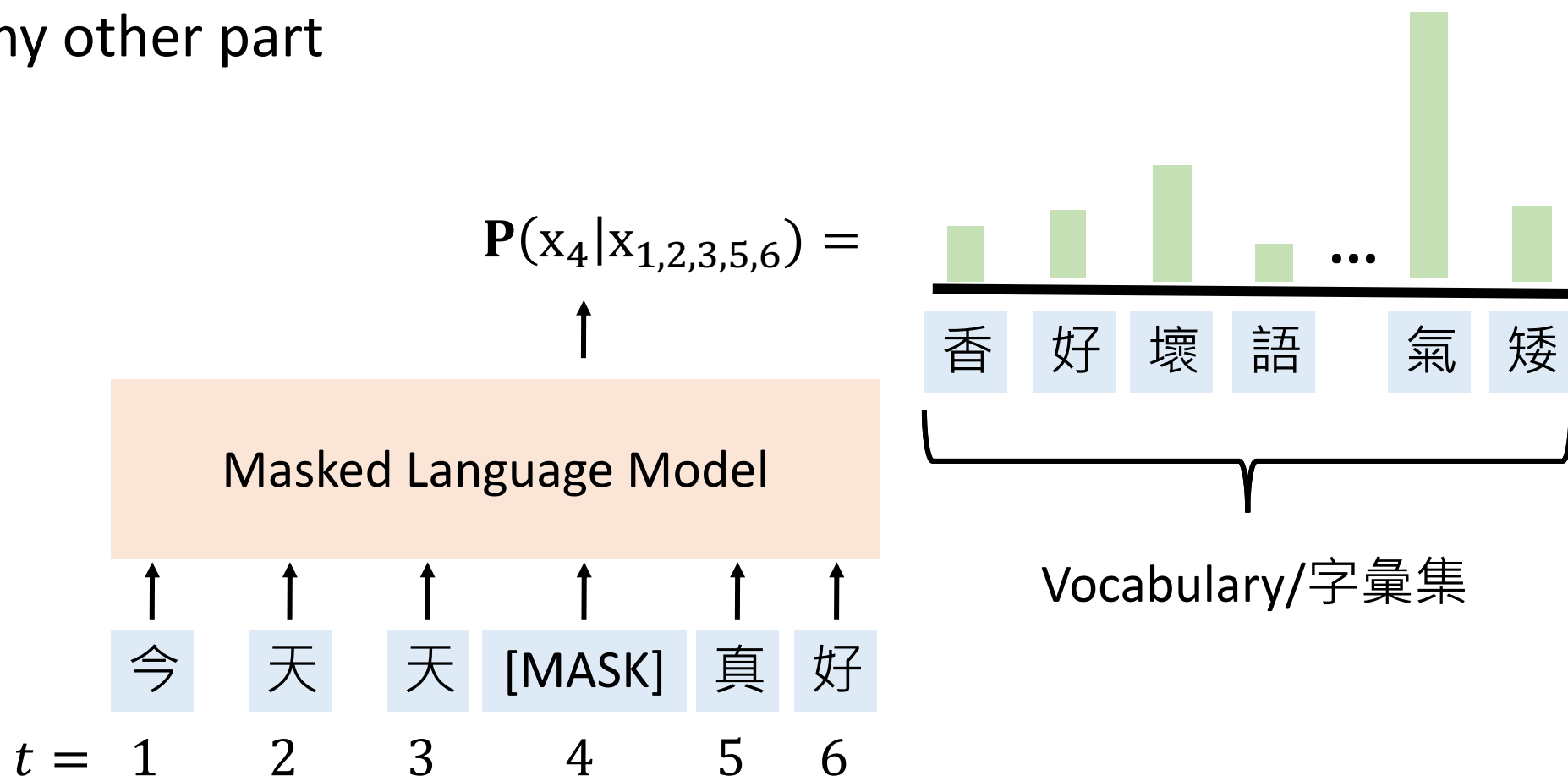
Background: Pre-trained Language Models

- Transformer-based ALMs: Composed of stacked layers of transformer layers



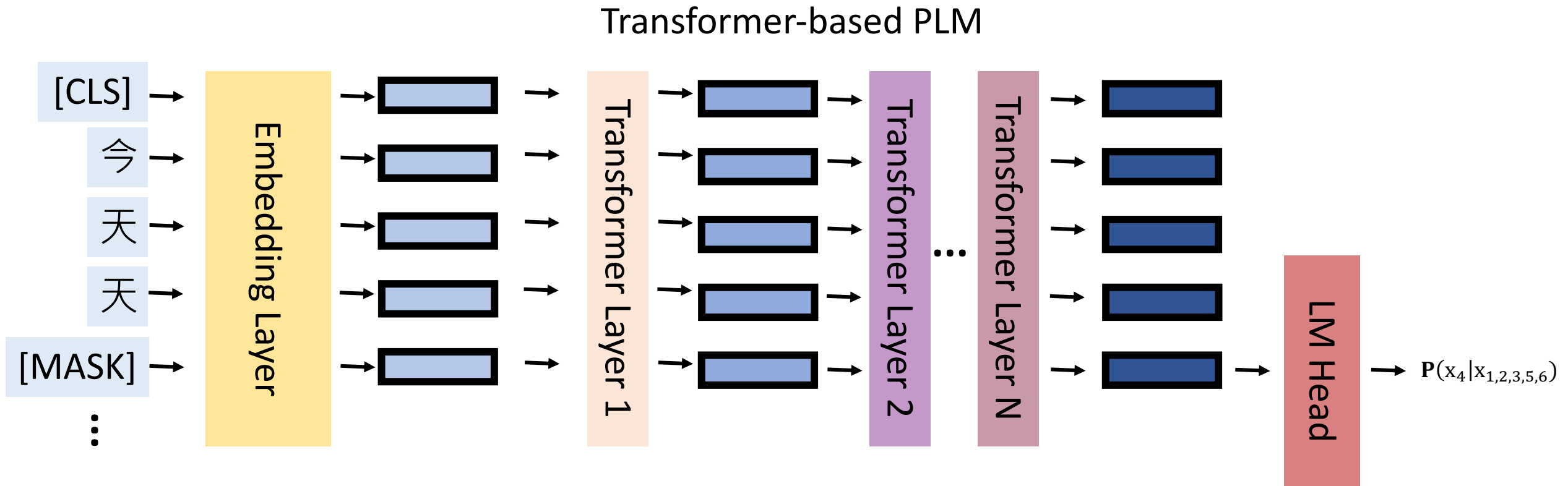
Background: Pre-trained Language Models

- Training a language model is self-supervised learning
- Self-supervised learning :Predicting any part of the input from any other part



Background: Pre-trained Language Models

- Transformer-based PLMs: Composed of stacked layers of transformer layers



Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - **Pre**-training/預訓練: Using a large corpora to train a neural language model
 - Autoregressive pre-trained: GPT 系列 (GPT, GPT-2, GPT-3)
 - MLM-based pre-trained: BERT 系列 (BERT, RoBERTa, ALBERT)

Pre-trained Language Model
(Pre-training)

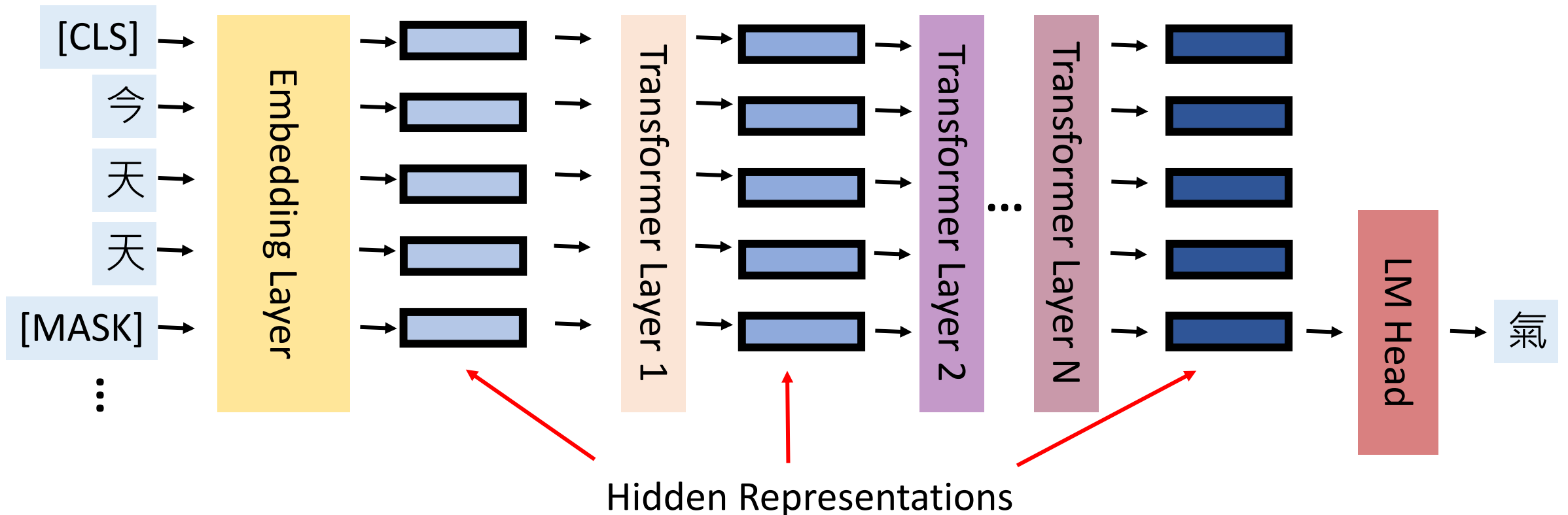


WIKIPEDIA
The Free Encyclopedia

Common
Crawl

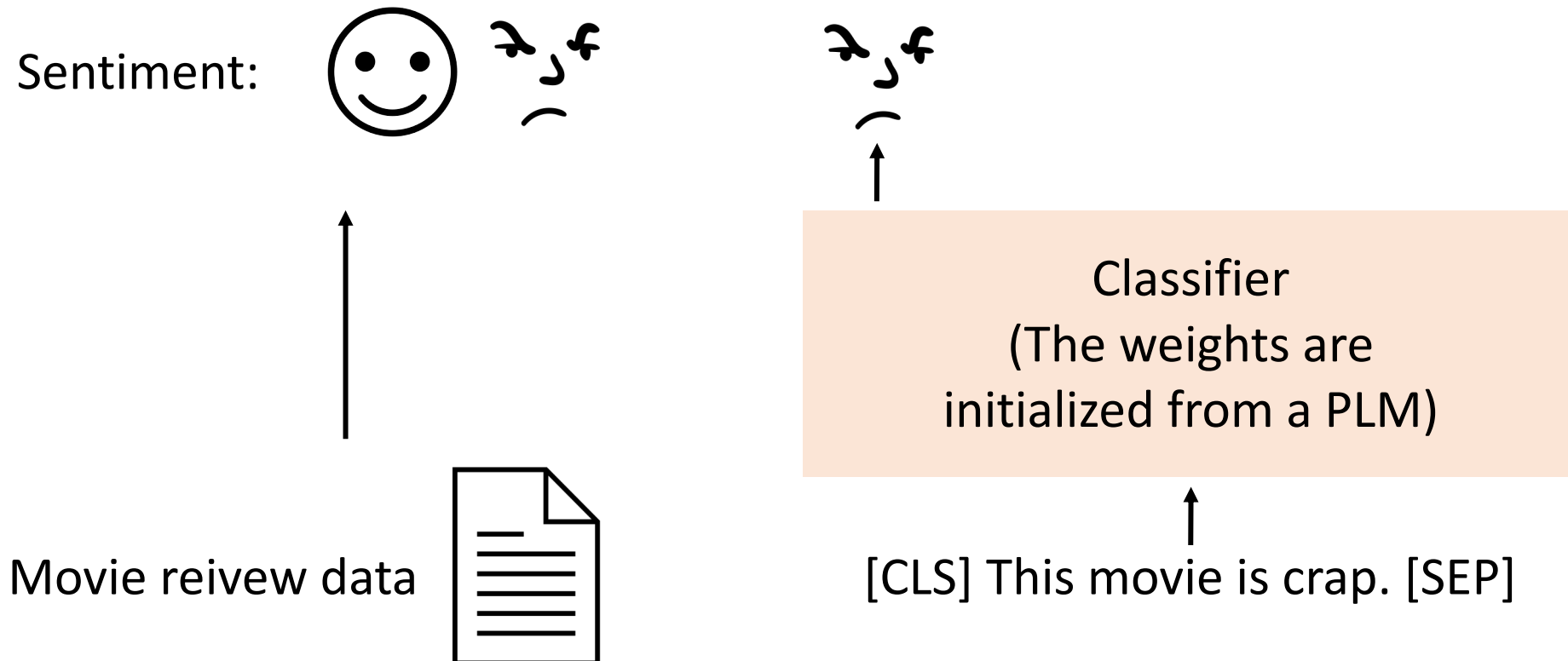
Background: Pre-trained Language Models

- We believe that after pre-training, the PLM learns some knowledge, encoded in its hidden representations, that can transfer to downstream tasks



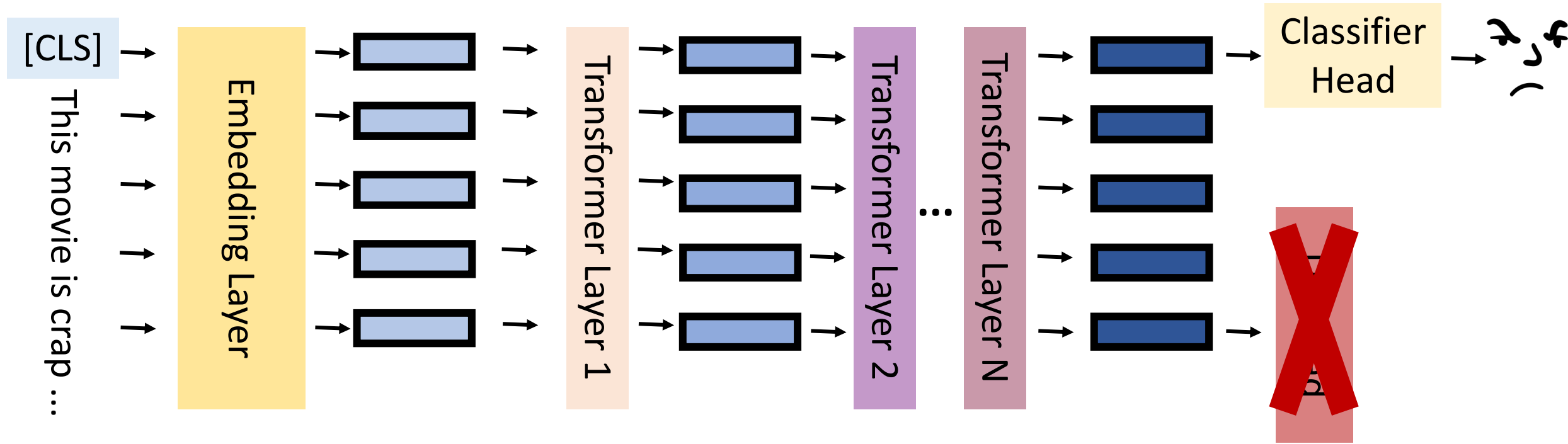
Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - (Standard) fine-tuning/微調: Using the pre-trained weights of the PLM to initialize a model for a **downstream task**



Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - (Standard) fine-tuning/微調: Using the pre-trained weights of the PLM to initialize a model for a **downstream task**



Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - Fine-tuning PLMs on downstream tasks achieves exceptional performance on many kinds of downstream tasks

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - PLMs are widely applied to many different scenarios in different realms

KFU NLP Team at SMM4H 2019 Tasks: Want to Extract Adverse Drugs

F A Simple Cross-Lingual Lemmatization and Morphology Tagging with Two-Stage Multilingual BERT Fine-Tuning

Zulfat Miftah

Kazar

Transfe

Yasuh

TMU Transformer System Using BERT for Re-ranking at BEA 2019 Grammatical Error Correction on Restricted Track

Incorporating medical knowledge in BERT for clinical relation extraction

S

Arpita Roy and Shimei Pan

Department of Information Systems

University of Maryland, Baltimore County

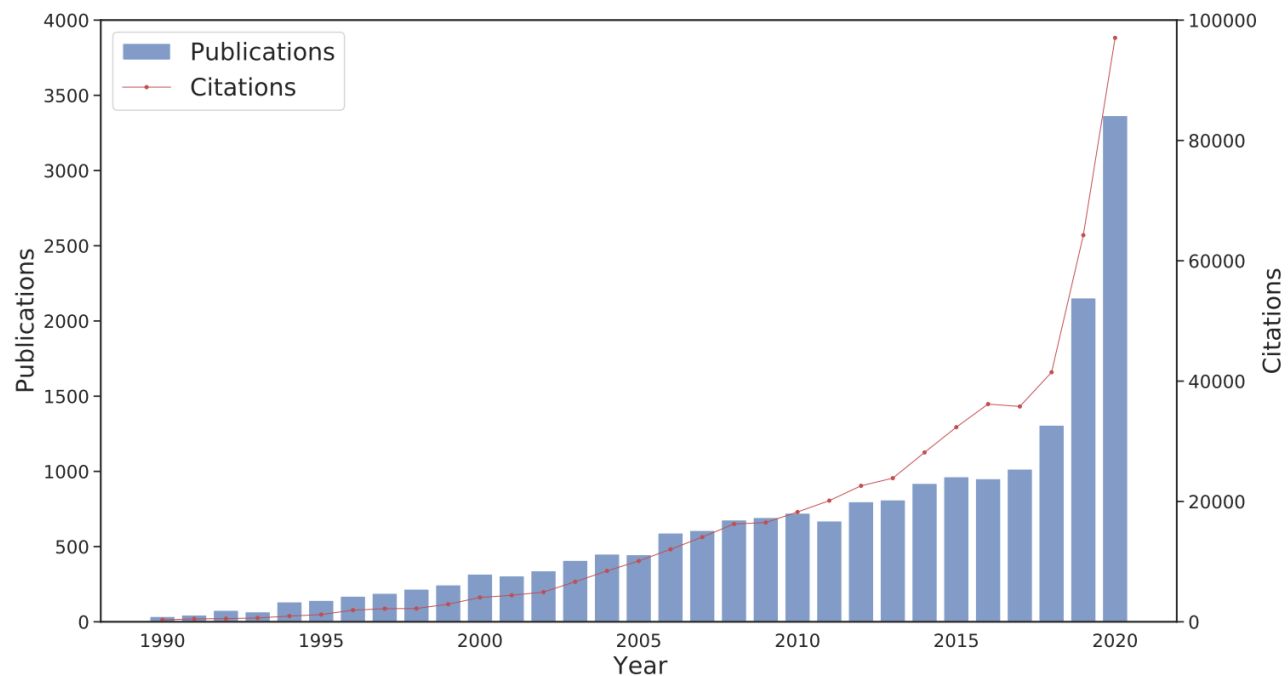
Maryland, USA

{ arpita2, shimei }@umbc.edu

machi

Background: Pre-trained Language Models

- Pre-trained Language Models (PLMs)/ 預訓練語言模型
 - PLMs are every where



(a) The number of publications on “language models” and their citations in recent years.

Background: Pre-trained Language Models

- PLMs has shown great success on a variety of benchmark datasets in NLP
- The next goal is to make PLMs fit in real-life use case
 - How unrealistic is PLMs nowadays?

Outline

- Background knowledge
- **The Problems of PLMs**
- The Solutions of Those Problems
- Closing Remarks

The Problems of PLMs

- Problem 1: **Data scarcity** in downstream tasks
- A large amount of labeled data is not easy to obtain for each downstream task

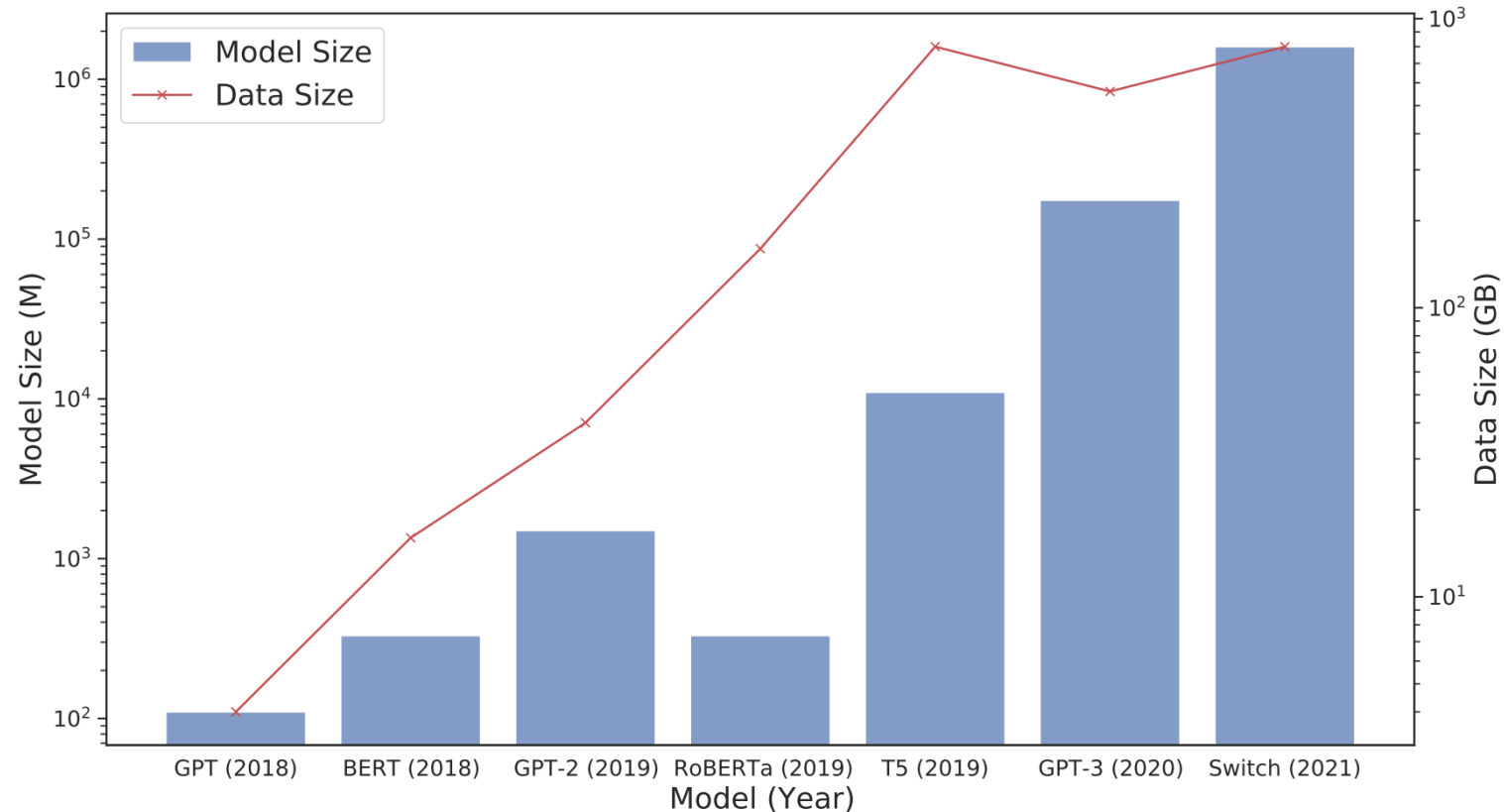


Pre-trained Language Model
(Fine-tuning)

MNLI-(m/mm)	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE
392k	363k	108k	67k	8.5k	5.7k	3.5k	2.5k

The Problems of PLMs

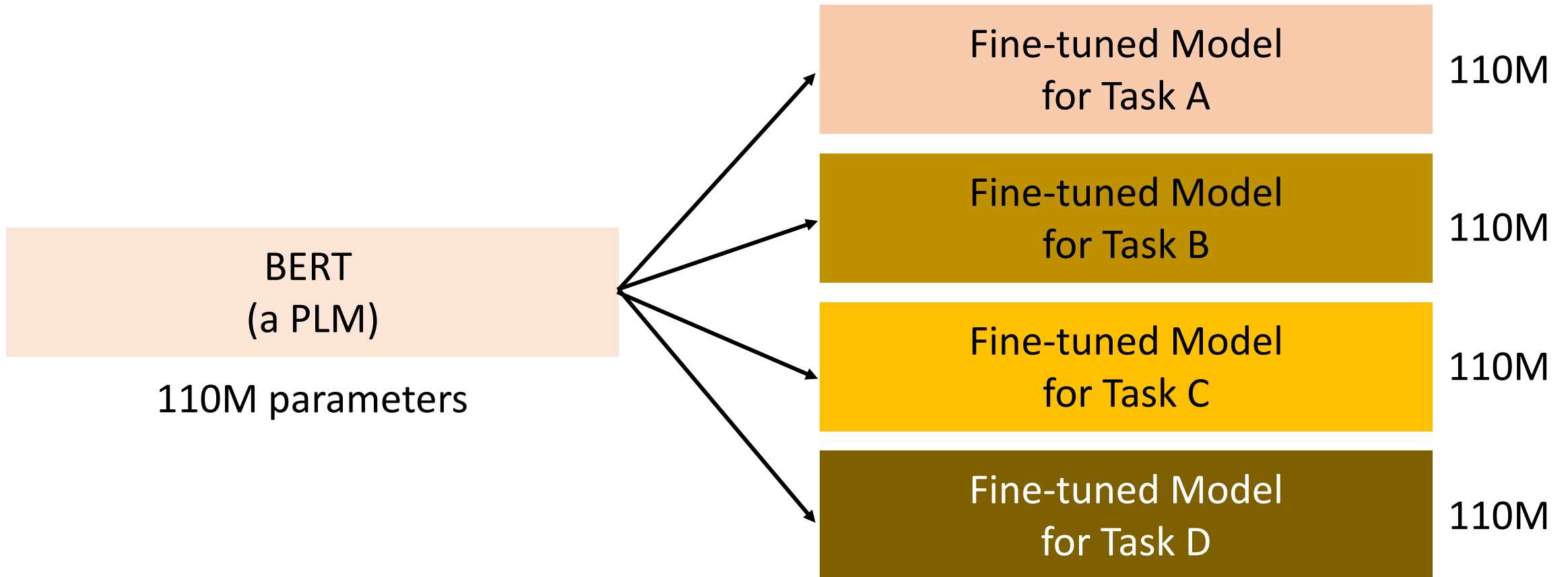
- Problem 2: The PLM is too big, and they are still getting bigger



(b) The model size and data size applied by recent NLP PTMs.

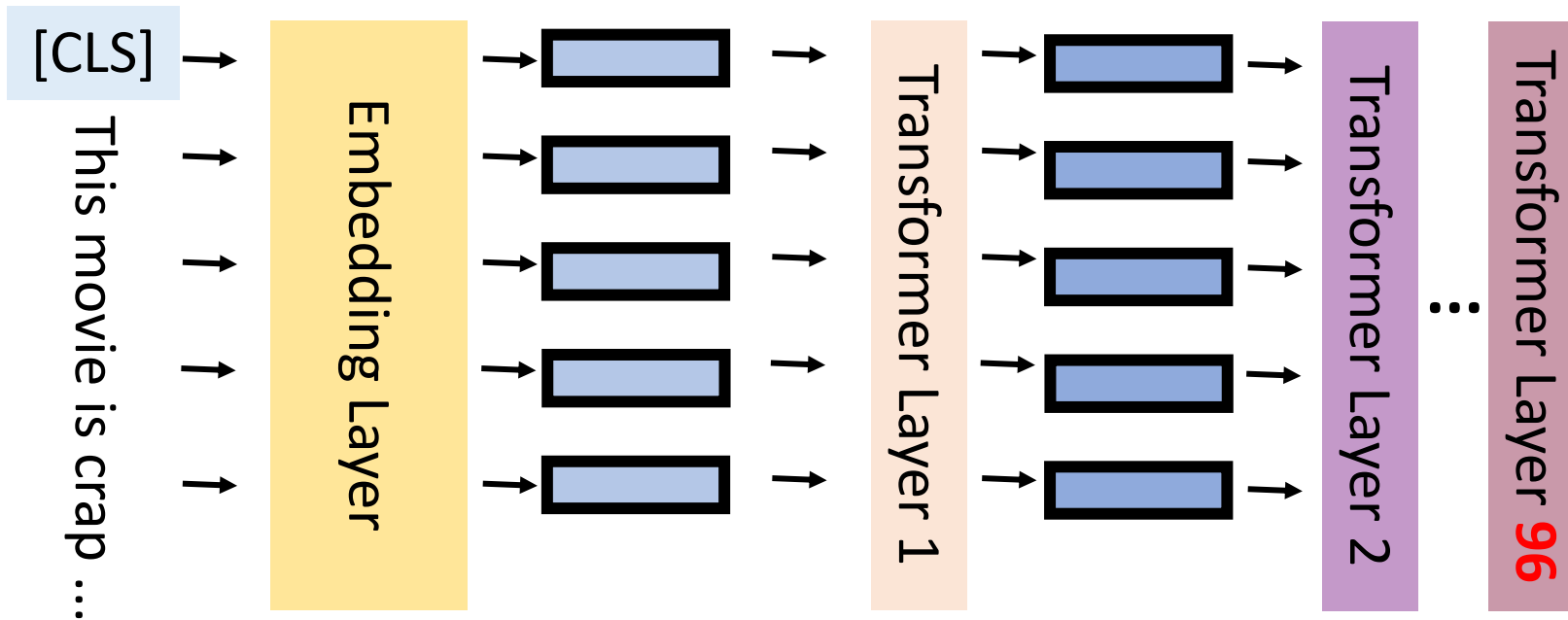
The Problems of PLMs

- Problem 2: The PLM is too big
 - Need a copy for each downstream task



The Problems of PLMs

- Problem 2: The PLM is too big
 - Inference takes too long
 - Consume too much space

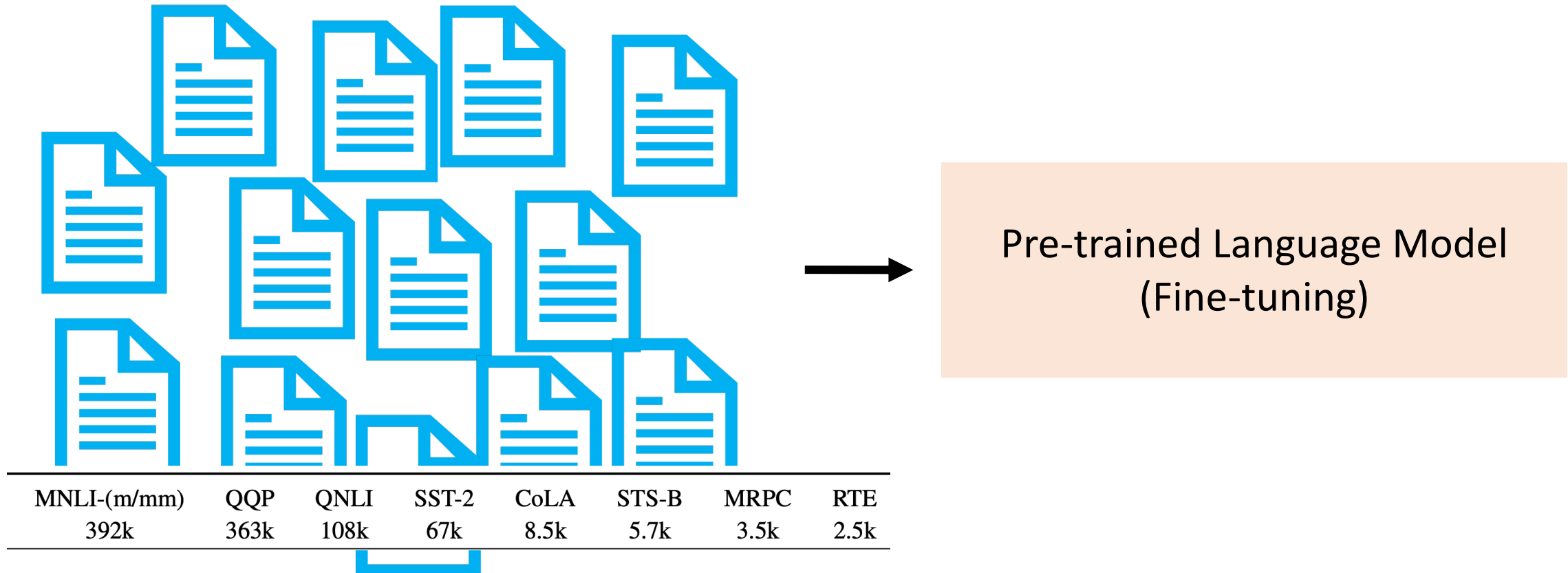


Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - PLMs Are Gigantic → Reducing the Number of Parameters
- Closing Remarks

The Problems of PLMs

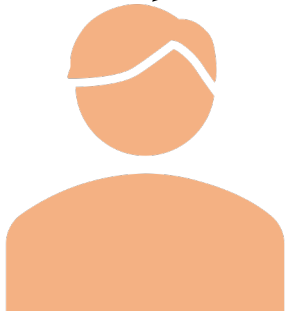
- Problem 1: **Data scarcity** in downstream tasks
- A large amount of labeled data is not easy to obtain for each downstream task



Outline

- Background knowledge
- The Problems of PLMs
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 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - Prompt Tuning
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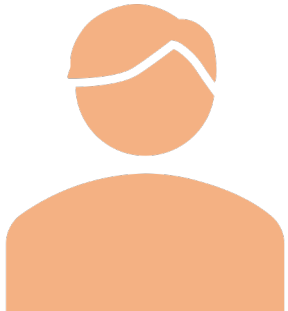
- [CLS] Jack likes dog. [SEP] Jack loves ice cream. [SEP] >>> **neutral**
- [CLS] The spring break is coming soon. [SEP] The spring break was over. [SEP] >>> **contradiction**
- [CLS] I am going to have dinner. [SEP] I am going to eat something. [SEP] >>> **entailment**
-
- [CLS] Mary likes pie. [SEP] Mary hates pie. [SEP] >>> ?



contradiction



- [CLS] The spring break is coming soon. [SEP] The spring break was over. [SEP] >>> **contradiction**
- [CLS] I am going to have dinner. [SEP] I am going to eat something. [SEP] >>> **entailment**
- [CLS] Mary likes pie. [SEP] Mary hates pie. [SEP] >>> ?

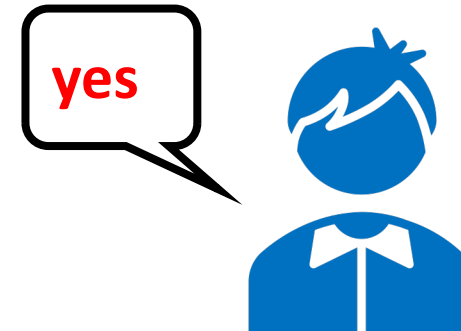
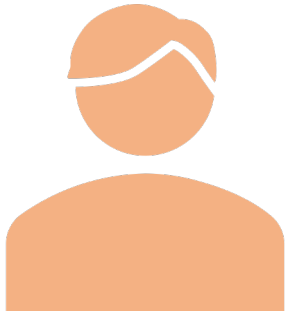


?????



Data-Efficient Fine-tuning: Prompt Tuning

- [CLS] The spring break is coming soon. Is it true that the spring break was over? >>> no
- [CLS] I am going to have dinner. Is it true that I am going to eat something? >>> yes
- [CLS] Mary likes pie. Is it true that Mary hates pie. [SEP] >>> ?



Natural language inference

Data-Efficient Fine-tuning: Prompt Tuning

- By converting the data points in the dataset into natural language prompts, the model may be easier to know what it should do

- [CLS] The spring break is coming soon. [SEP] The spring break was over. [SEP] >>> **contradiction**
- [CLS] I am going to have dinner. [SEP] I am going to eat something. [SEP] >>> **entailment**
- [CLS] Mary likes pie. [SEP] Mary hates pie. [SEP] >>> ?

- [CLS] The spring break is coming soon. **Is it true that** the spring break was over? >>> **no**
- [CLS] I am going to have dinner. **Is it true that** I am going to eat something? >>> **yes**
- [CLS] Mary likes pie. **Is it true that** Mary hates pie. [SEP] >>> ?

Data-Efficient Fine-tuning: Prompt Tuning

- Format the downstream task as a language modelling task with pre-defined templates into natural language **prompts**

prompt

noun [C]

UK  /prompt/ US  /pra:mpt/

prompt *noun* [C] (COMPUTER)



a sign on a computer screen that shows that the computer is ready to receive your instructions

(電腦螢幕上的) **提示符** (顯示電腦已經準備好接受指令)

prompt *noun* [C] (ACTOR'S HELP)



words that are spoken to an actor who has forgotten what he or she is going to say during the performance of a play

(給演員的) **提詞，提白**



Data-Efficient Fine-tuning: Prompt Tuning

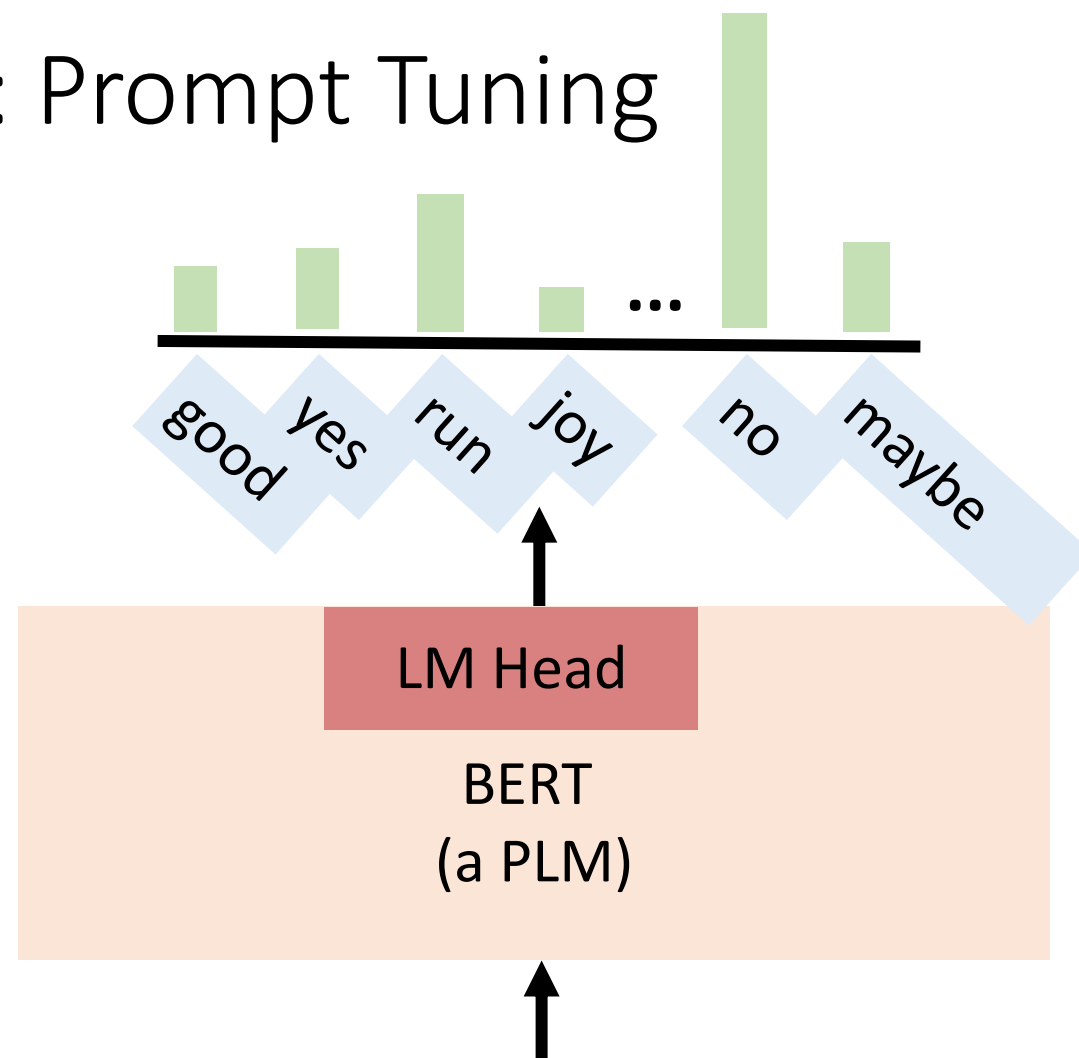
- What you need in prompt tuning

1. A prompt template
2. A PLM
3. A verbalizer

Premise	Mary likes pie.
Hypothesis	Mary hates pie.
Label	2

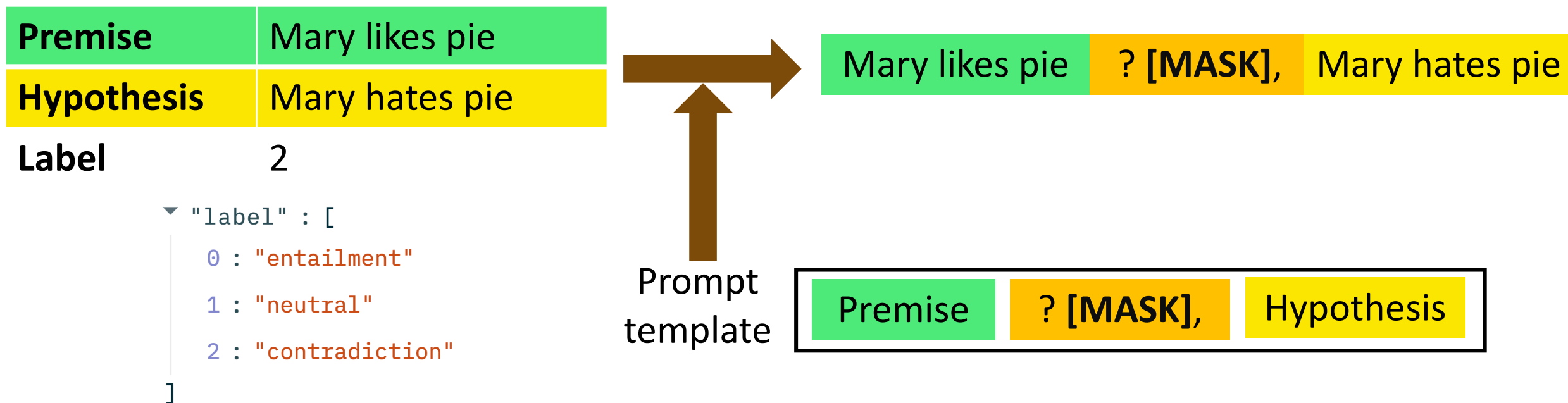
```
▼ "label" : [  
  0 : "entailment"  
  1 : "neutral"  
  2 : "contradiction"  
]
```

Prompt
template:



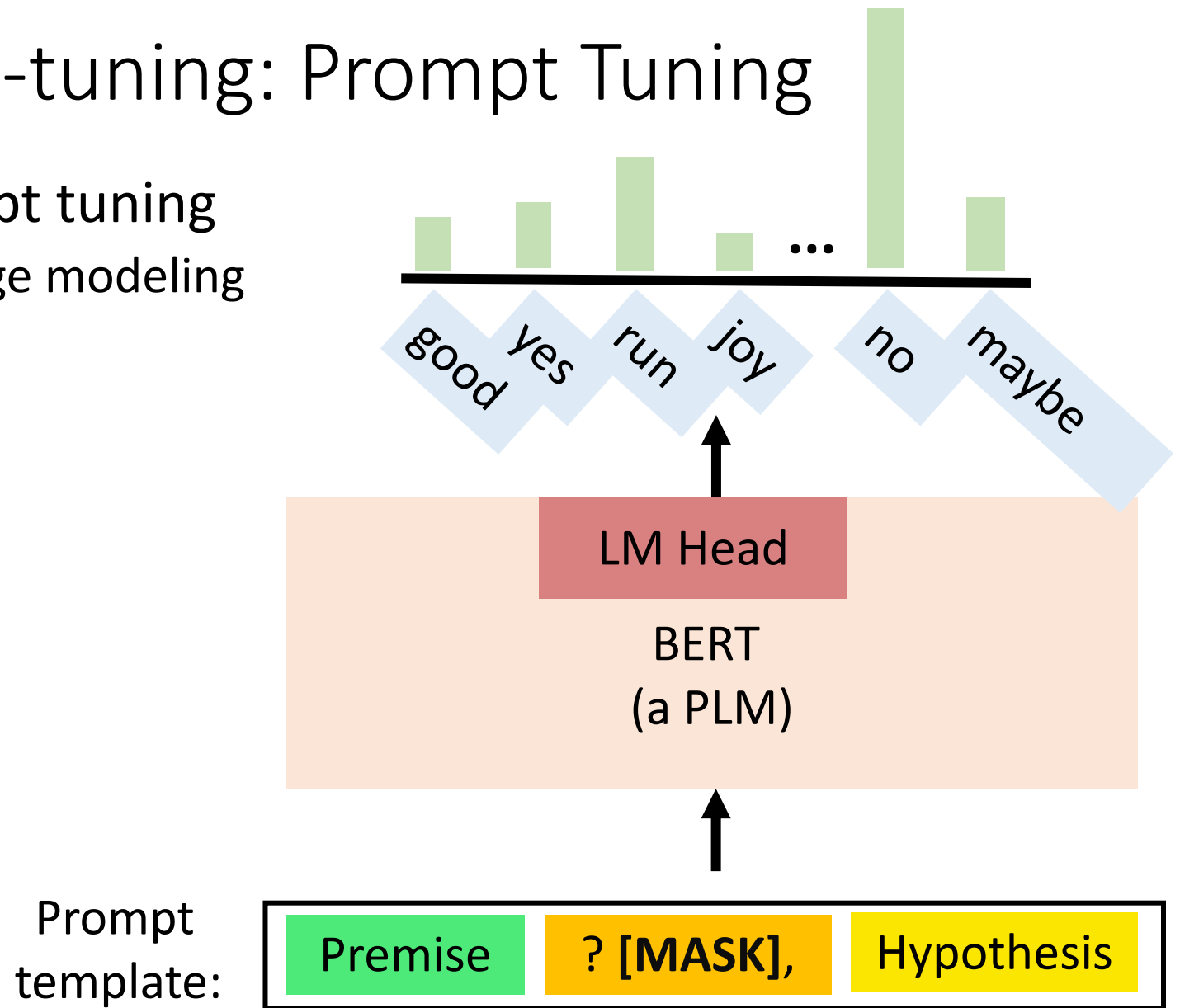
Data-Efficient Fine-tuning: Prompt Tuning

- What you need in prompt tuning
 1. A prompt template: convert data points into a natural language prompt



Data-Efficient Fine-tuning: Prompt Tuning

- What you need in prompt tuning
 2. A PLM: perform language modeling

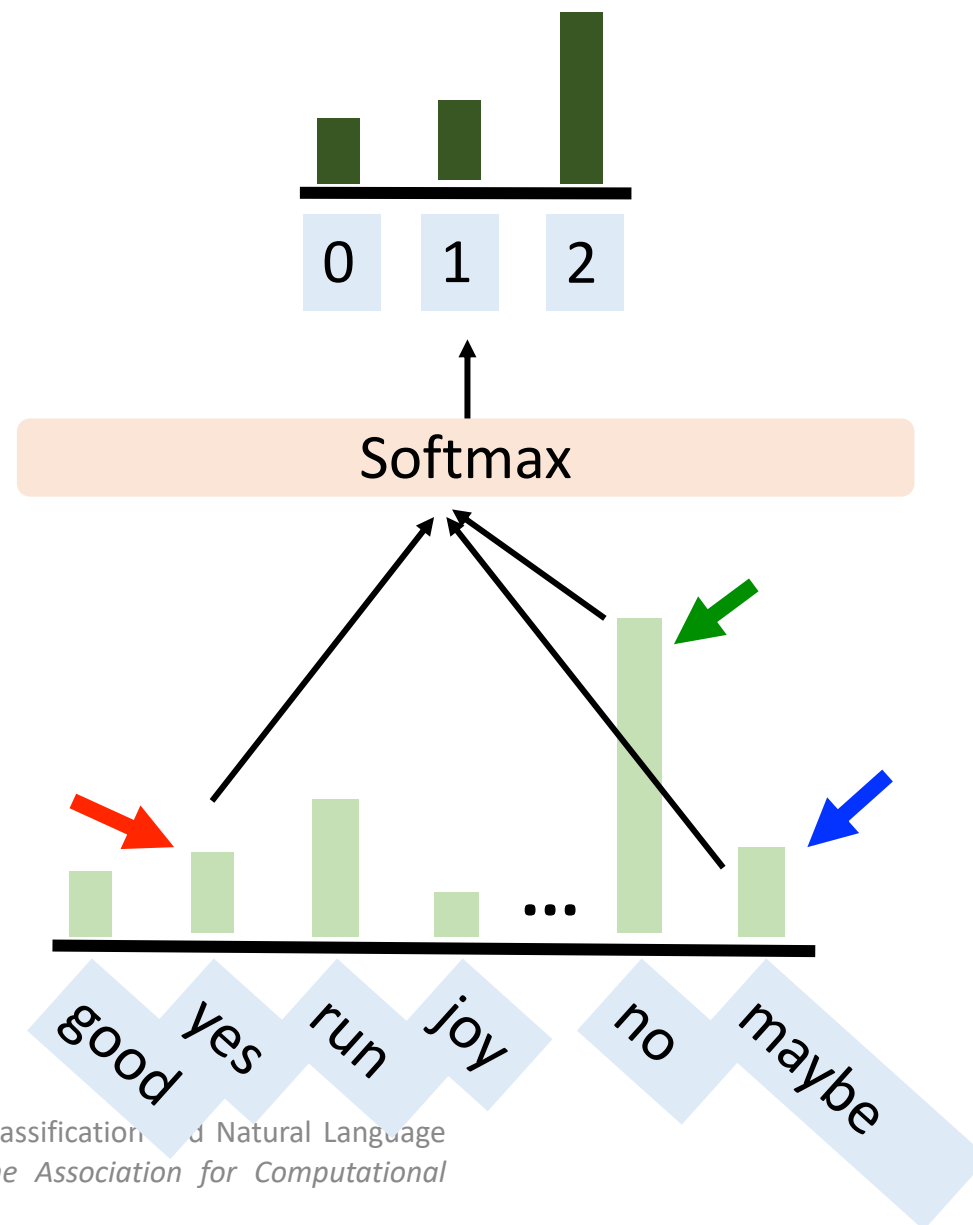
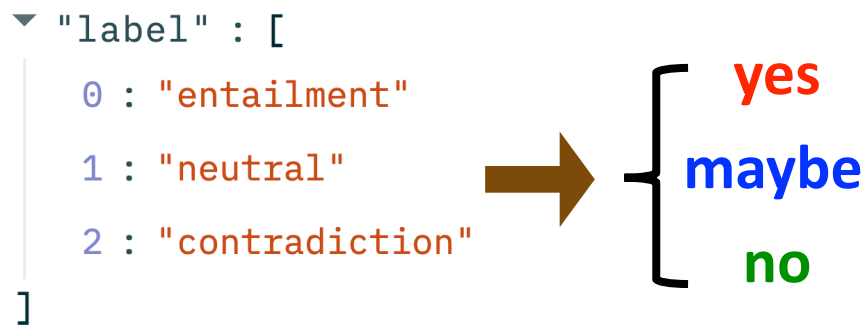


Data-Efficient Fine-tuning: Prompt Tuning

- What you need in prompt tuning

3. A verbalizer: A mapping between the label and the vocabulary

- Which vocabulary should represents the class “entailment”



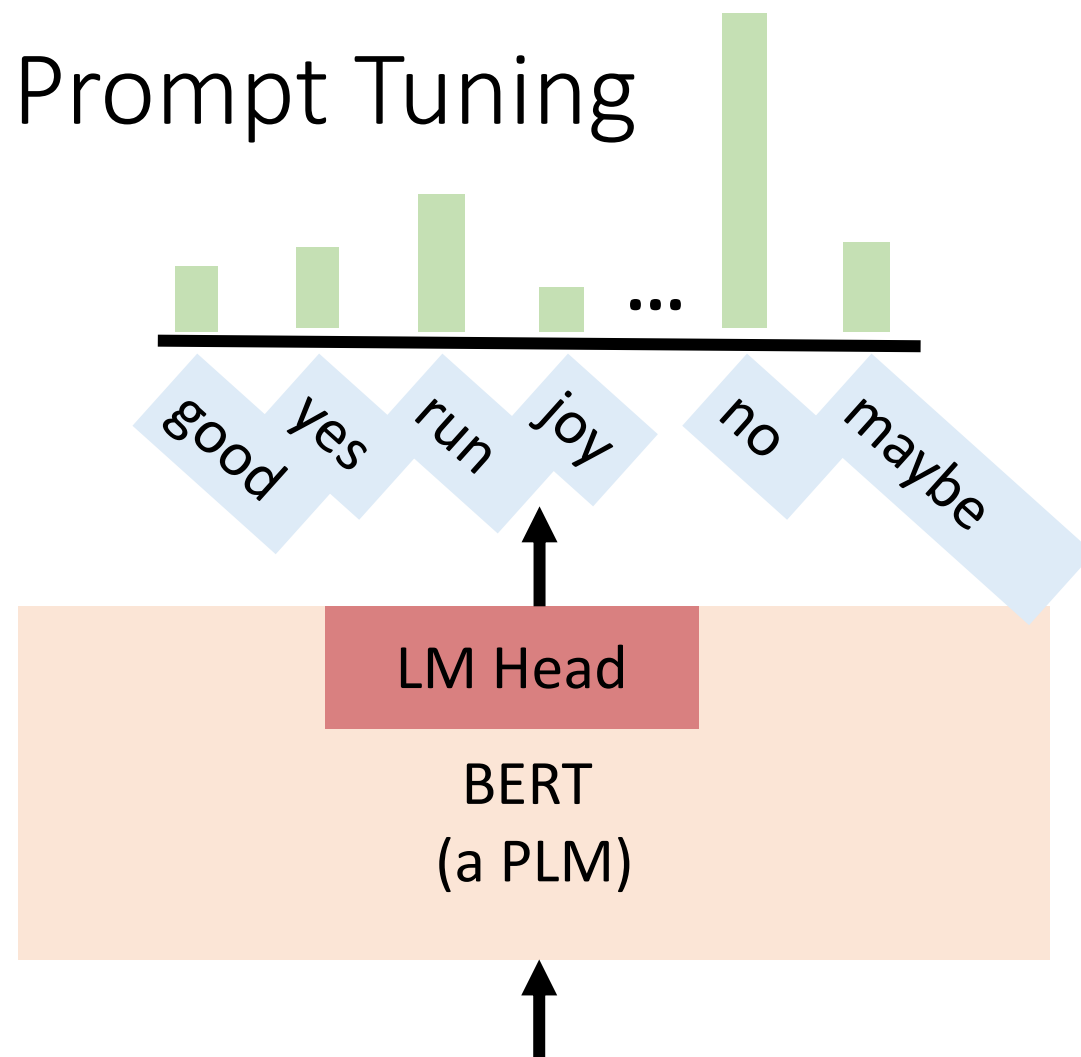
Data-Efficient Fine-tuning: Prompt Tuning

- Prompt tuning
 - The whole PLM will be fine-tuned

Premise	Mary likes pie.
Hypothesis	Mary hates pie.
Label	2

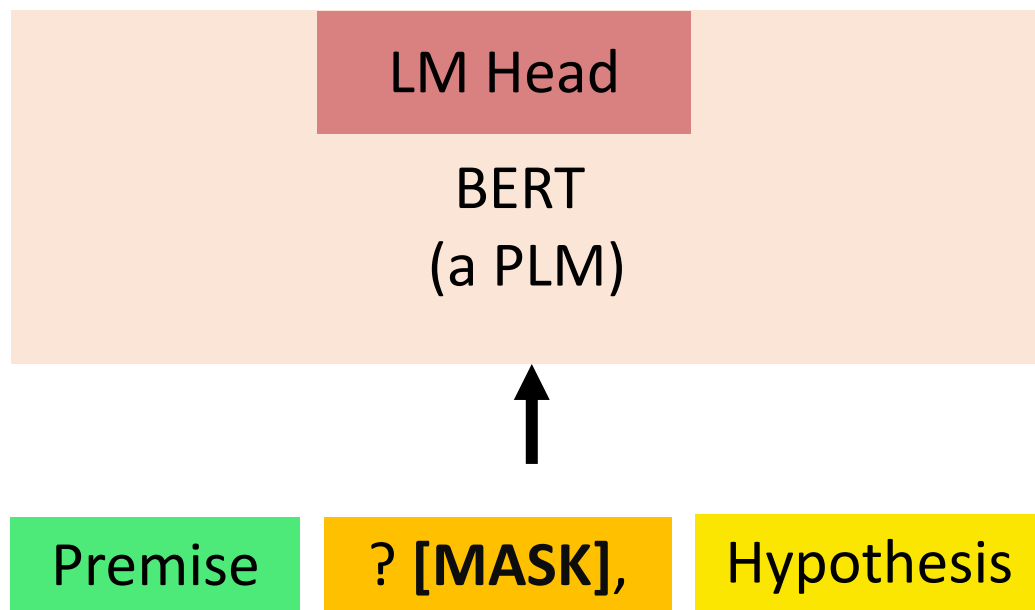
```
▼ "label" : [  
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Prompt
template:

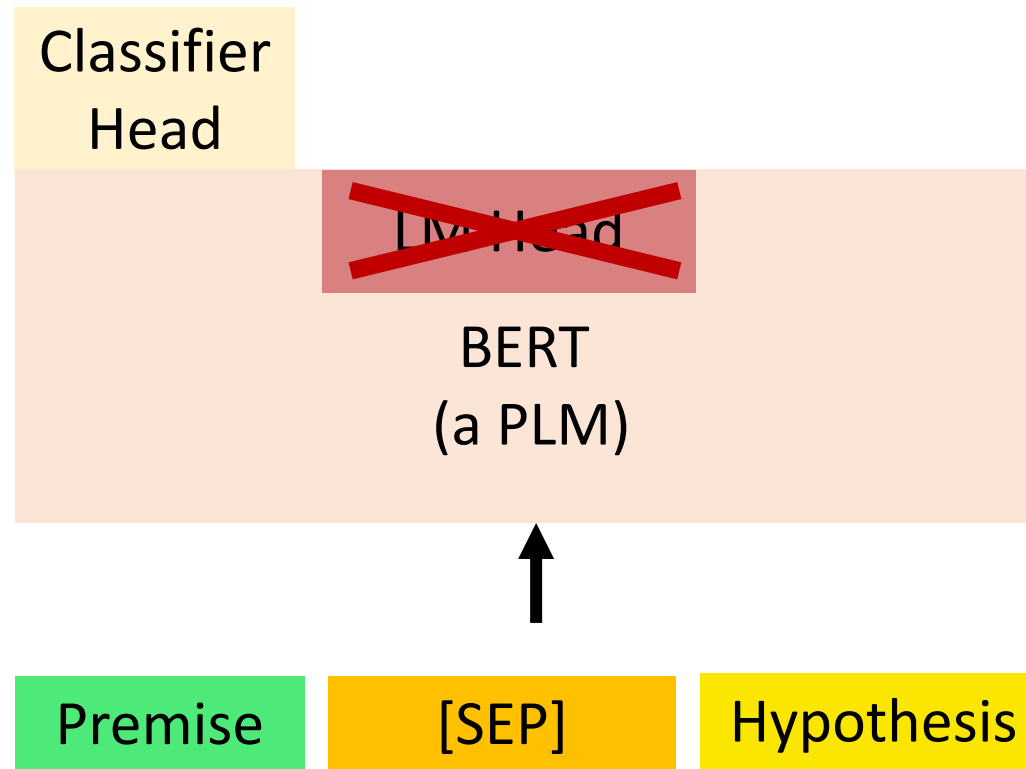


Data-Efficient Fine-tuning: Prompt Tuning

- Prompt tuning



- Standard fine-tuning

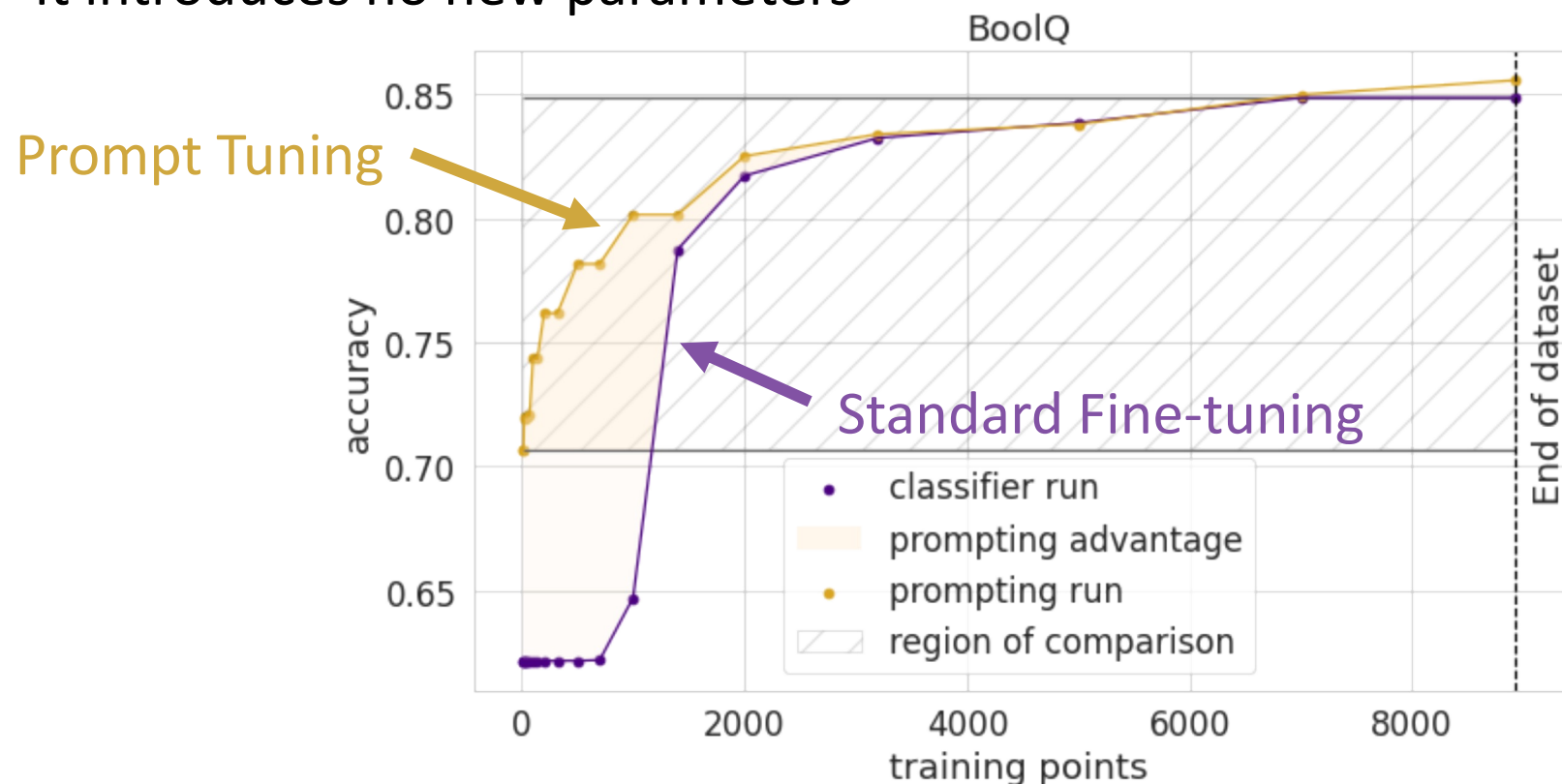


* I omit the [CLS] at the beginning and the [SEP] at the end

Schick, Timo, and Hinrich Schütze. "Exploiting Cloze-Questions for Few-Shot Text Classification and Natural Language Inference." *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*. 2021.

Data-Efficient Fine-tuning: Prompt Tuning

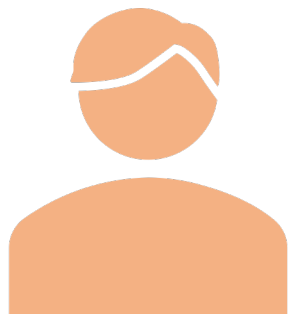
- Prompt tuning has better performance under data scarcity because
 - It incorporates human knowledge
 - It introduces no new parameters



Data-Efficient Fine-tuning: Prompt Tuning

Lets see how prompts can help us under different level of data scarcity

- [CLS] The spring break is coming soon. Is it true that the spring break was over? >>> no
- [CLS] I am going to have dinner. Is it true that I am going to eat something? >>> yes
- [CLS] Mary likes pie. Is it true that Mary hates pie. [SEP] >>> ?



Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - Few-shot Learning
 - PLMs Are Gigantic → Reducing the Number of Parameters
- Closing Remarks

Data-Efficient Fine-tuning: Few-shot Learning

- Few-shot learning: We have some labeled training data
 - "Some" ≈ 10 幾筆



Labeled Training data



Pre-trained Language Model

Data-Efficient Fine-tuning: Few-shot Learning

- Good news: GPT-3 can be used for few-shot setting
- Bad news: GPT-3 is not freely available and contains 175B parameters

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:
2  sea otter => loutre de mer
3  peppermint => menthe poivrée
4  plush girafe => girafe peluche
5  cheese => .....
```

← task description

← examples

← prompt

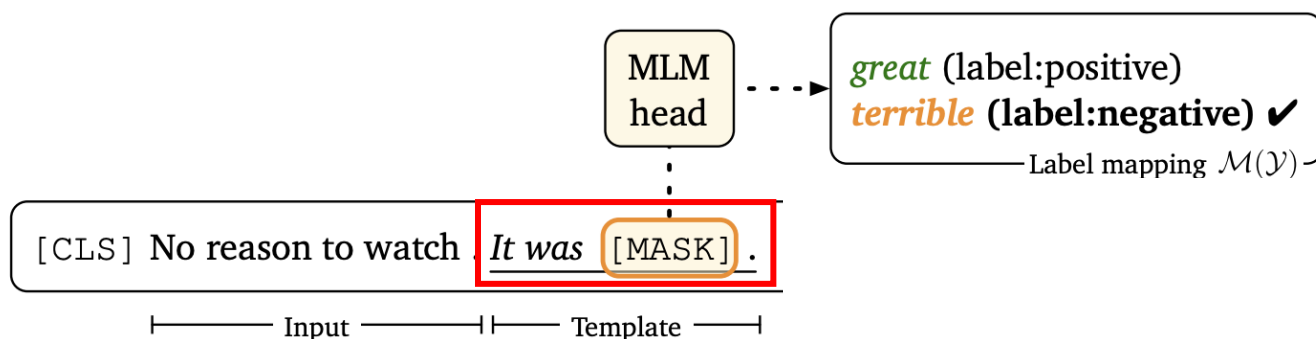
Data-Efficient Fine-tuning: Few-shot Learning

- Can we use smaller(?) PLMs and make them to perform well in few-shot learning?

- LM-BFF: better few-shot fine-tuning of language models

¹ Alternatively, language models' best friends forever.

- Core concept: **prompt** + demonstration



Data-Efficient Fine-tuning: Few-shot Learning

- LM-BFF
 - Prompt tuning: No new parameters are introduced during fine-tuning
 - Automatic template searching

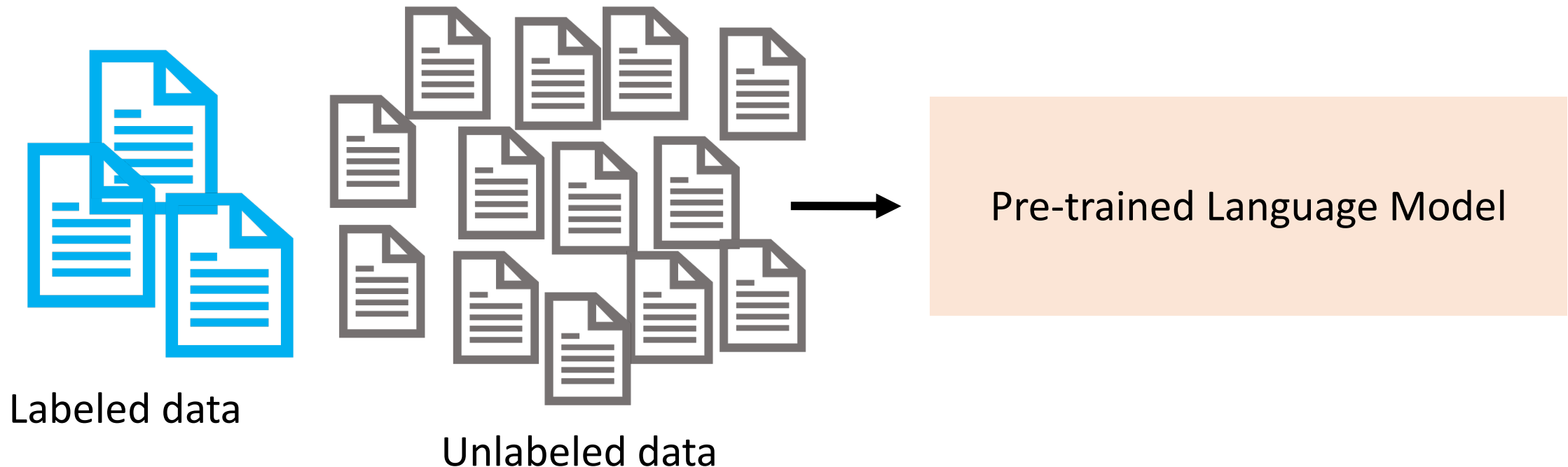
	MNLI (acc)	MNLI-mm (acc)	SNLI (acc)	QNLI (acc)	RTE (acc)	MRPC (F1)	QQP (F1)	STS-B (Pear.)
Majority [†]	32.7	33.0	33.8	49.5	52.7	81.2	0.0	-
Prompt-based zero-shot [‡]	50.8	51.7	49.5	50.8	51.3	61.9	49.7	-3.2
K=16 {	Fine-tuning	45.8 (6.4)	47.8 (6.8)	48.4 (4.8)	60.2 (6.5)	54.4 (3.9)	76.6 (2.5)	60.7 (4.3)
	Prompt-based FT (man)	68.3 (2.3)	70.5 (1.9)	77.2 (3.7)	64.5 (4.2)	69.1 (3.6)	74.5 (5.3)	71.0 (7.0)
	+ demonstrations	70.7 (1.3)	72.0 (1.2)	79.7 (1.5)	69.2 (1.9)	68.7 (2.3)	69.8 (1.8)	73.5 (5.1)
	Prompt-based FT (auto)	68.3 (2.5)	70.1 (2.6)	77.1 (2.1)	68.3 (7.4)	73.9 (2.2)	76.2 (2.3)	75.0 (3.3)
	+ demonstrations	70.0 (3.6)	72.0 (3.1)	77.5 (3.5)	68.5 (5.4)	71.1 (5.3)	78.1 (3.4)	76.4 (6.2)
	Fine-tuning (full) [†]	89.8	89.5	92.6	93.3	80.9	91.4	91.9

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Data-Efficient Fine-tuning: Semi-supervised Learning

- Semi-Supervised learning: We have some labeled training data and a large amount of unlabeled data



Data-Efficient Fine-tuning: Semi-supervised Learning

- Semi-Supervised learning: We have some labeled training data and a large amount of unlabeled data

It's Not Just Size That Matters: Small Language Models Are Also **Few-Shot Learners**

Timo Schick^{1,2} and Hinrich Schütze¹

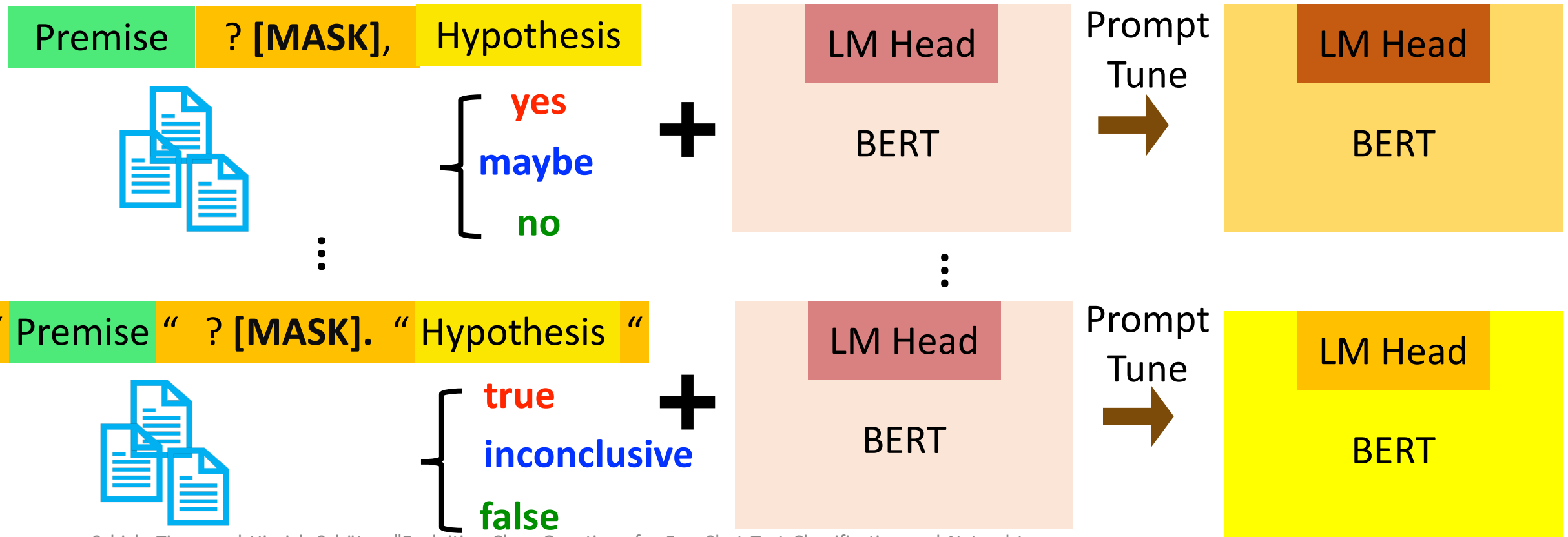
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Data-Efficient Fine-tuning: Semi-supervised Learning

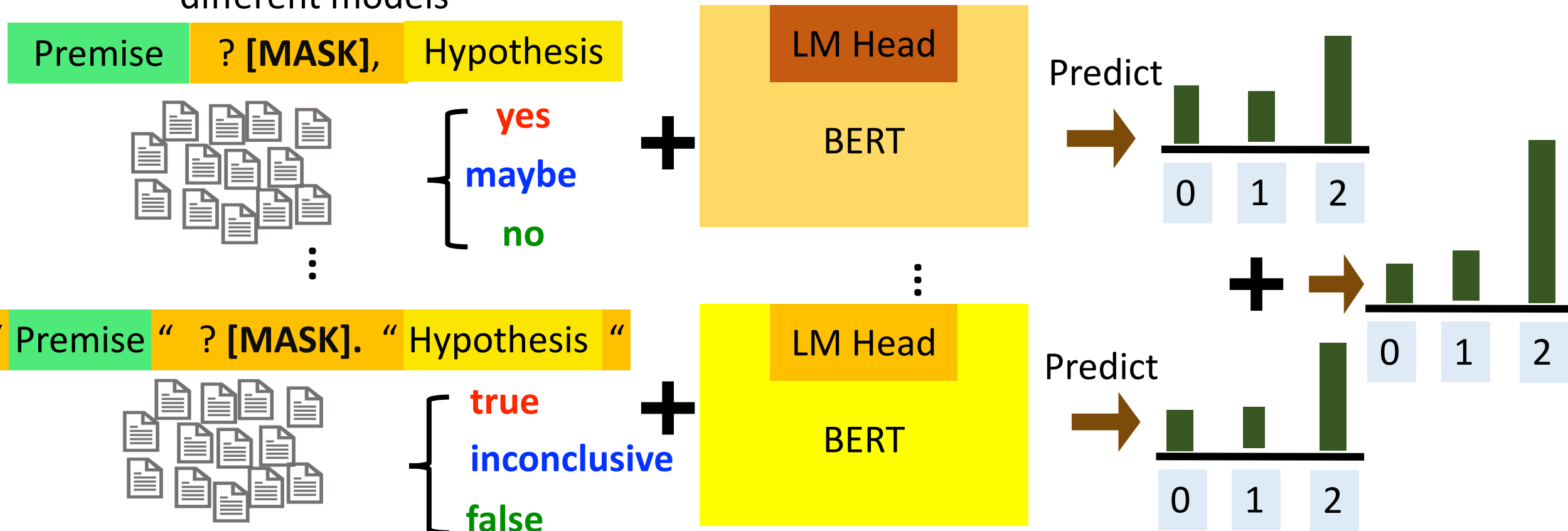
- Pattern-Exploiting Training (PET)
 - Step 1: Use different prompts and verbalizer to prompt-tune different PLMs on the labeled dataset



Data-Efficient Fine-tuning: Semi-supervised Learning

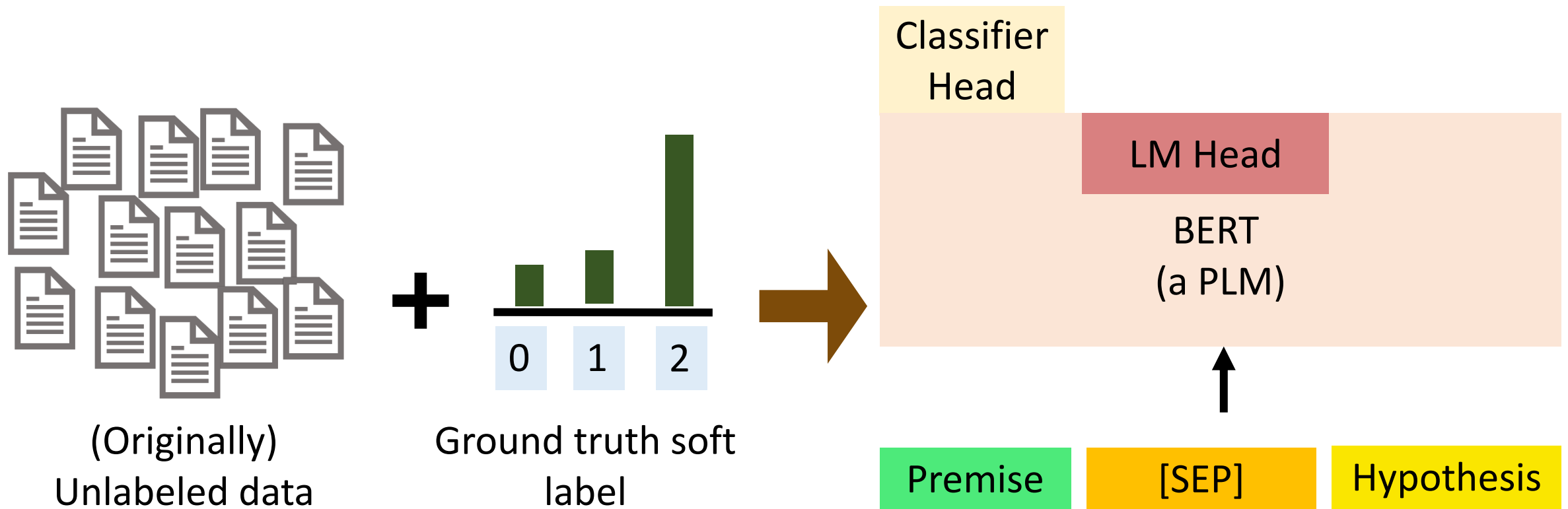
- Pattern-Exploiting Training (PET)

- Step 2: Predict the unlabeled dataset and combine the predictions from different models



Data-Efficient Fine-tuning: Semi-supervised Learning

- Pattern-Exploiting Training (PET)
 - Step 3: Use a PLM with classifier head to train on the soft-labeled data set



Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - Zero-shot Learning
 - PLMs Are Gigantic → Reducing the Number of Parameters
- Closing Remarks

Data-Efficient Fine-tuning(?): Zero-shot

- Zero-shot inference: inference on the downstream task without any training data
- If you don't have training data, then we need a model that can zero-shot inference on downstream tasks



Training data



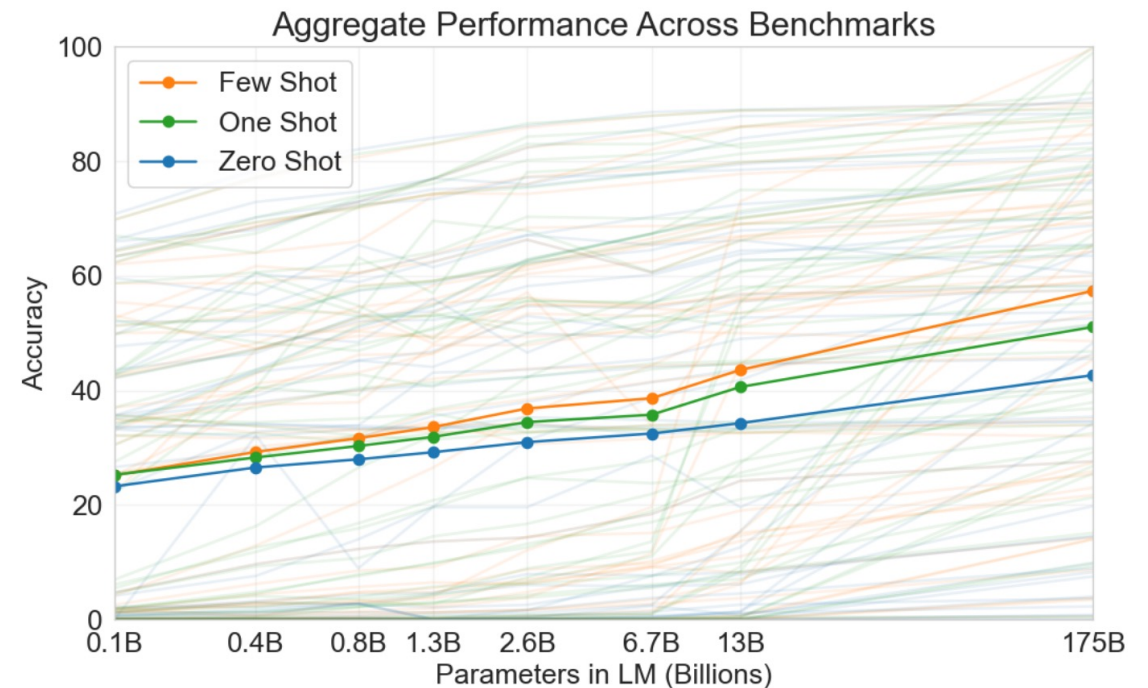
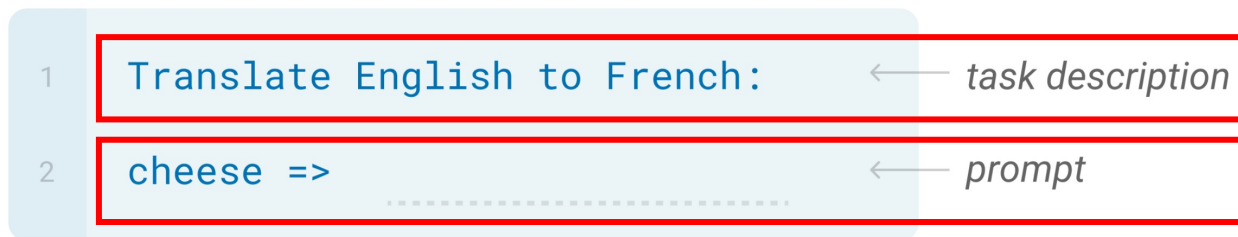
Pre-trained Language Model

Data-Efficient Fine-tuning(?): Zero-shot

- GPT-3 shows that zero-shot (with task description) is possible
 - Only if your model is large enough

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



Data-Efficient Fine-tuning(?): Zero-shot

- Where does this zero-shot ability spring from?
 - Hypothesis: during pre-training, the training datasets implicitly contains a mixture of different tasks
 - QA

Q: I got 4 papers. Should I expect this load in the future?

A: The average monthly load for reviewers should be much closer to 2, but in certain periods (close to large conferences), it's possible that the load is higher.
 - Summarization

Finetuned Language Models are Zero-Shot Learners

Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, Quoc V Le

29 Sept 2021 (modified: 10 Feb 2022) ICLR 2022 Oral Readers:  Everyone [Show Bibtex](#) [Show Revisions](#)

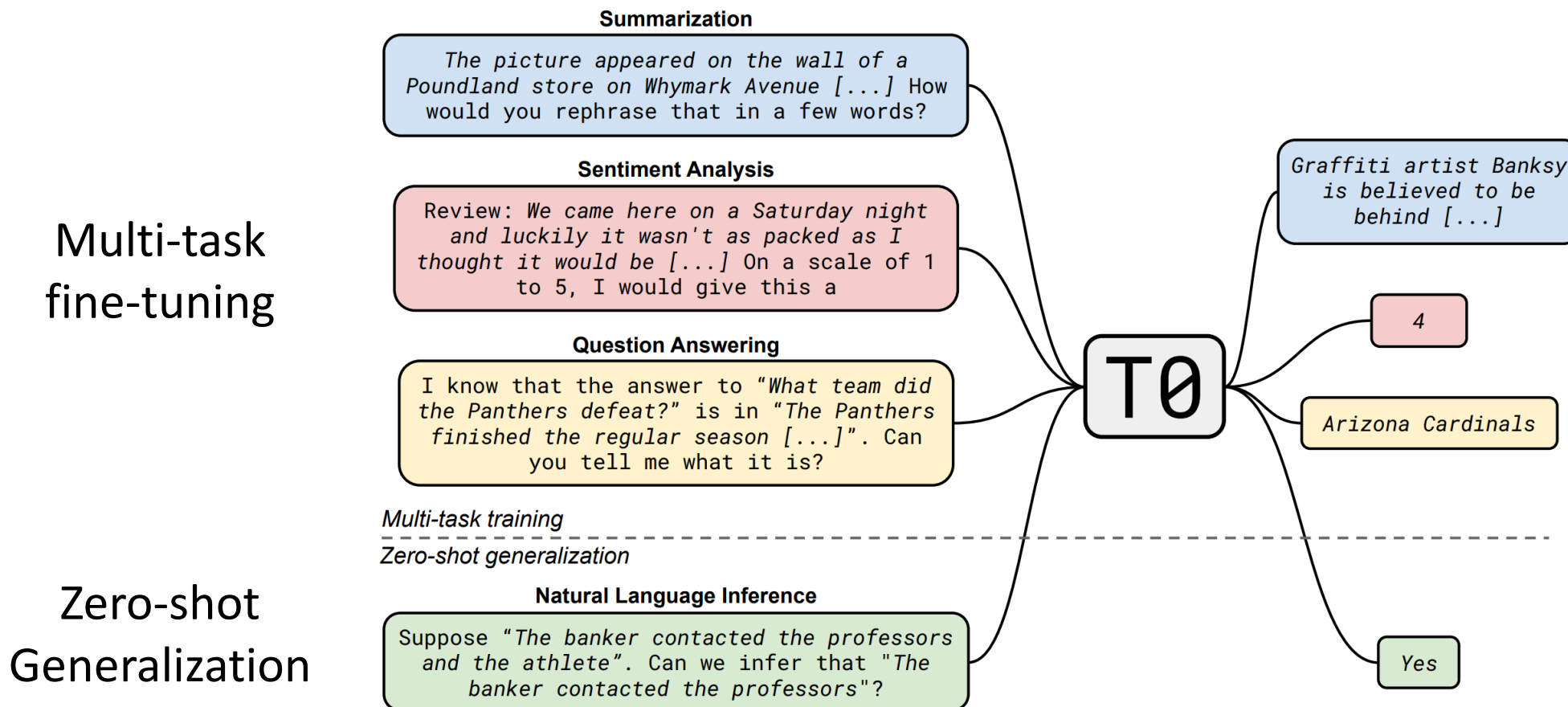
Keywords: natural language processing, zero-shot learning, language models

Abstract: This paper explores a simple method for improving the zero-shot learning abilities of language models. We show that instruction tuning—finetuning language models on a collection of datasets described via instructions—substantially improves zero-shot performance on unseen tasks. We take a 137B parameter pretrained language model and instruction tune it on over 60 NLP datasets verbalized via natural language instruction templates. We evaluate this instruction-tuned model, which we call FLAN, on unseen task types. FLAN substantially improves the performance of its unmodified counterpart and surpasses zero-shot 175B GPT-3 on 20 of 25 datasets that we evaluate. FLAN even outperforms few-shot GPT-3 by a large margin on ANLI, RTE, BoolQ, AI2-ARC, OpenbookQA, and StoryCloze. Ablation studies reveal that number of finetuning datasets, model scale, and natural language instructions are key to the success of instruction tuning.

One-sentence Summary: "Instruction tuning", which finetunes language models on a collection of tasks described via instructions, substantially boosts zero-shot performance on unseen tasks.

Data-Efficient Fine-tuning(?): Zero-shot

- Hypothesis: multi-task training enables zero-shot generalization
 - Why not train a model with multi-task learning on a bunch of dataset?



Data-Efficient Fine-tuning(?): Zero-shot

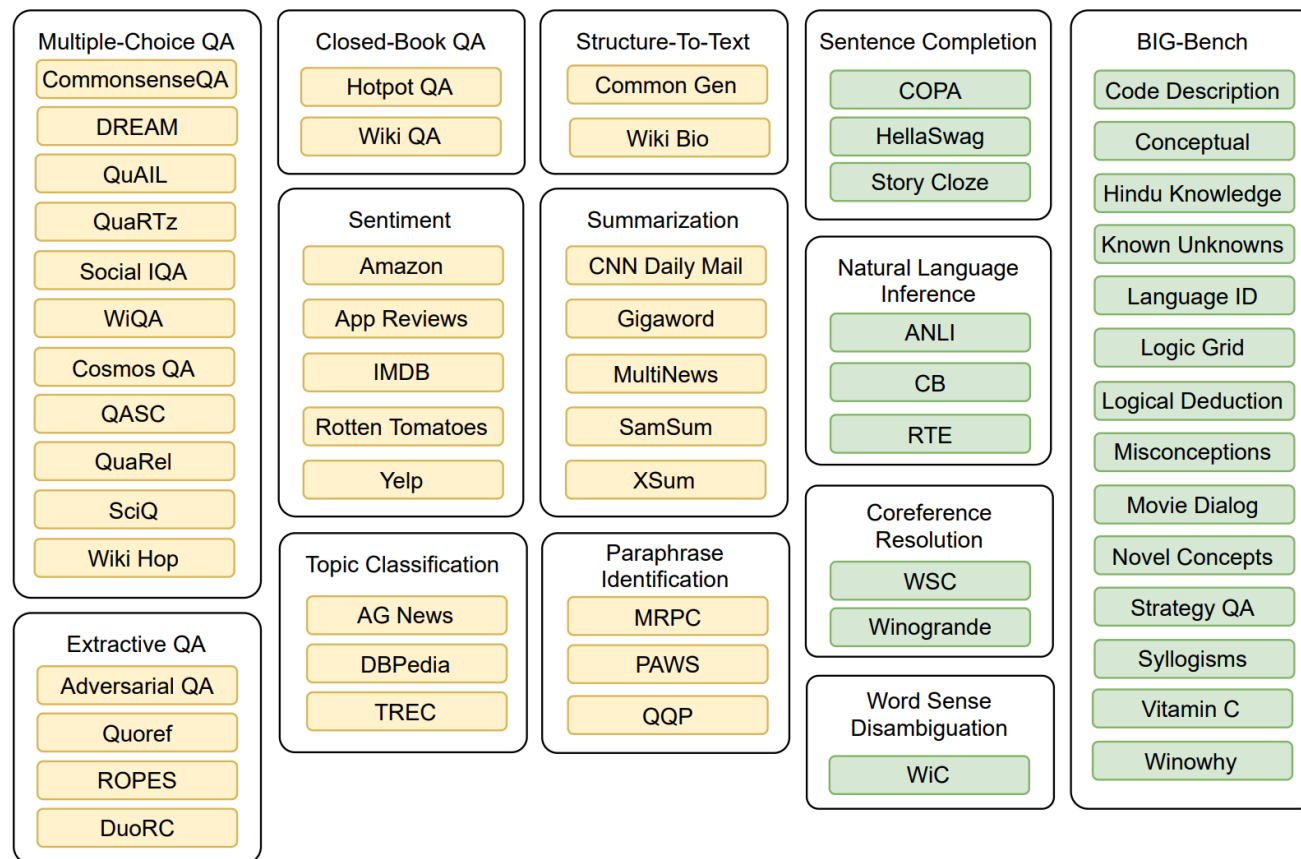
- Multi-task fine-tuning using a PLM
 - Convert the task into a natural language **prompts**
 - Example: Natural Language Inference

```
▼ "label" : [  
  0 : "entailment"  
  1 : "neutral"  
  2 : "contradiction"  
]
```

	premise	hypothesis	label
NLI dataset	Conceptually cream skimming has two basic dimensions - product and geography.	Product and geography are what make cream skimming work.	1
Natural language prompt	{{premise}} Based on the previous passage, is it true that "{{hypothesis}}"? Yes, no, or maybe?		
	{{premise}} Based on that information, is the claim: "{{hypothesis}}", {{"true"}}, {{"false"}}, or {{"inconclusive"}}?		
	Given that {{premise}} Does it follow that {{hypothesis}} Yes, no, or maybe?		

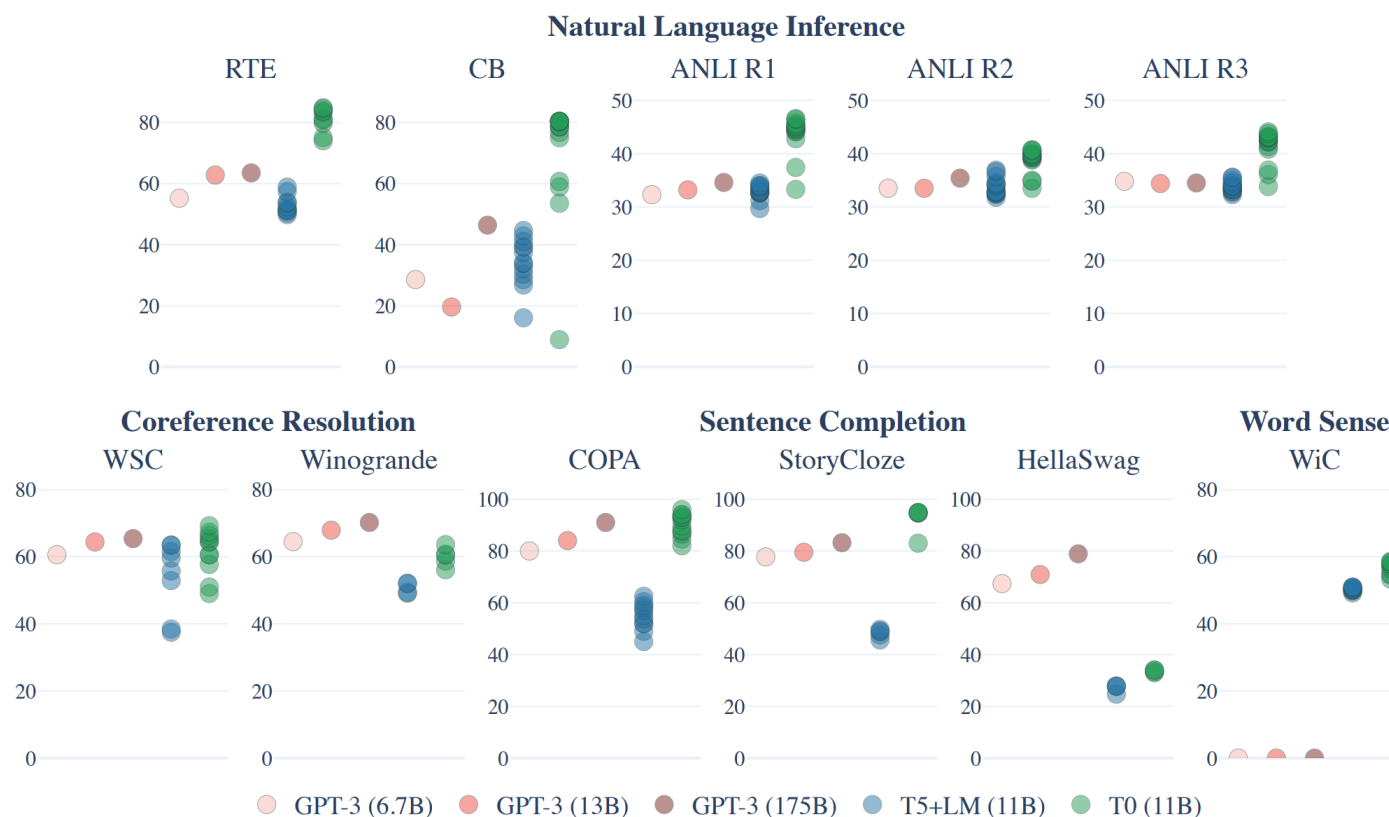
Data-Efficient Fine-tuning(?): Zero-shot

- Fine-tuning with some types of tasks and zero-shot inference on other types of tasks



Data-Efficient Fine-tuning(?): Zero-shot

- Sometimes achieves performance better than GPT-3 (175B parameters) with **only 11B** parameters



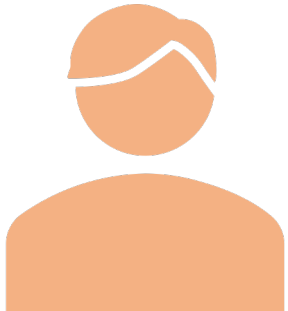
Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - Summary
 - PLMs Are Gigantic → Reducing the Number of Parameters
- Closing Remarks

Data-Efficient Fine-tuning: Summary

- Use natural language prompts and add scenario-specific designs

- [CLS] The spring break is coming soon. Is it true that the spring break was over? >>> no
- [CLS] I am going to have dinner. Is it true that I am going to eat something? >>> yes
- [CLS] Mary likes pie. Is it true that Mary hates pie. [SEP] >>> ?

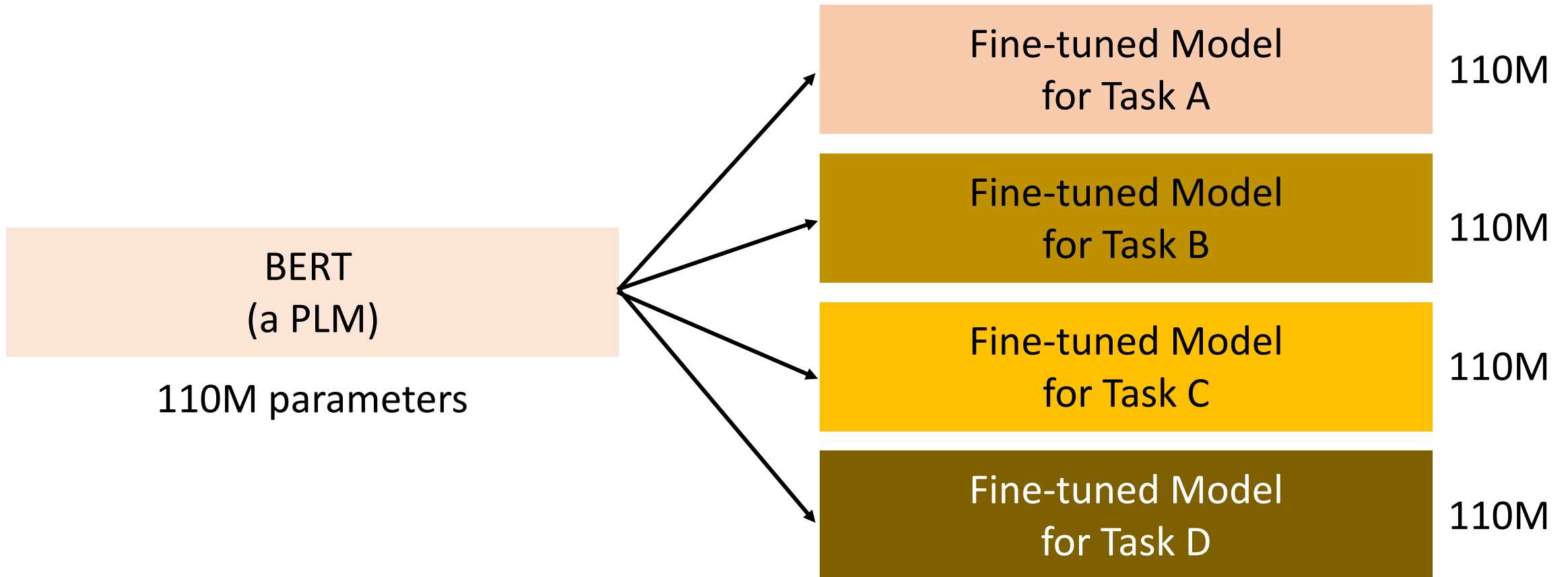


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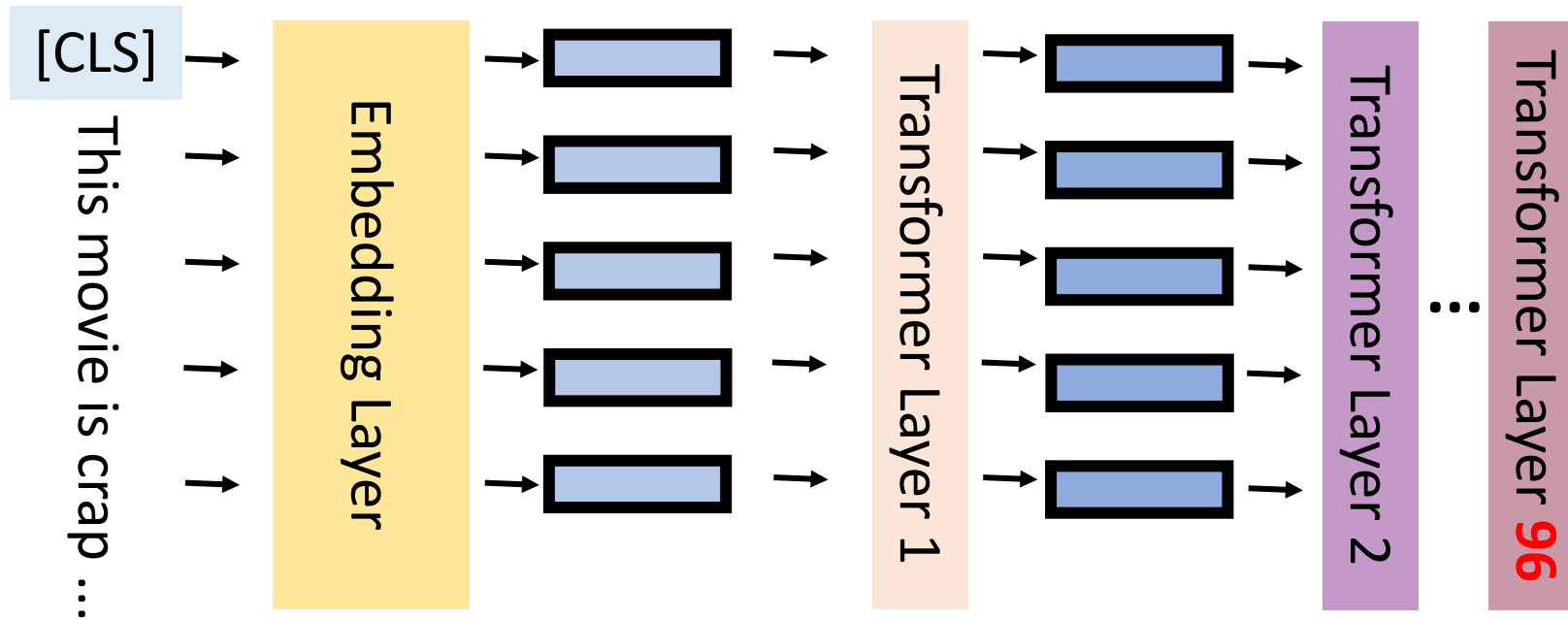
The Problems of PLMs

- Problem 2: The PLM is too big
 - Need a copy for each downstream task



The Problems of PLMs

- Problem 2: The PLM is too big
 - Inference takes too long
 - Consume too much space

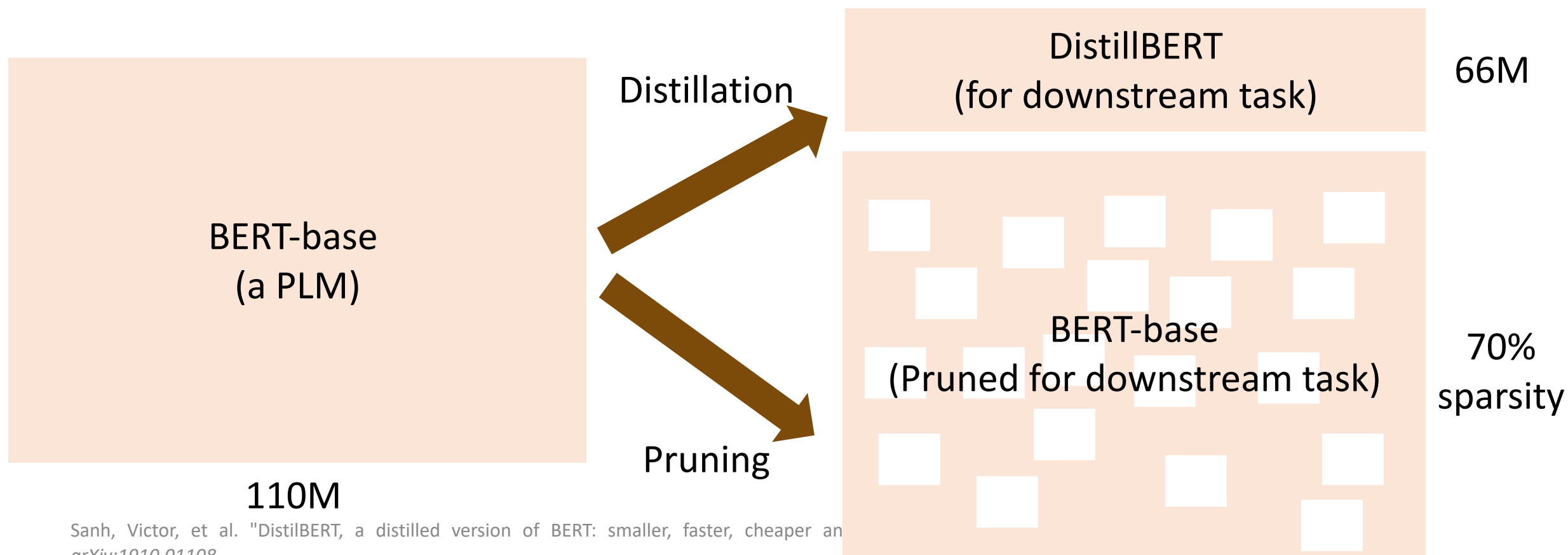


Reducing the Number of Parameters

- Problem: PLM is too large (in terms of numbers of parameters, model size, and the storage needed to store the model)
- Solution: Reduce the number of parameters
 - Smaller pre-trained model?

Reducing the Number of Parameters

- Pre-train a large model, but use a smaller model for the downstream tasks



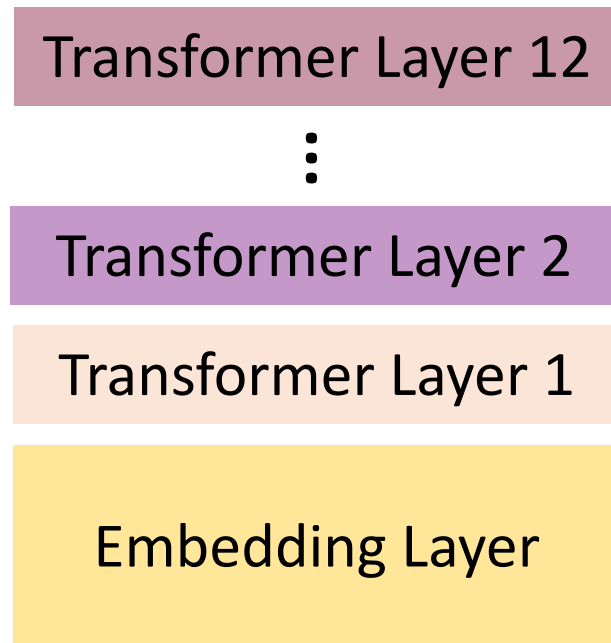
Sanh, Victor, et al. "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and
arXiv:1910.01108

(2019).

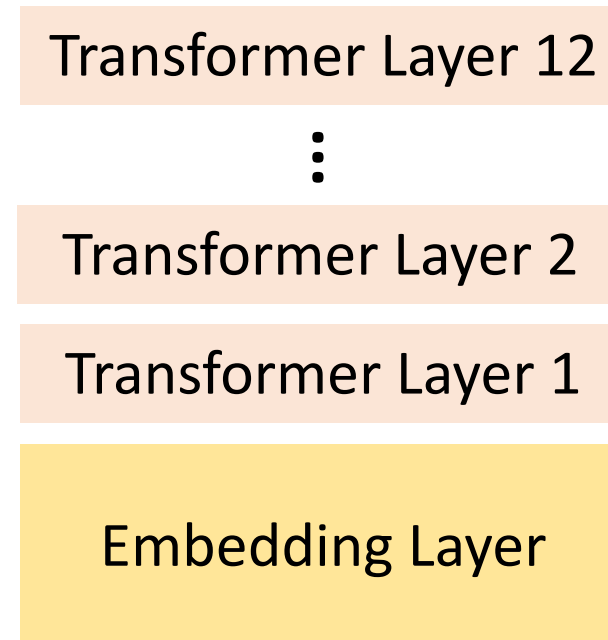
Lai, Cheng-I. Jeff, et al. "Parp: Prune, adjust and re-prune for self-supervised speech recognition." *Advances in Neural Information Processing Systems* 34 (2021)

Reducing the Number of Parameters

- Share the parameters among the transformer layers



BERT:110M parameters



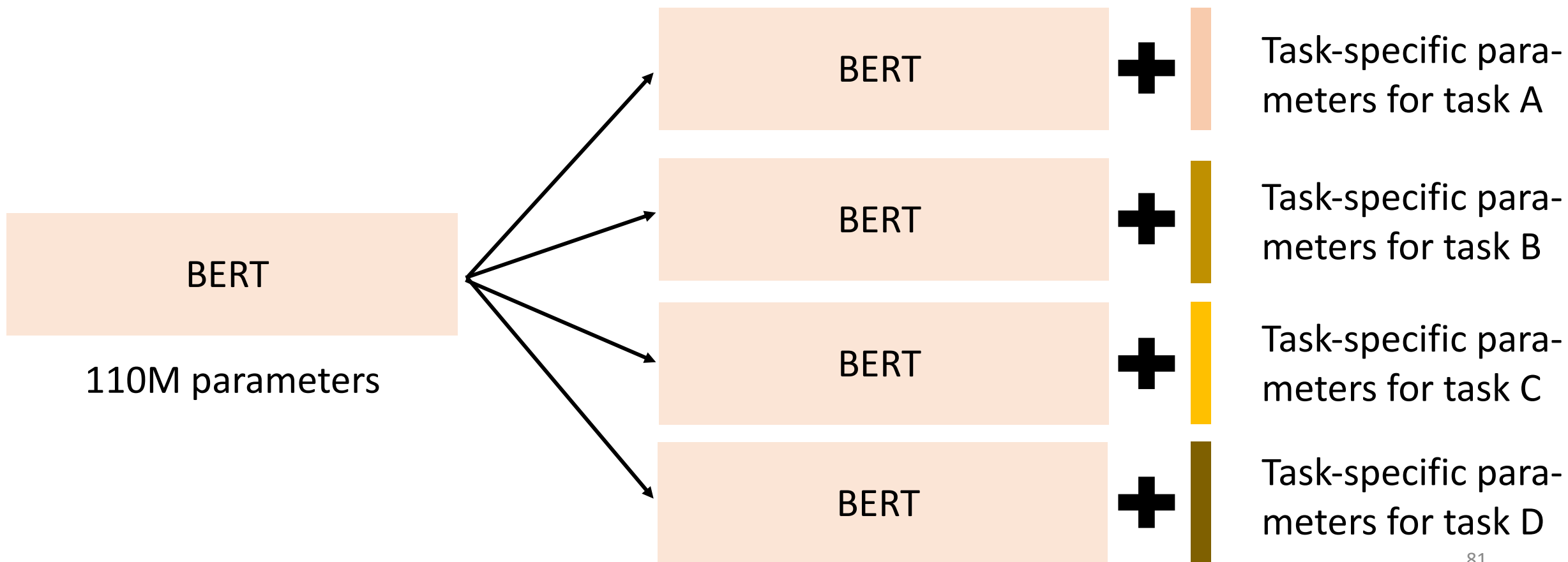
ALBERT:12M parameters

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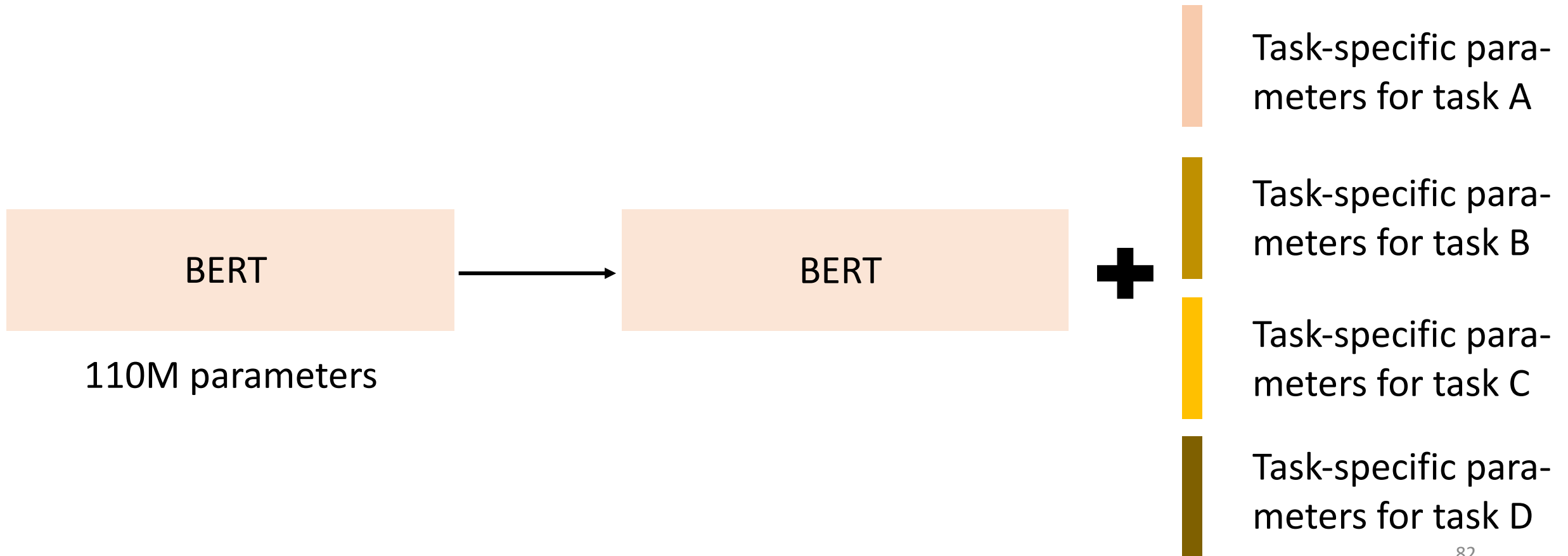
Parameter-Efficient Fine-tuning

- Use a small amount of parameters for each downstream task



Parameter-Efficient Fine-tuning

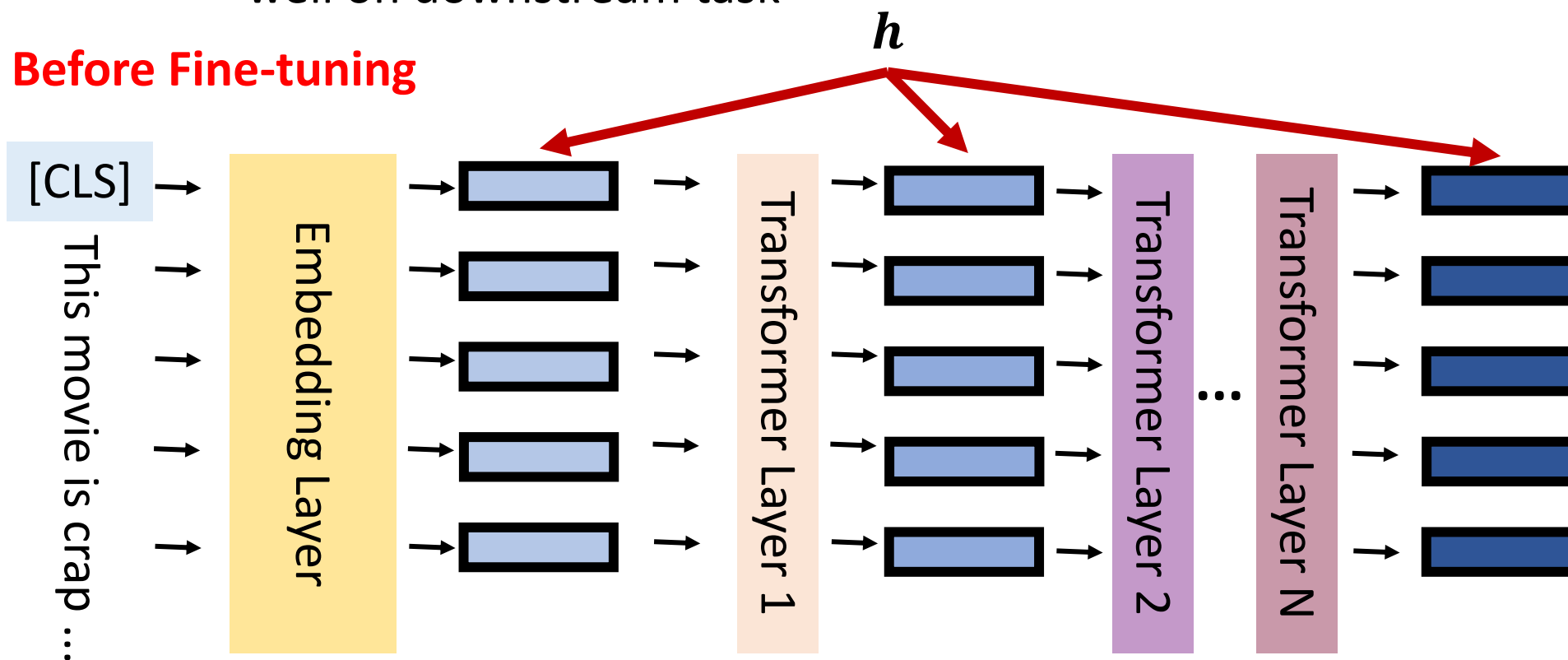
- Use a small amount of parameters for each downstream task



Parameter-Efficient Fine-tuning

- What is standard fine-tuning really doing?
 - Modify the hidden representations ($\mathbf{h}$) of the PLM such that it can perform well on downstream task

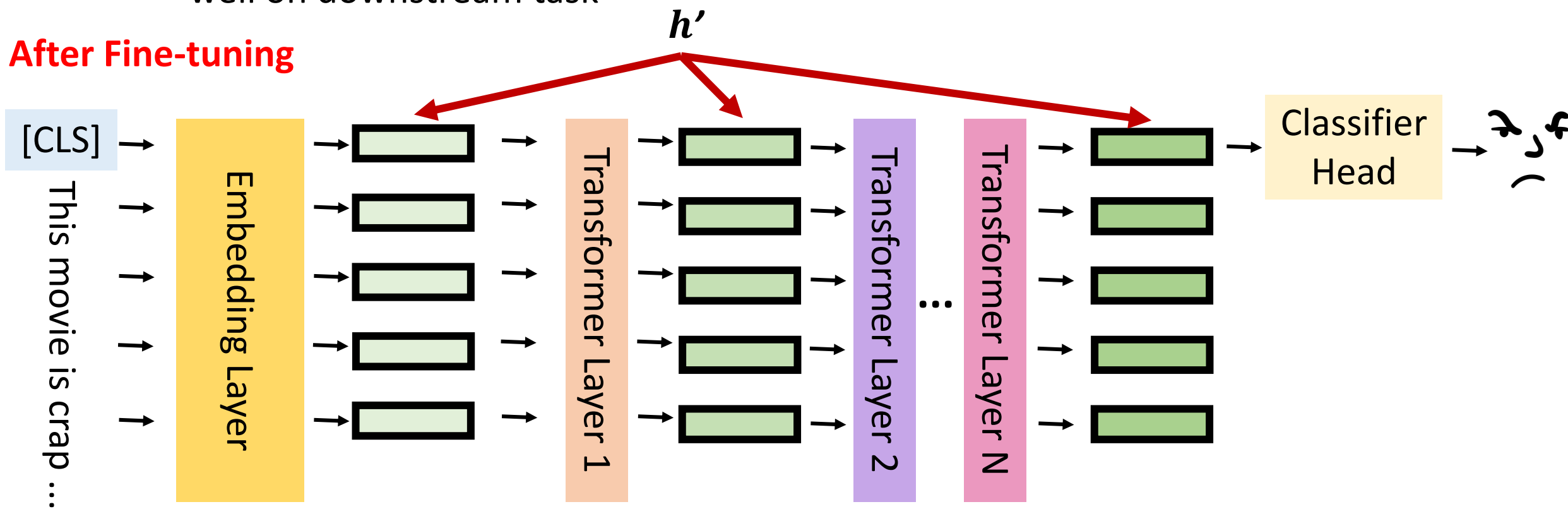
Before Fine-tuning



Parameter-Efficient Fine-tuning

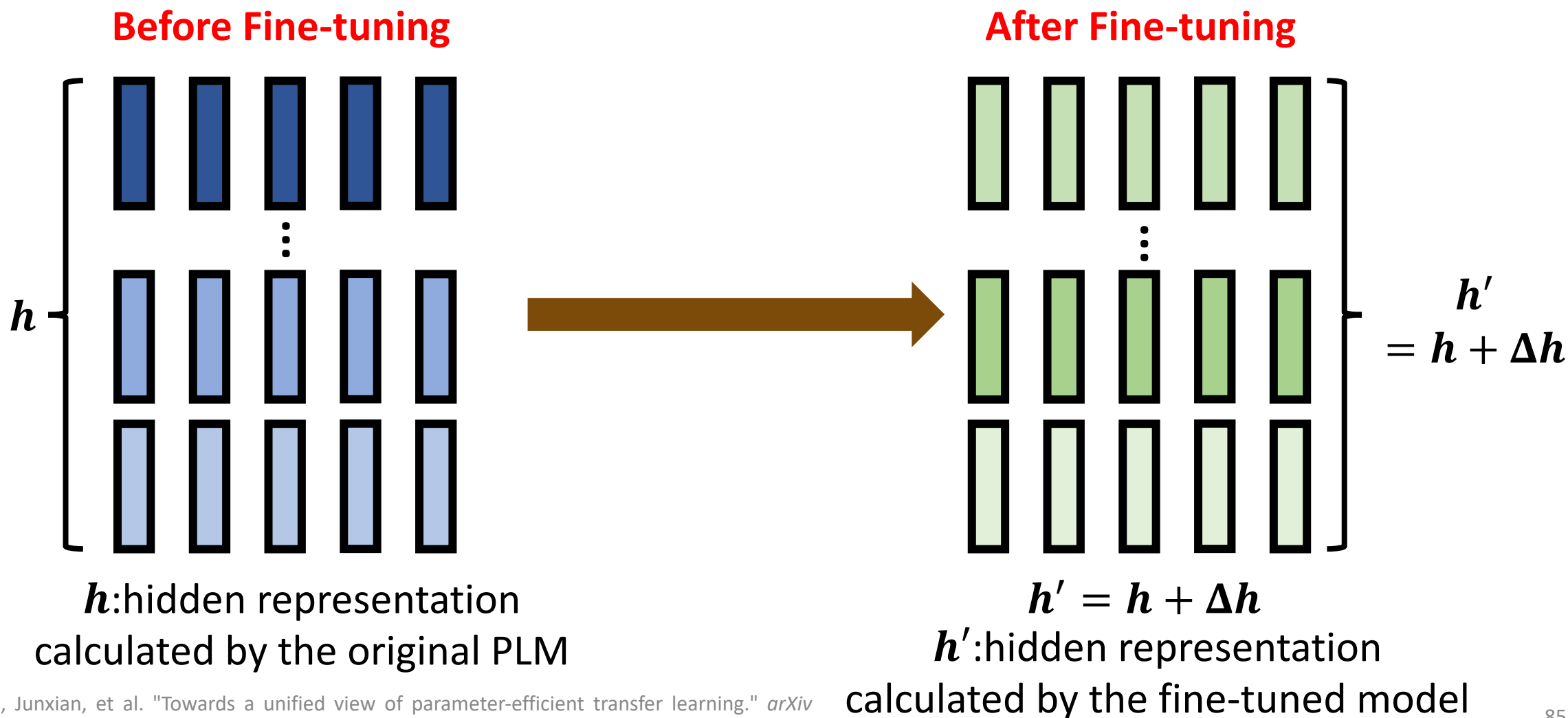
- What is standard fine-tuning really doing?
 - Modify the hidden representations ($\mathbf{h}$) of the PLM such that it can perform well on downstream task

After Fine-tuning



Parameter-Efficient Fine-tuning

- Fine-tuning = modifying the hidden representation based on a PLM

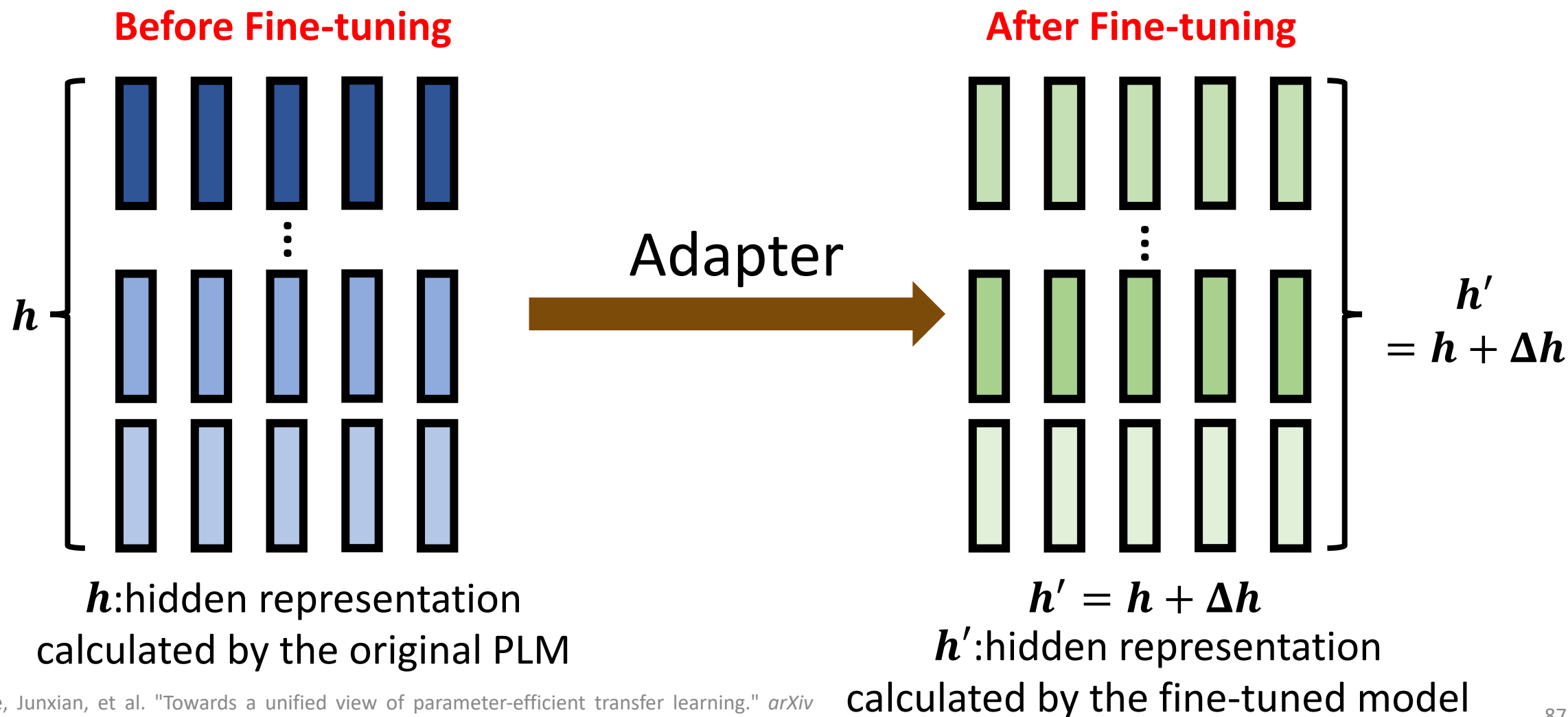


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 - Adapter
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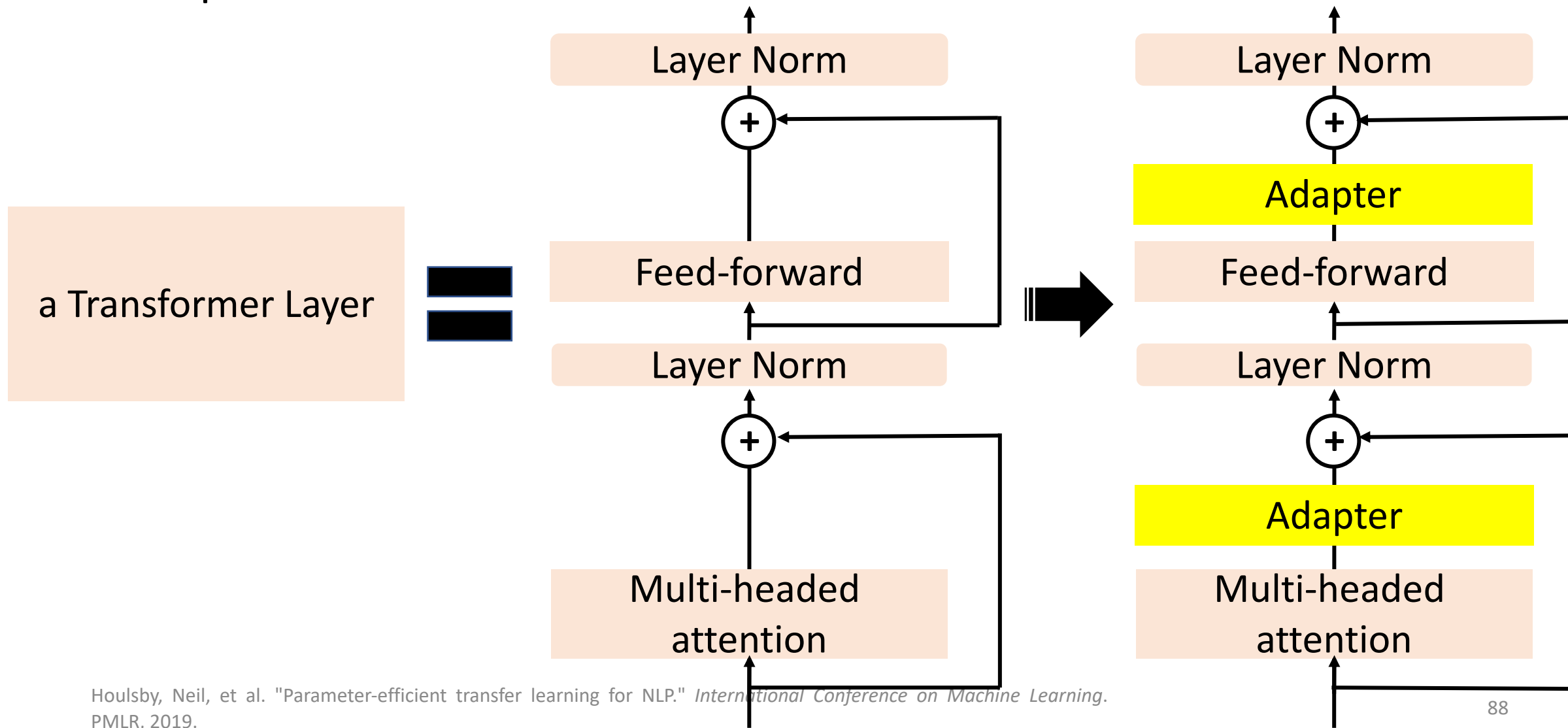
Parameter-Efficient Fine-tuning: Adapter

- Use special submodules to modify hidden representations!



Parameter-Efficient Fine-tuning: Adapter

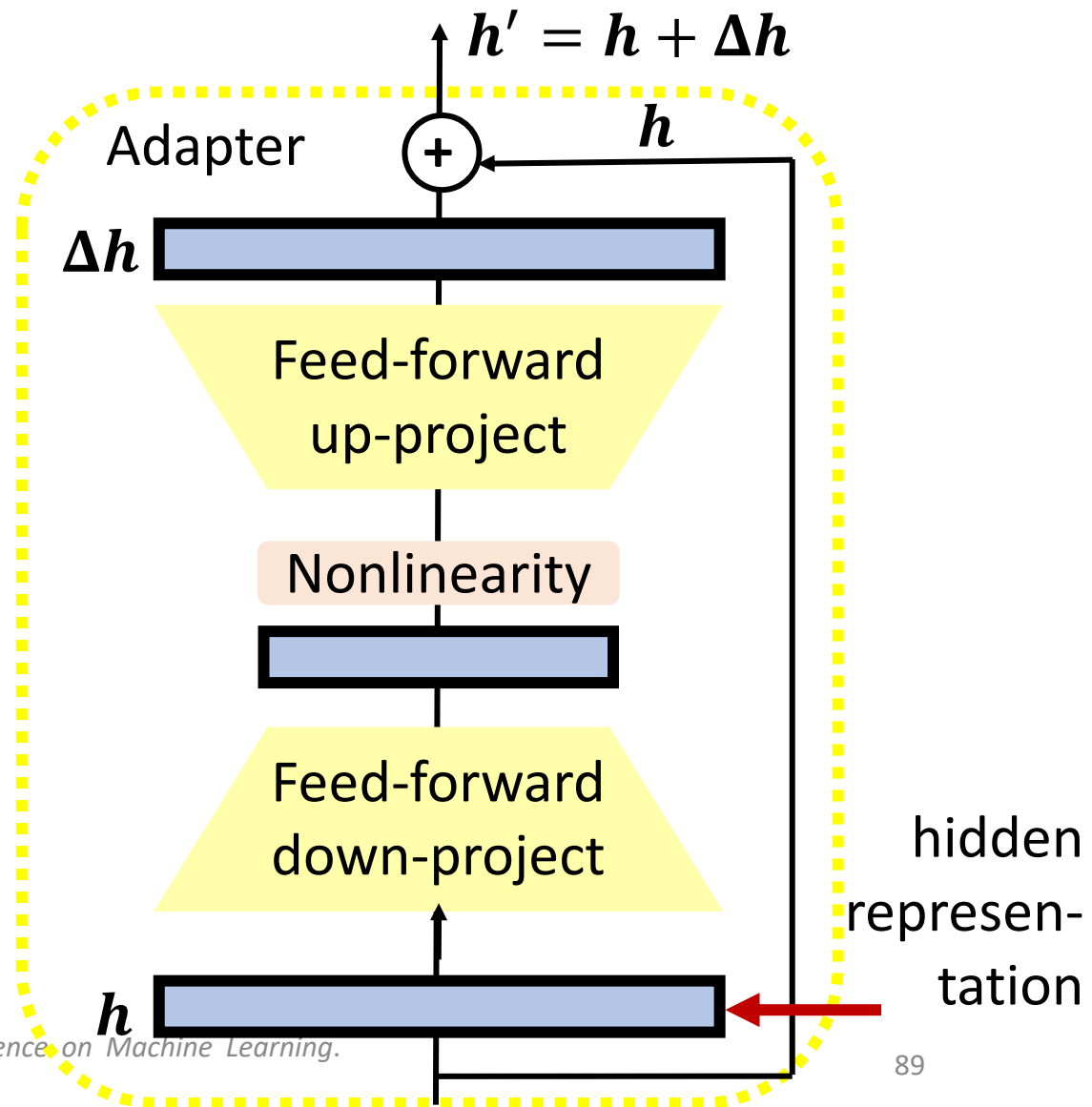
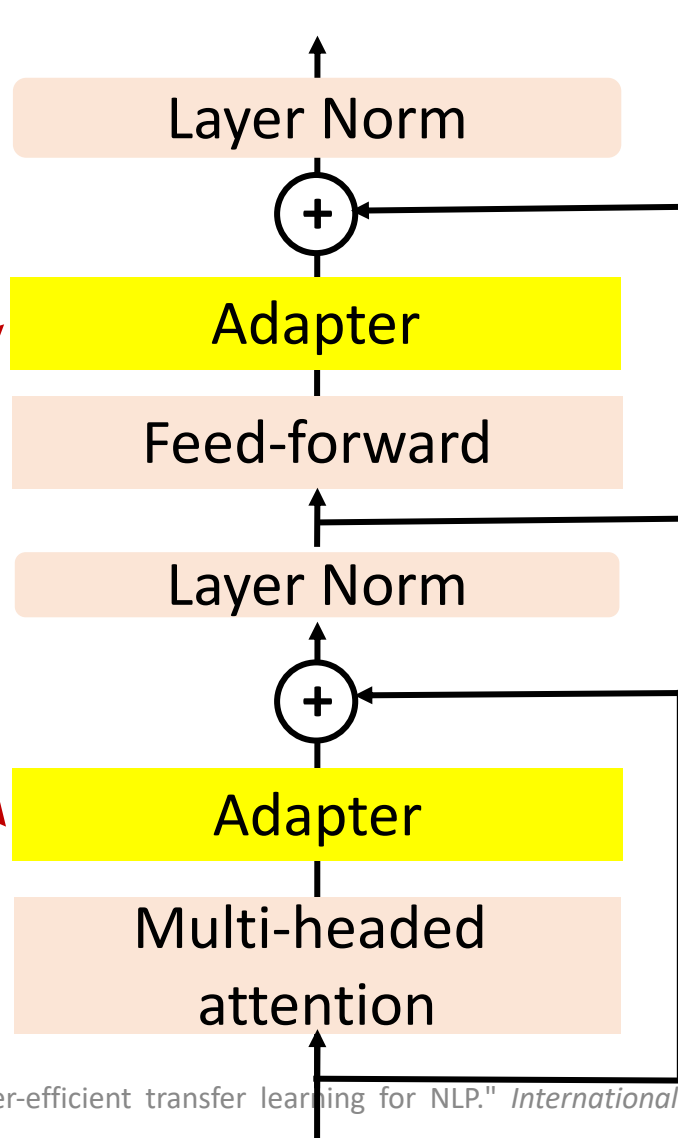
- Adapters: small trainable submodules inserted in transformers



Parameter-Efficient Fine-tuning: Adapter

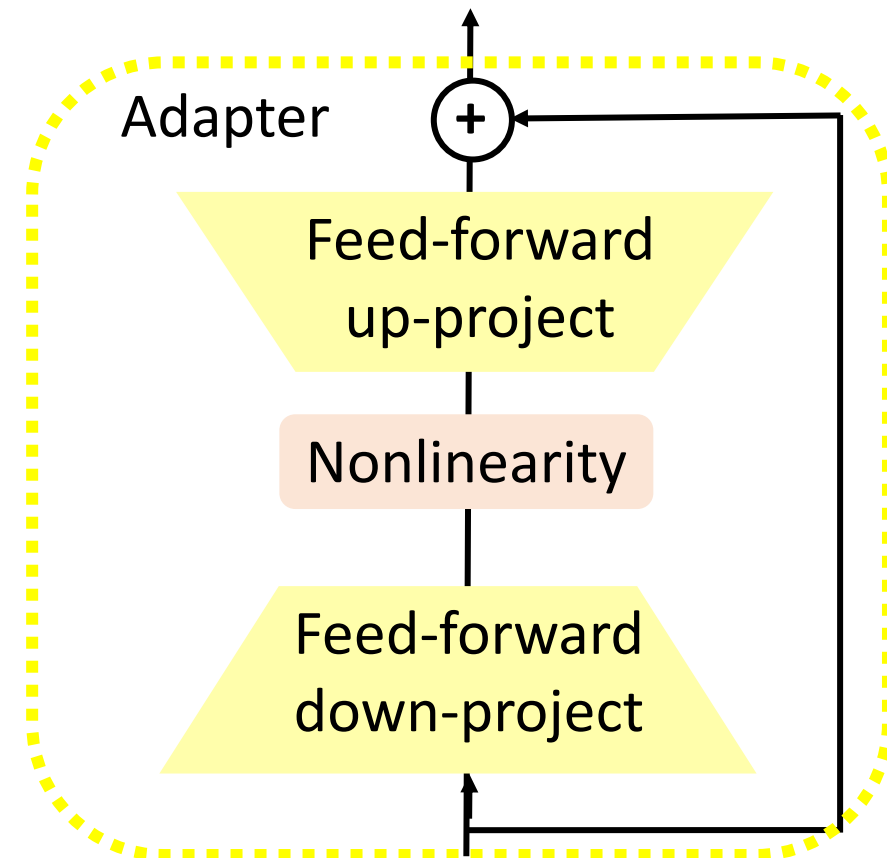
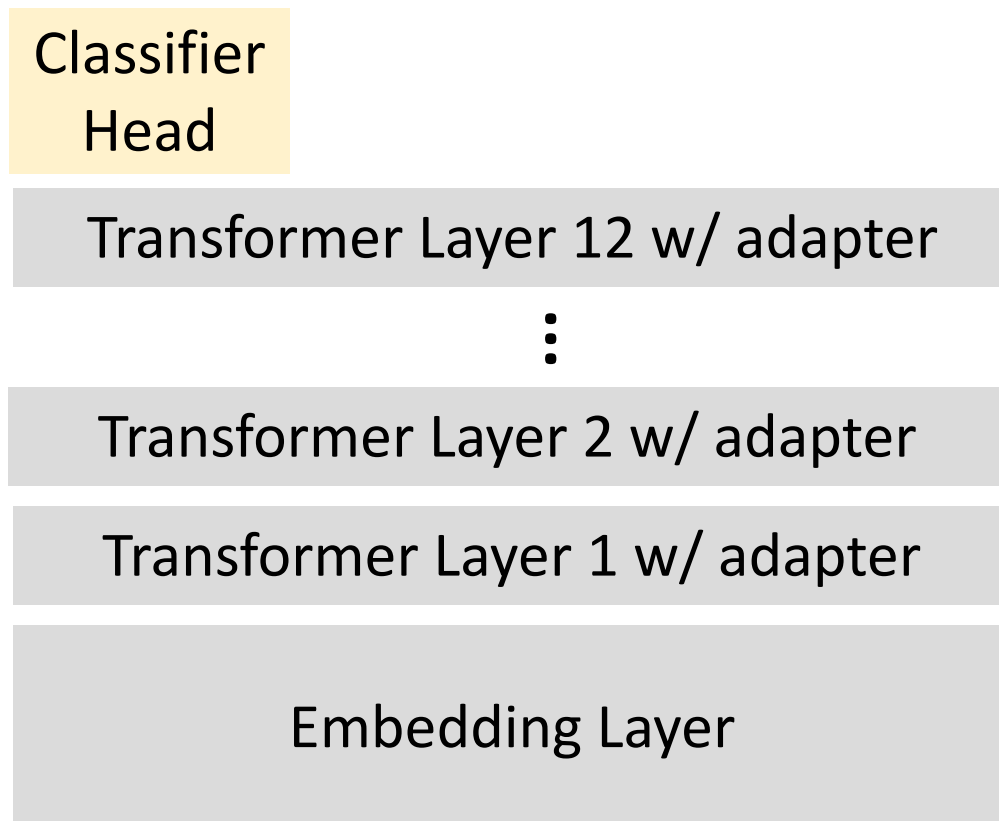
- Adapters

Inside of the
transformer
layer, only
adapters are
updated



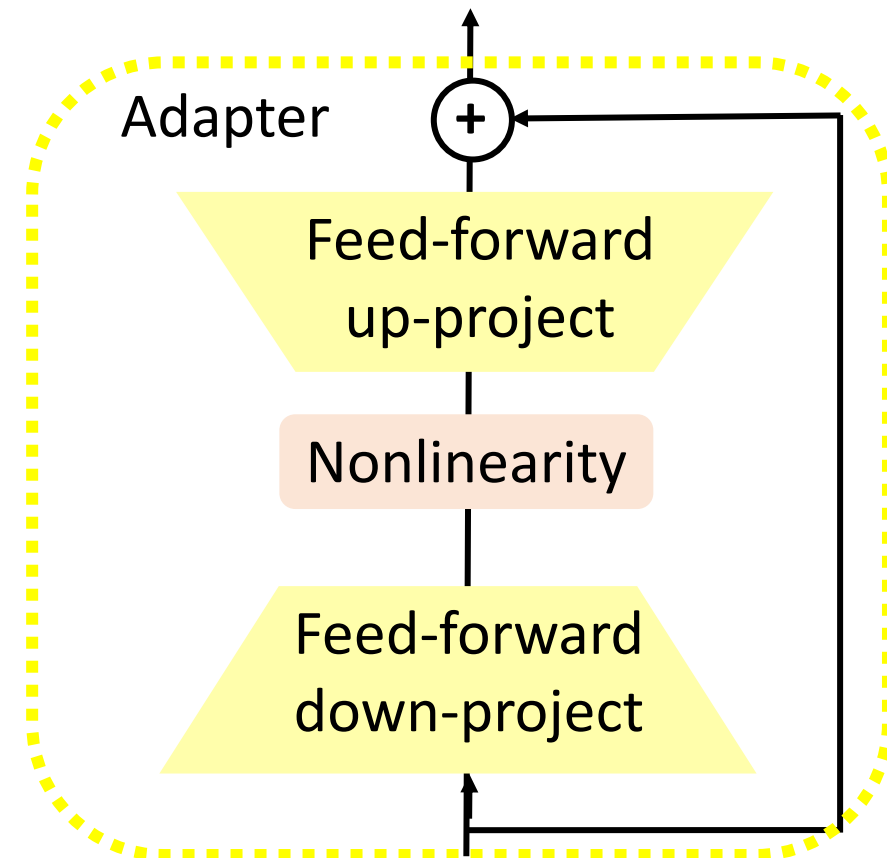
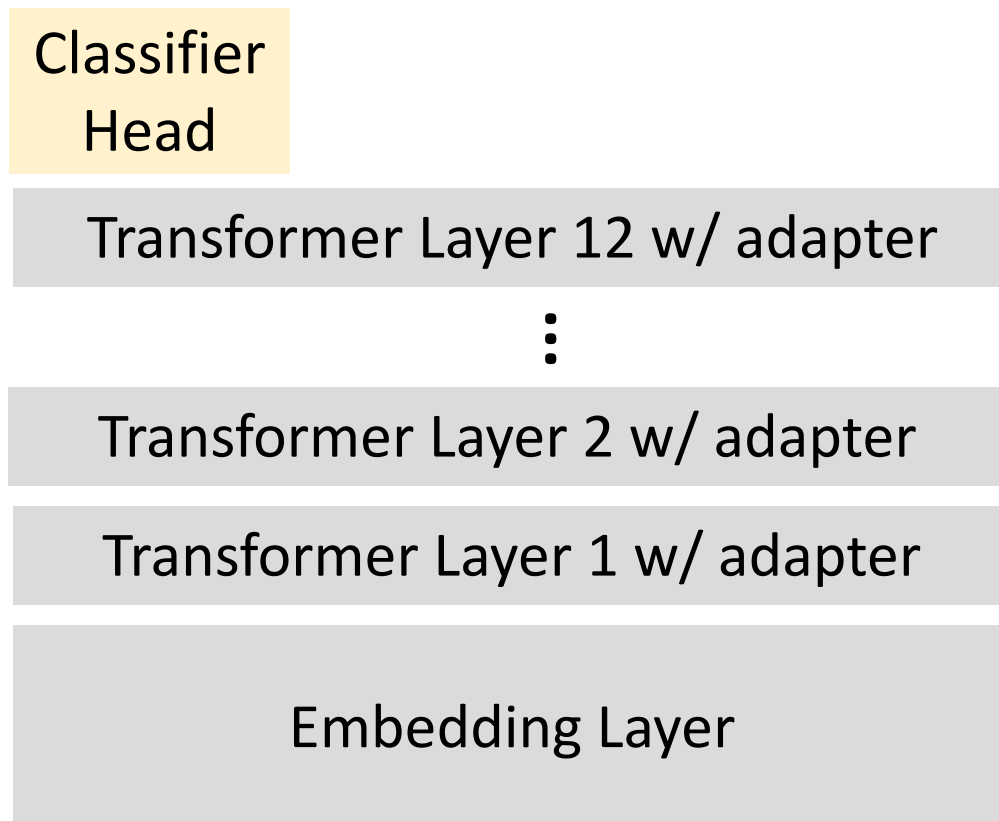
Parameter-Efficient Fine-tuning: Adapter

- Adapters: During fine-tuning, only update the adapters and the classifier head



Parameter-Efficient Fine-tuning: Adapter

- Adapters: All downstream tasks share the PLM; the adapters in each layer and the classifier heads are the task-specific modules



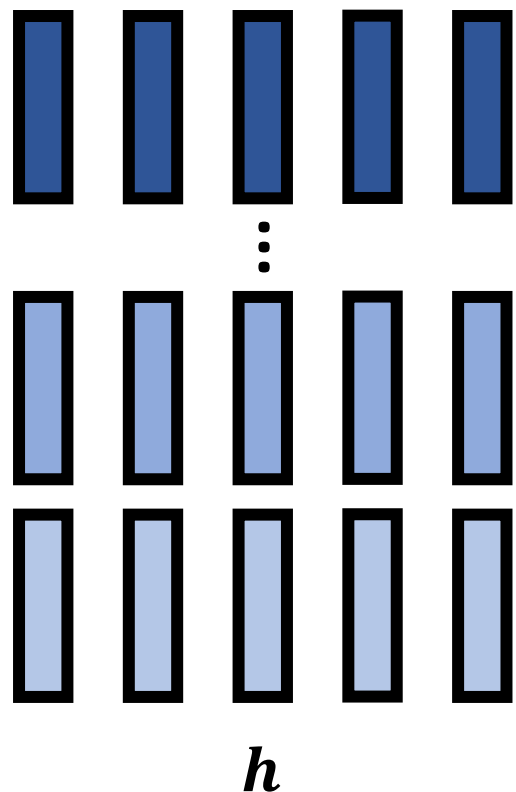
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Parameter-Efficient Fine-tuning: LoRA

- Use special submodules to modify hidden representations!

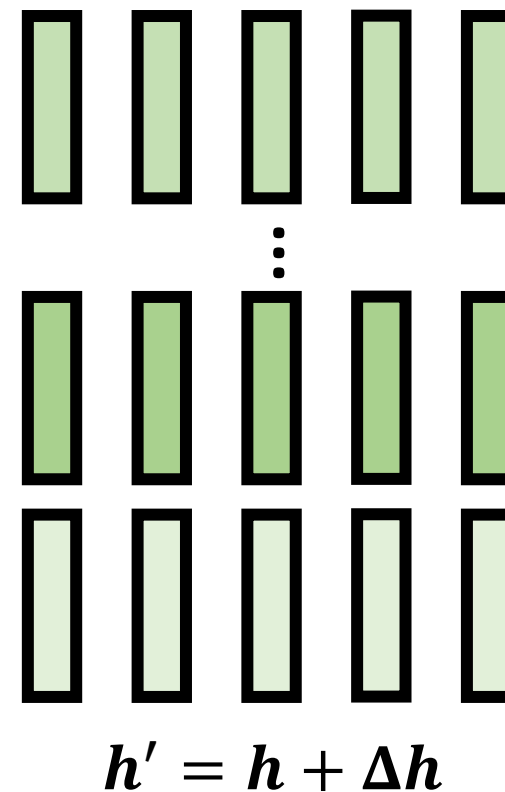
Before Fine-tuning



LoRA

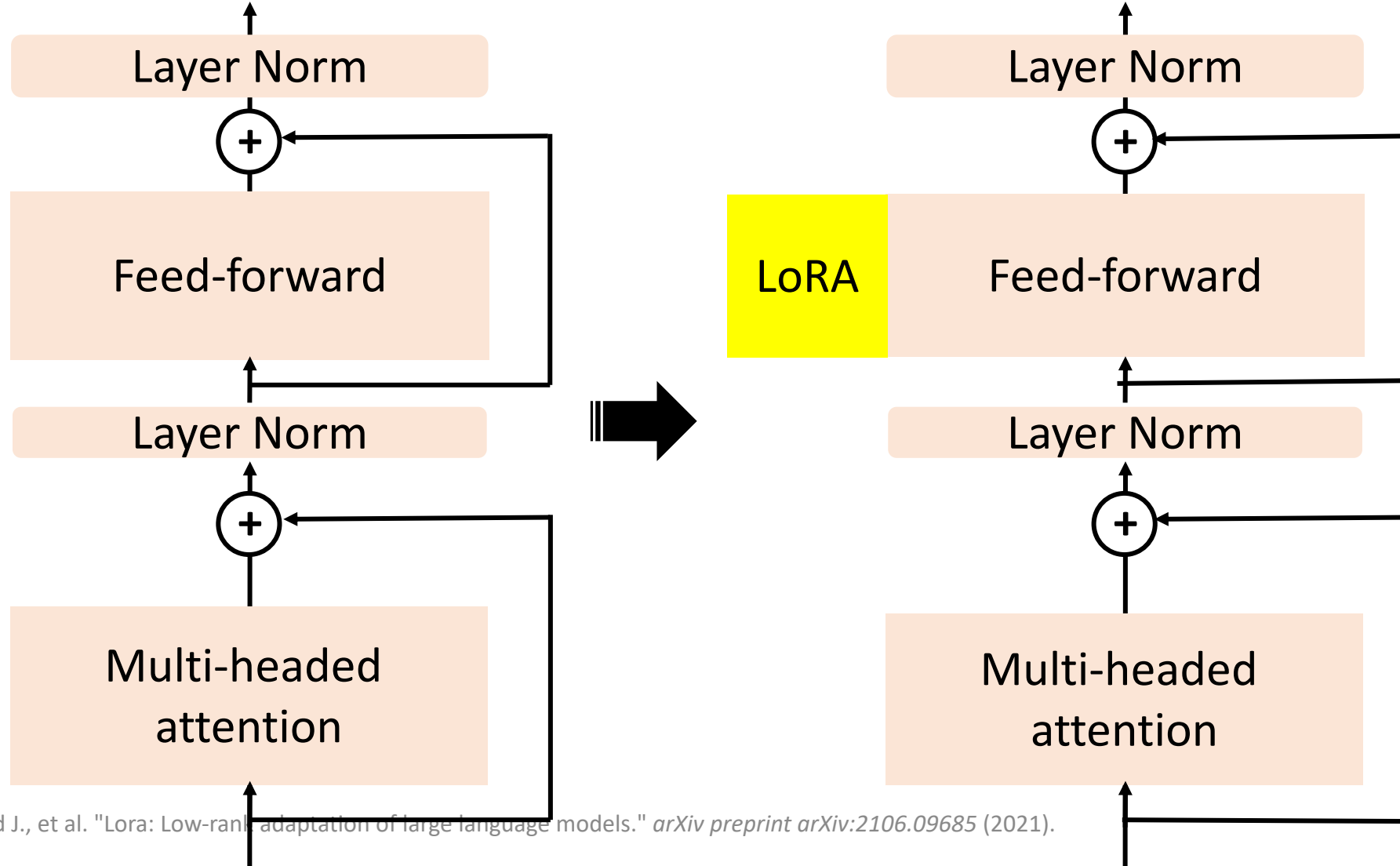


After Fine-tuning



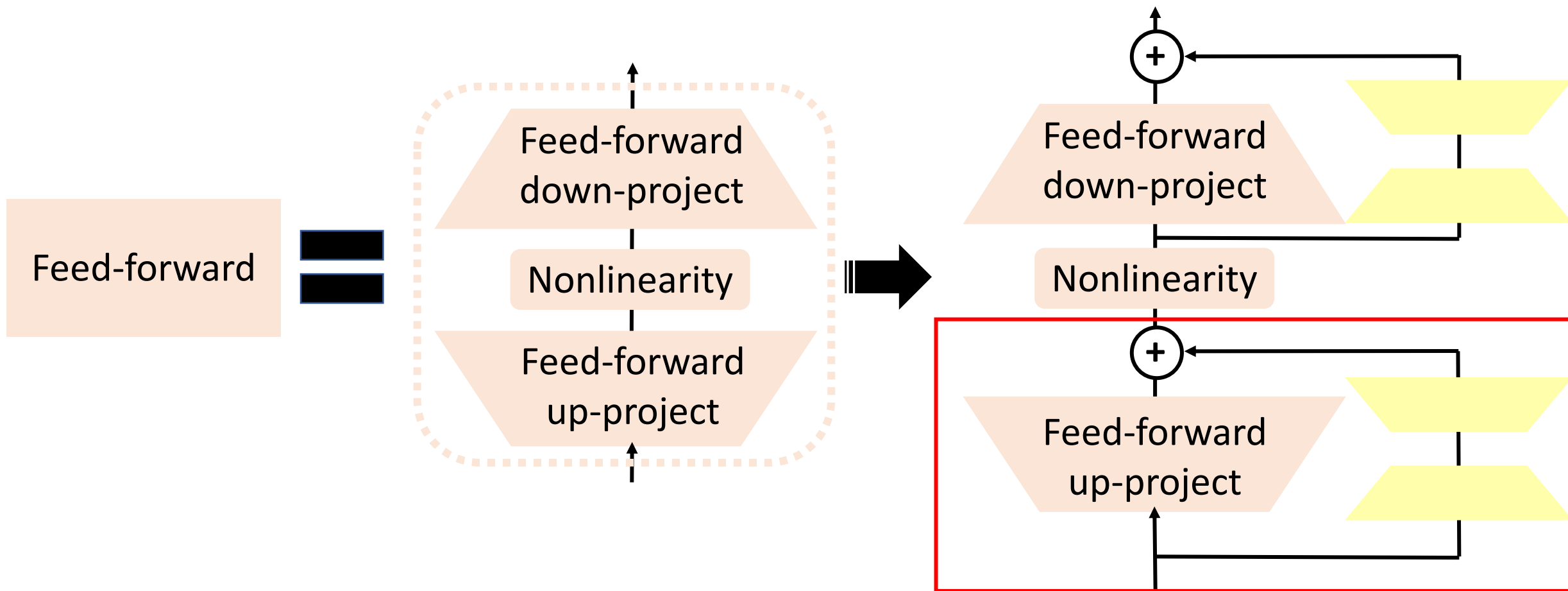
Parameter-Efficient Fine-tuning: LoRA

- LoRA: Low-Rank Adaptation of Large Language Models



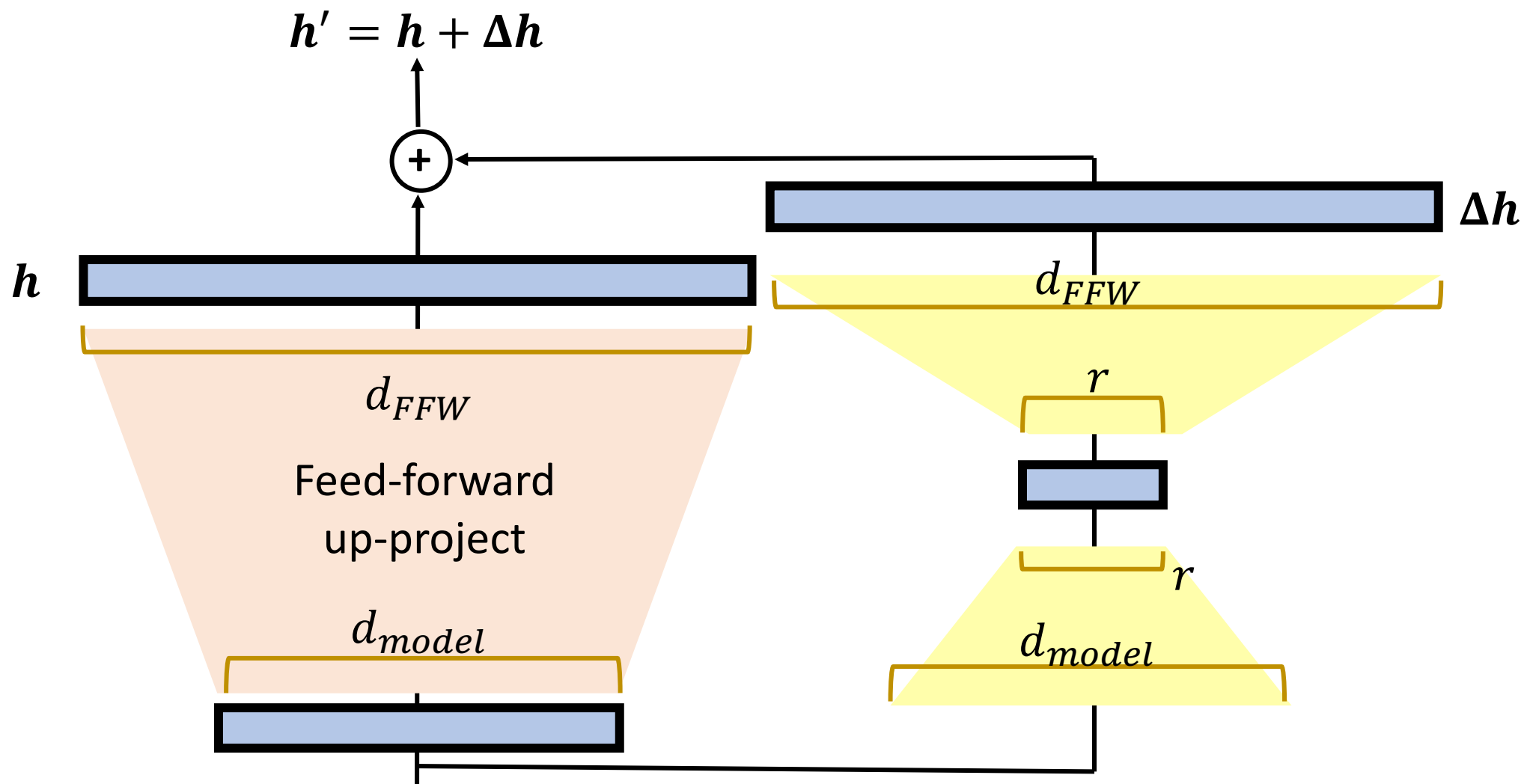
Parameter-Efficient Fine-tuning: LoRA

- LoRA



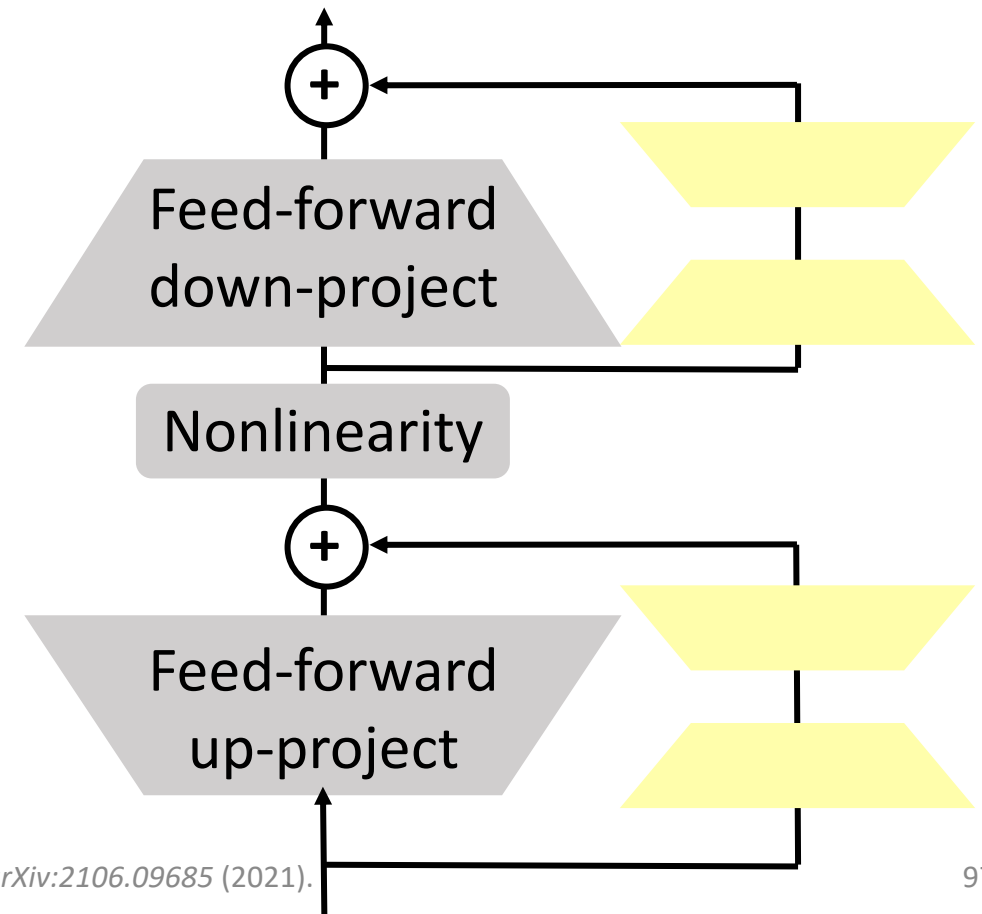
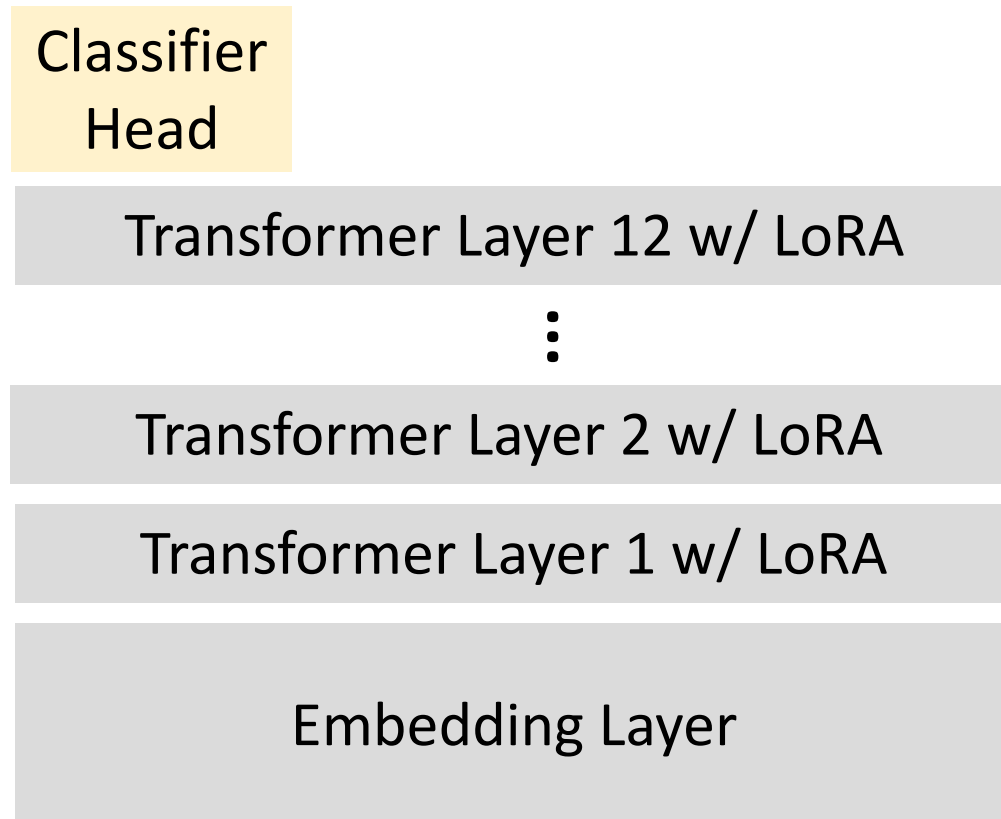
Parameter-Efficient Fine-tuning: LoRA

- LoRA



Parameter-Efficient Fine-tuning: LoRA

- LoRA: All downstream tasks share the PLM; the LoRA in each layer and the classifier heads are the task-specific modules



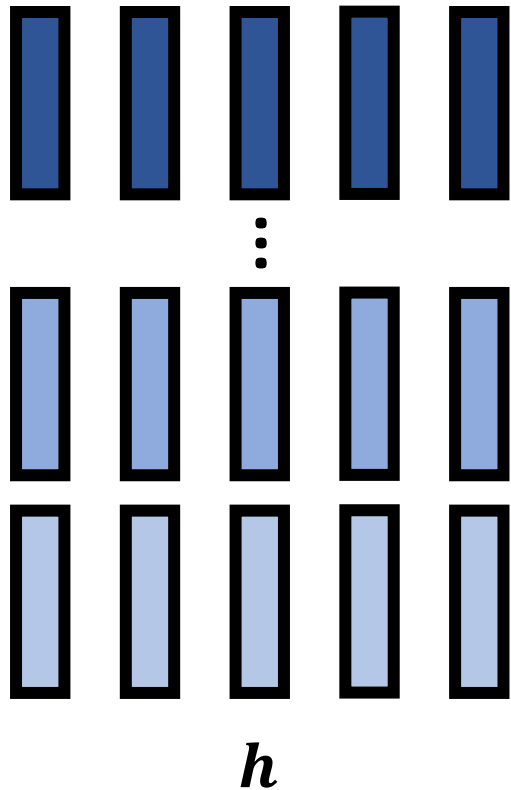
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Parameter-Efficient Fine-tuning: Prefix Tuning

- Use special submodules to modify hidden representations!

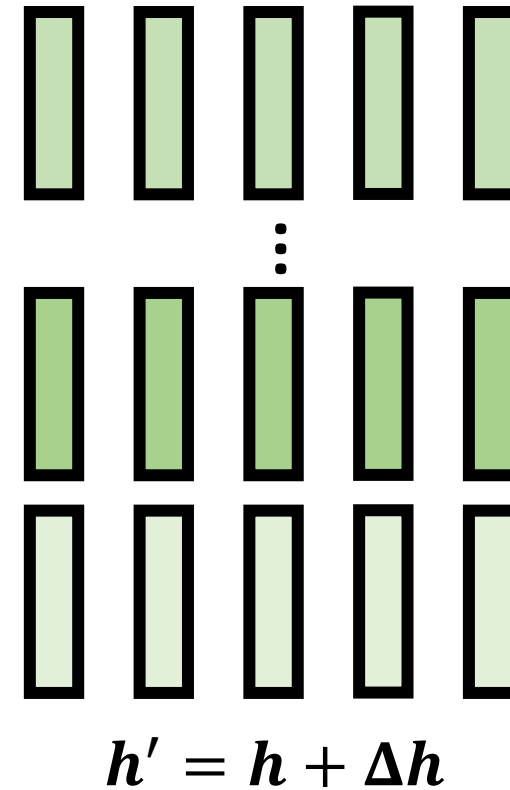
Before Fine-tuning



Prefix
Tuning

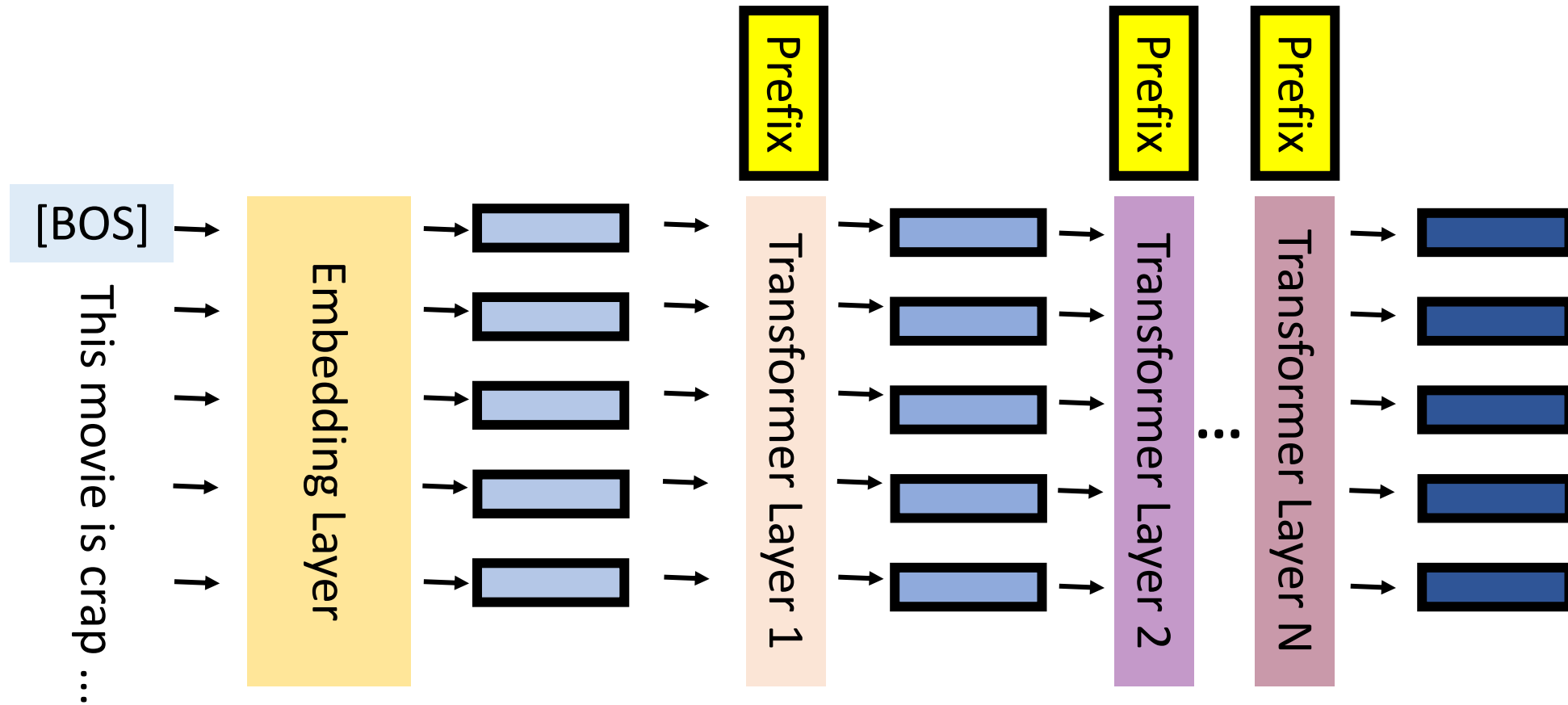


After Fine-tuning



Parameter-Efficient Fine-tuning: Prefix Tuning

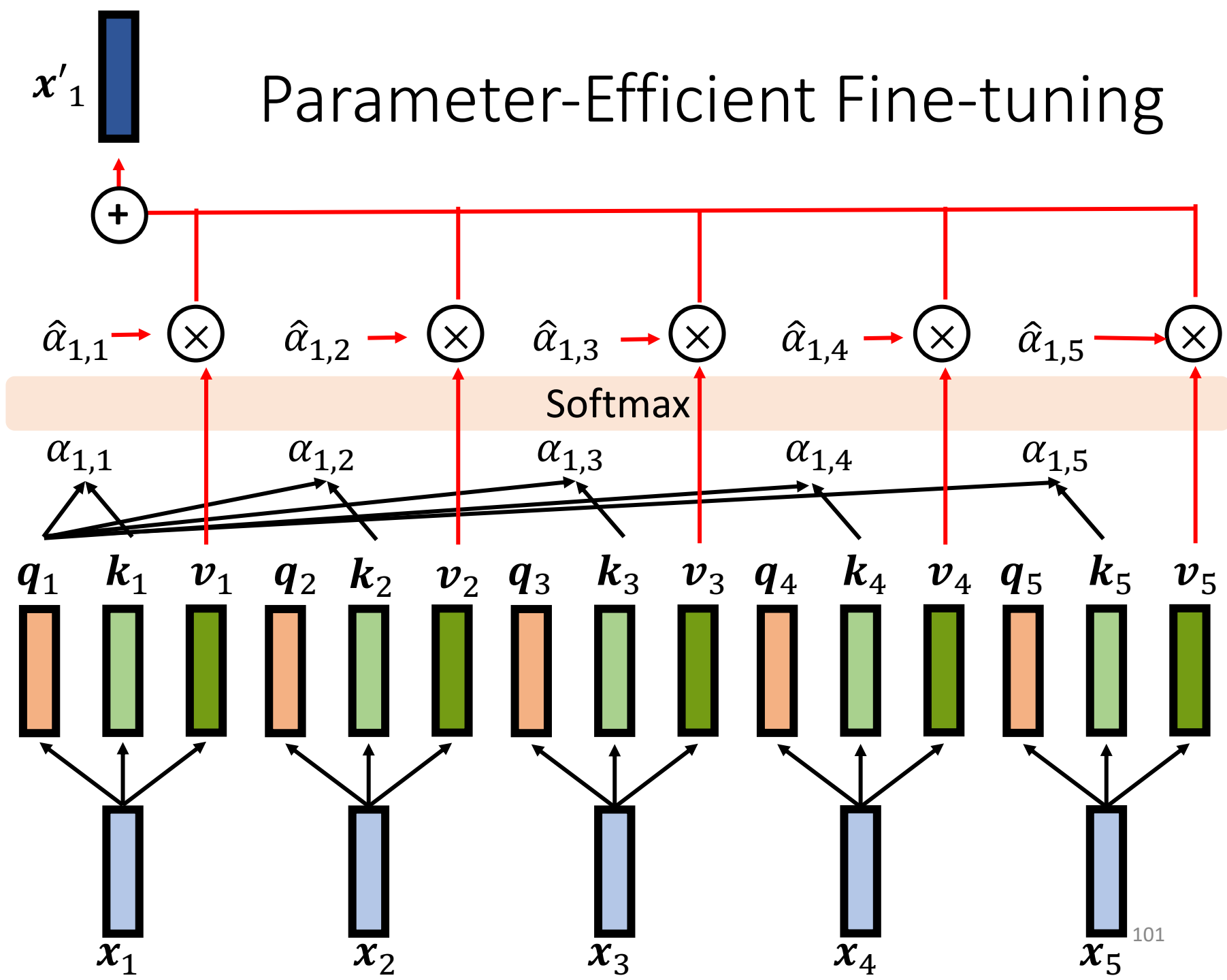
- Prefix Tuning: Insert trainable prefix in each layer



x'_1

Parameter-Efficient Fine-tuning

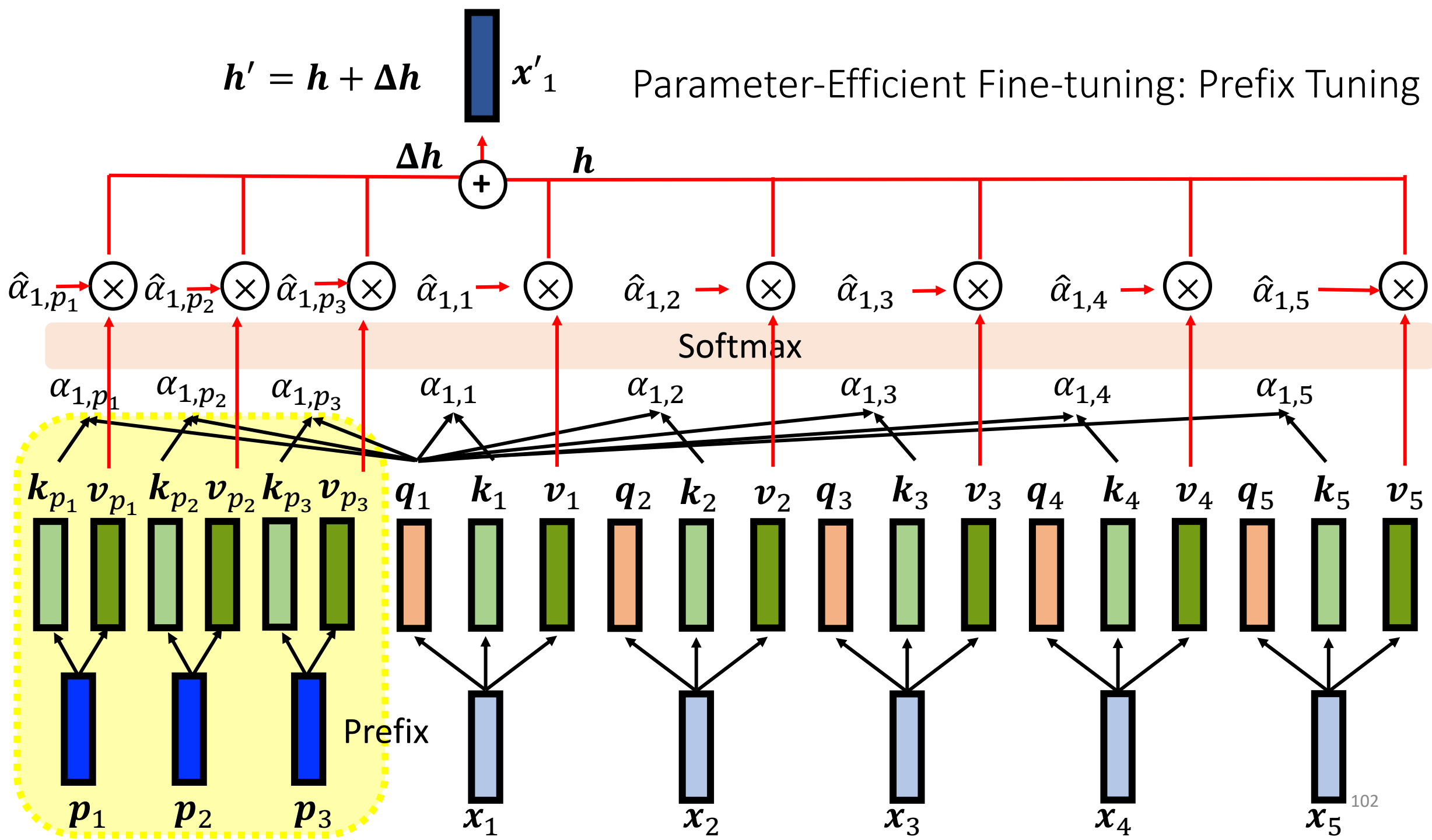
Standard
Self-Attention



$$h' = h + \Delta h$$

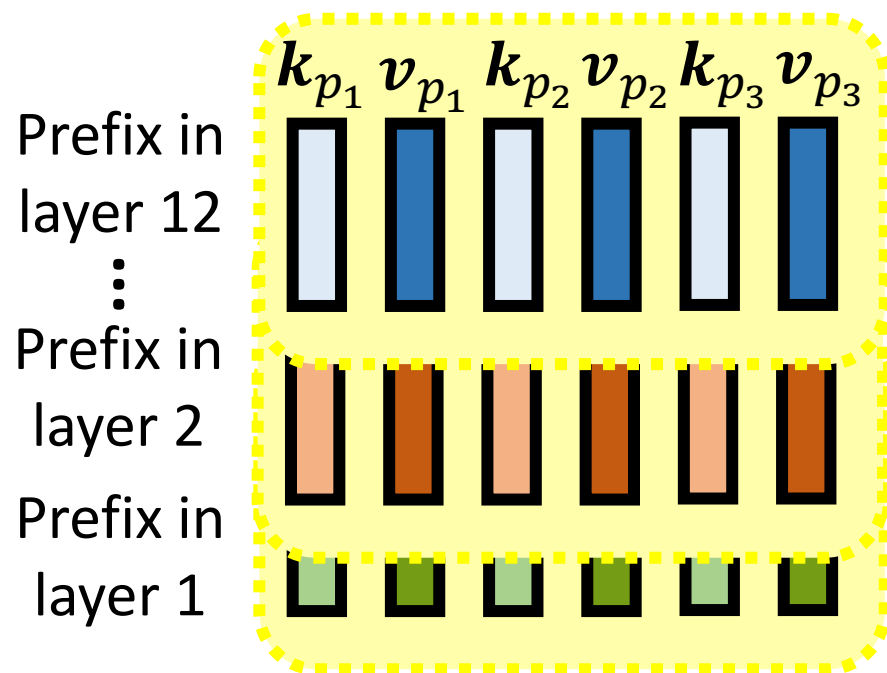
 x'_1

Parameter-Efficient Fine-tuning: Prefix Tuning



Parameter-Efficient Fine-tuning: Prefix Tuning

- Prefix Tuning: Only the prefix (key and value) are updated during fine-tuning

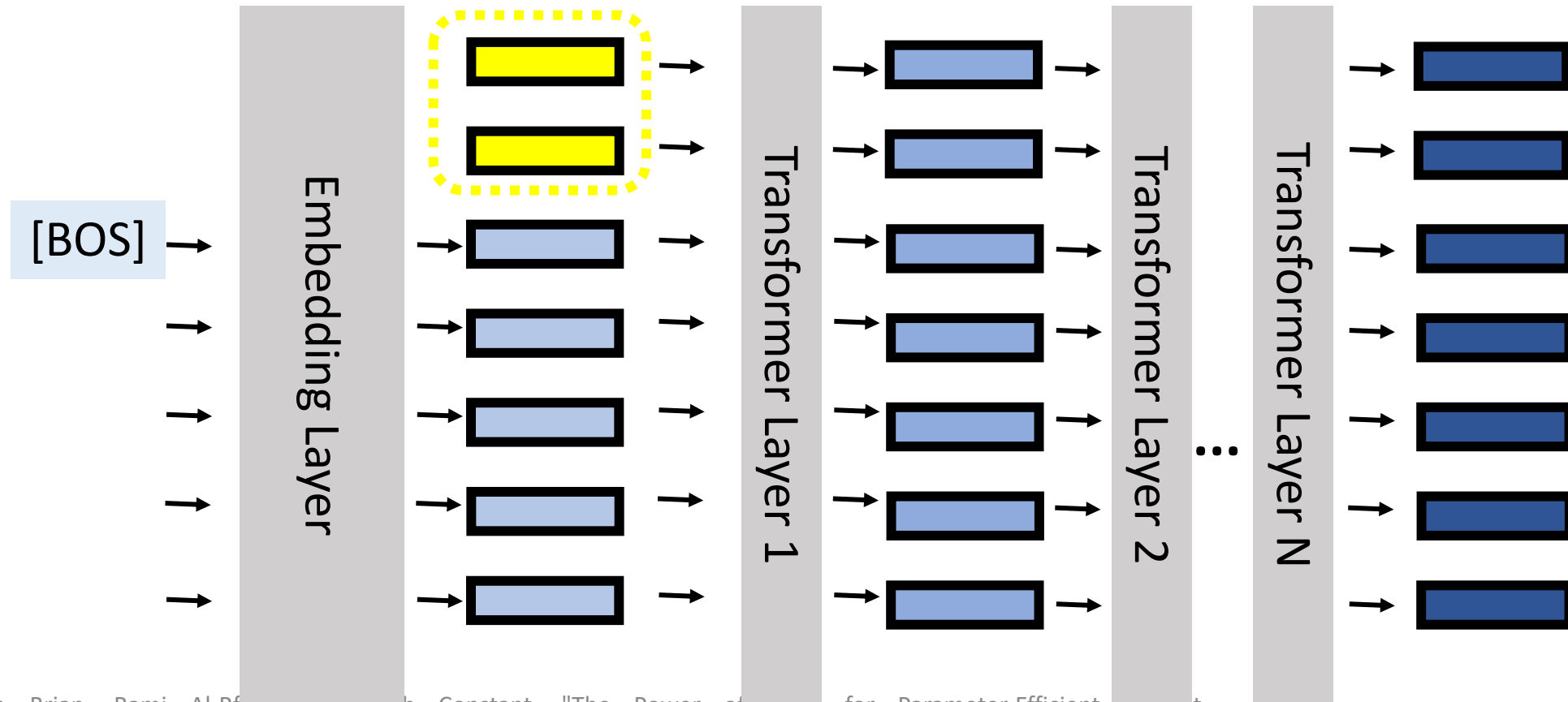


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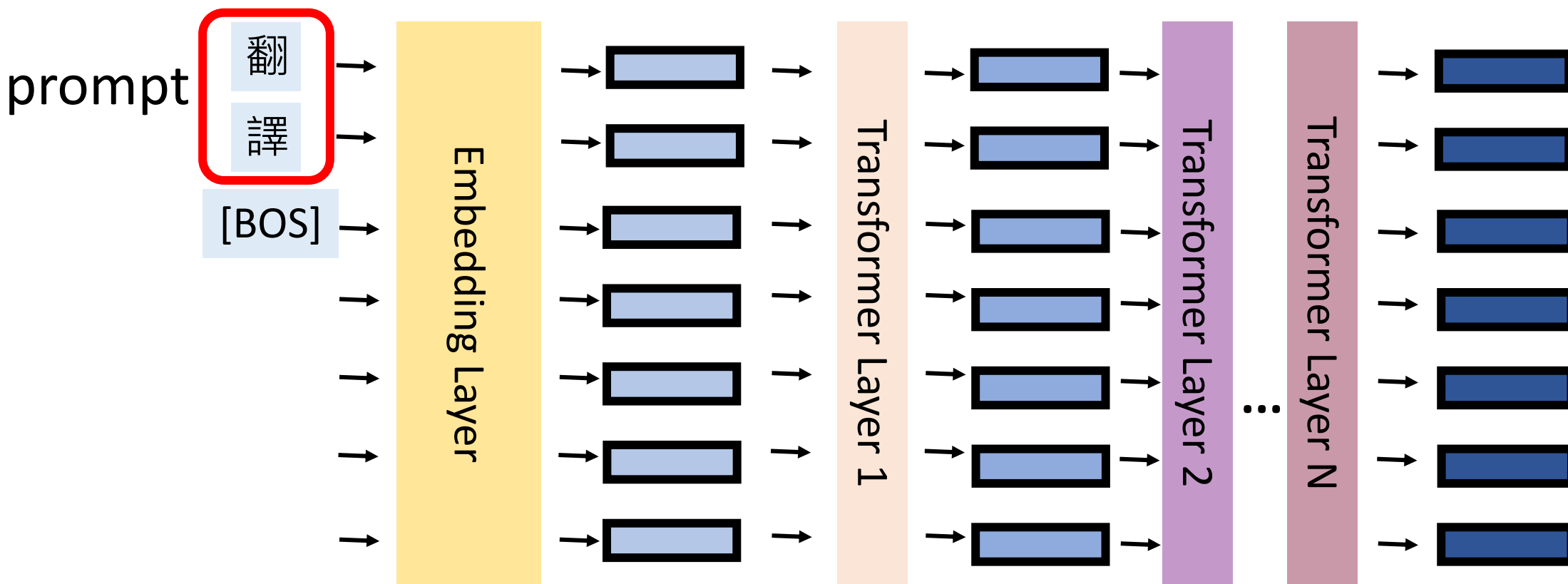
Parameter-Efficient Fine-tuning: Soft Prompting

- Soft Prompting
 - Prepend the prefix embedding at the input layer



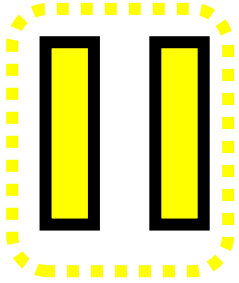
Parameter-Efficient Fine-tuning: Soft Prompting

- Soft Prompting can be considered as the ***soften*** version of prompting
 - (Hard) prompting: add words in the input sentence (fine-tune the model while fixing the prompts)

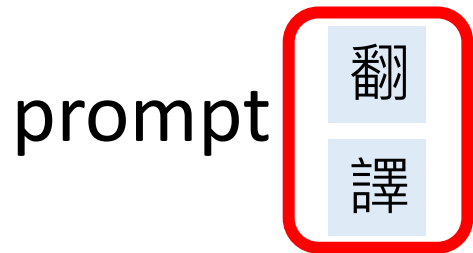


Parameter-Efficient Fine-tuning

- Soft Prompts: vectors (can be initialized from some word embeddings)



- Hard Prompts: words (that are originally in the vocabulary)



Parameter-Efficient Fine-tuning

- Benefit 1: Drastically decreases the task-specific parameters

	Adapter	LoRA	Prefix Tuning	Soft Prompt
Task-specific parameters*	$\Theta(d_{model}rL)$	$\Theta(d_{model}rL)$	$\Theta(\textcolor{red}{d}_{model}nL)$	$\Theta(d_{model}n)$
Percent Trainable	<5%	<0.1%	<0.1%	<0.05%
Illustration				

*not including the classifier head

Parameter-Efficient Fine-tuning

- Benefit 2: Less easier to overfit on training data; better out-of-domain performance

	Dataset	Domain	Model	Soft Prompt	Δ
	SQuAD	Wiki	94.9 $\pm$ 0.2	94.8 $\pm$ 0.1	-0.1
In-domain dataset	TextbookQA	Book	54.3 $\pm$ 3.7	66.8 $\pm$ 2.9	+12.5
	BioASQ	Bio	77.9 $\pm$ 0.4	79.1 $\pm$ 0.3	+1.2
	RACE	Exam	59.8 $\pm$ 0.6	60.7 $\pm$ 0.5	+0.9
	RE	Wiki	88.4 $\pm$ 0.1	88.8 $\pm$ 0.2	+0.4
	DuoRC	Movie	68.9 $\pm$ 0.7	67.7 $\pm$ 1.1	-1.2
	DROP	Wiki	68.9 $\pm$ 1.7	67.1 $\pm$ 1.9	-1.8
OOD test dataset					

Parameter-Efficient Fine-tuning

- Benefit 3: Fewer parameters to fine-tune; a good candidate when training with small dataset

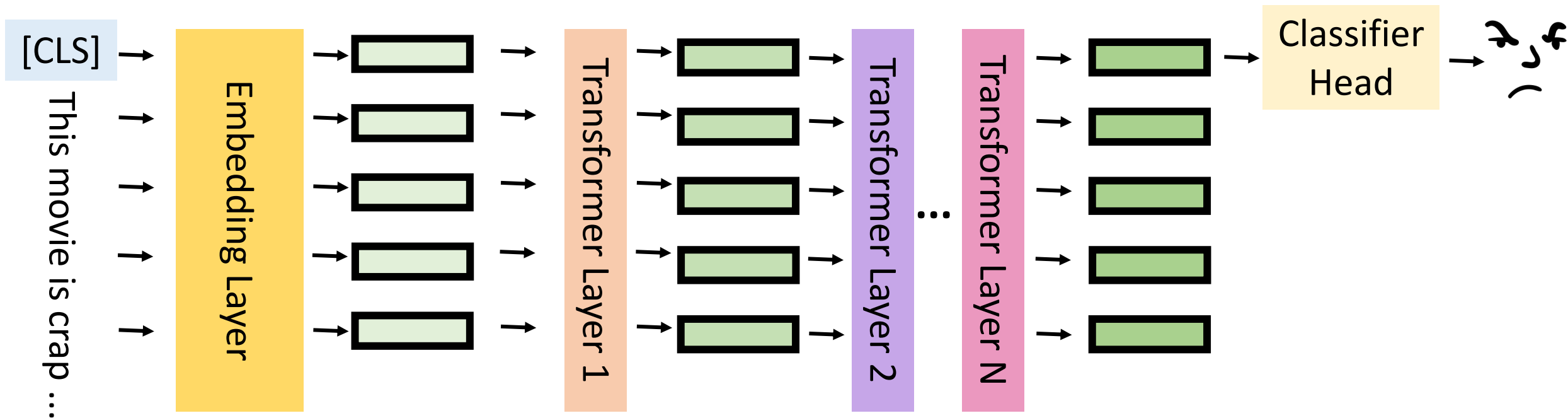
Model	low-resource				high-resource			
	CHEMPROT (4169)	ACL-ARC (1688)	SCIERC (3219)	HYP. (515)	RCT (180k)	AGNEWS (115k)	HELPFUL. (115k)	IMDB (20k)
RoBa.-ft [†]	81.9 _{1.0}	63.0 _{5.8}	77.3 _{1.9}	86.6 _{0.9}	87.2 _{0.1}	93.9 _{0.2}	65.1 _{3.4}	95.0 _{0.2}
RoBa.-ft*	81.7 _{0.8}	65.0 _{3.6}	78.5 _{1.8}	88.9 _{3.3}	87.0 _{0.1}	93.7 _{0.2}	69.1 _{0.6}	95.2 _{0.1}
RoBa.-adapter ₂₅₆	82.9 _{0.6}	67.5 _{4.3}	80.8 _{0.7}	90.4 _{4.2}	87.1 _{0.1}	93.8 _{0.1}	69.0 _{0.4}	95.7 _{0.1}

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 - Early Exit
- Closing Remarks

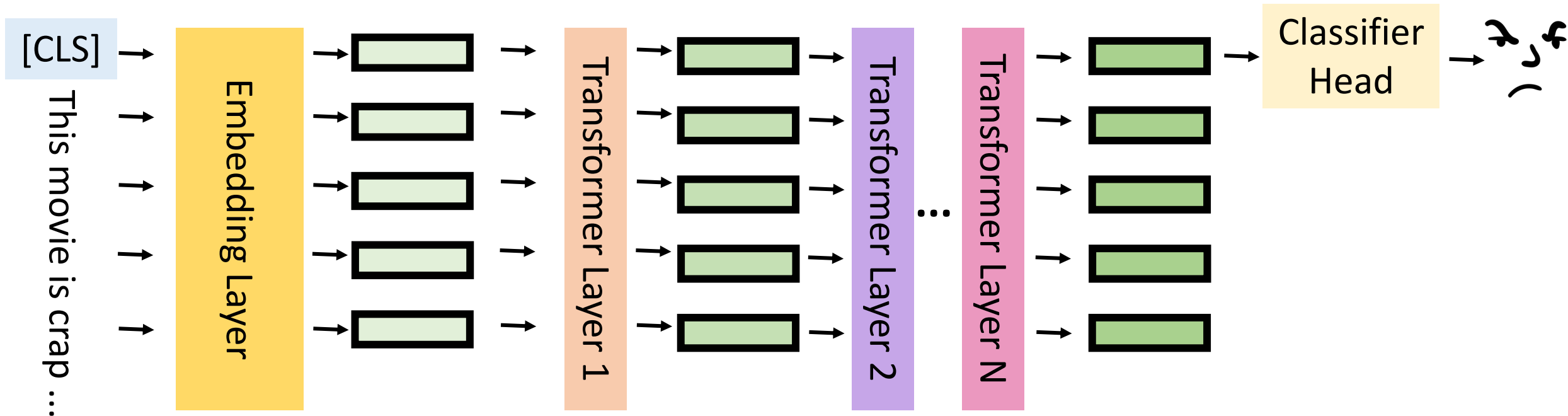
Early Exit

- Problem 1: The PLM is too big
 - Inference takes too long



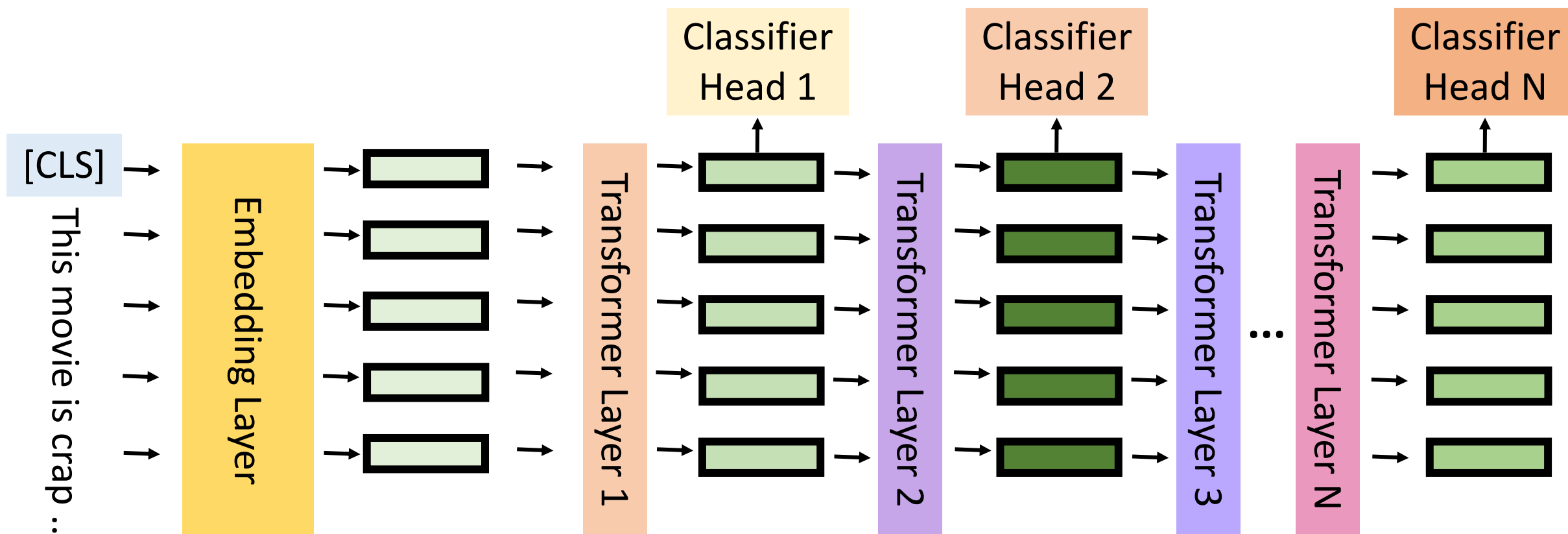
Early Exit

- Inference **using the whole** model takes too long
- Simpler data may require lesser effort to obtain the answer
- Reduce the number of layers used during inference



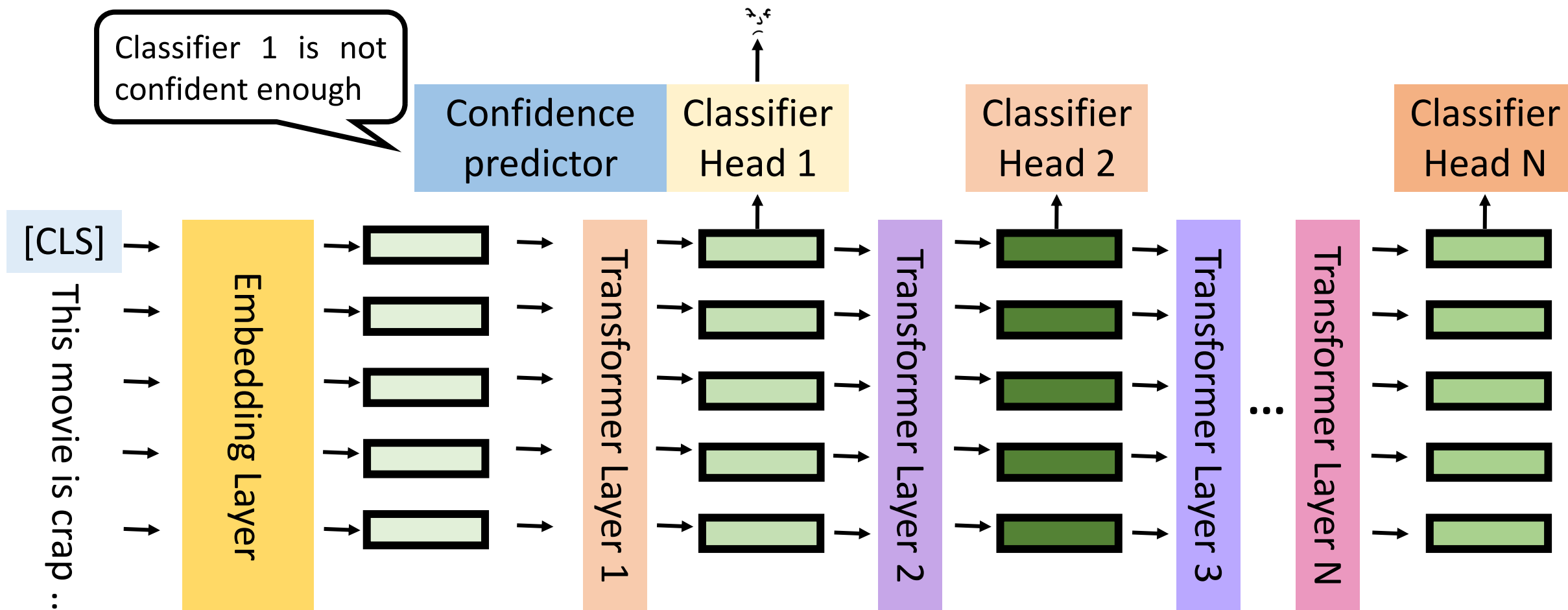
Early Exit

- Add a classifier at each layer



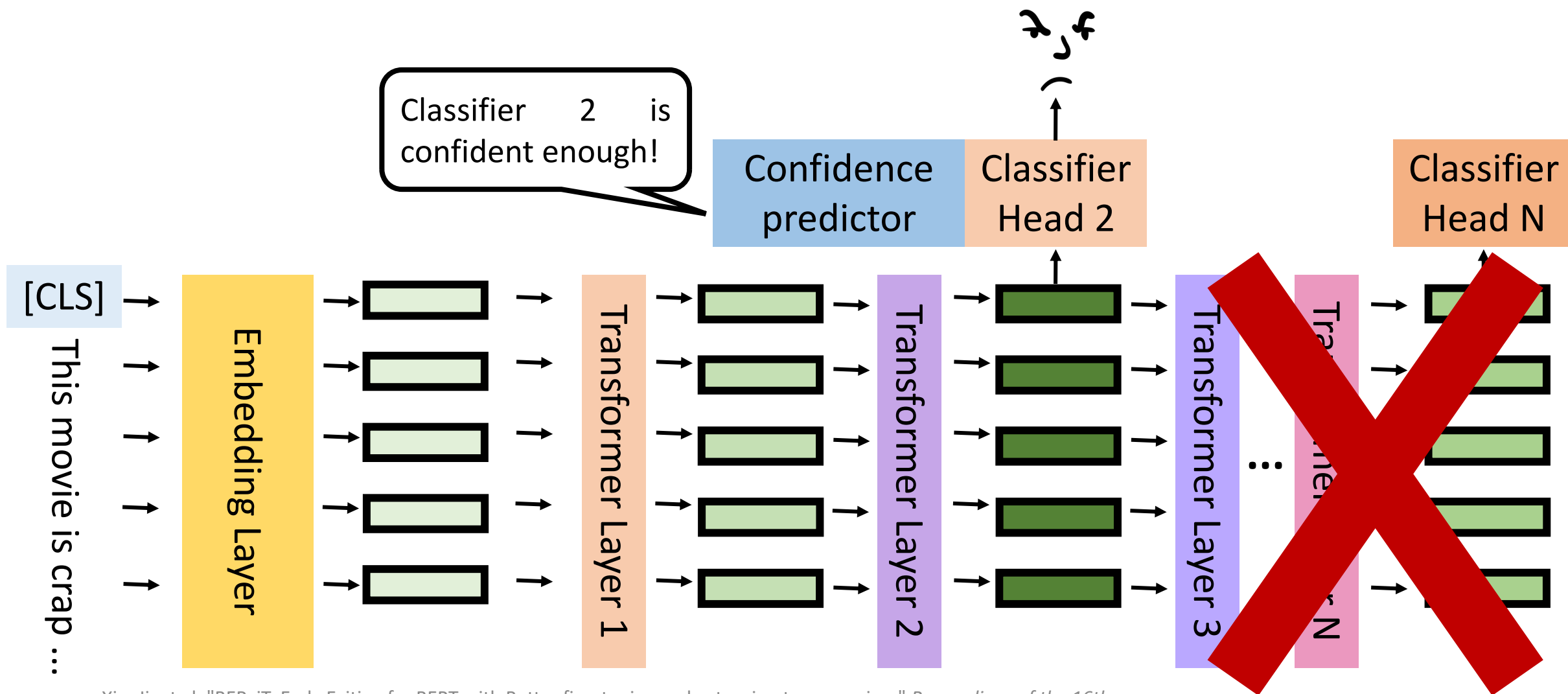
Early Exit

- How do we know which classifier to use?



Early Exit

- How do we know which classifier to use?



Early Exit

- Early exit reduces the inference time while keeping the performance

	RTE		MRPC		SST-2		QNLI		QQP		MNLI-(m/mm)		STS-B	
	Score	Layer	Score	Layer	Score	Layer	Score	Layer	Score	Layer	Score	Layer	Score	Layer
BERT_{BASE}														
RAW	66.4	12	88.9	12	93.5	12	90.5	12	71.2	12	84.6/83.4	12	85.8	12
ALT	101%	−44%	99%	−30%	98%	−65%	99%	−42%	99%	−56%	99%/99%	−37%	95%	−50%
	99%	−54%	97%	−56%	96%	−79%	98%	−63%	97%	−75%	97%/97%	−57%	91%	−67%
	96%	−64%	94%	−74%	94%	−87%	95%	−71%	93%	−84%	93%/92%	−72%	85%	−75%
BERT_{LARGE}														
RAW	70.1	24	89.3	24	94.9	24	92.7	24	72.1	24	86.7/85.9	24	86.5	24
ALT	95%	−33%	99%	−32%	100%	−32%	97%	−62%	98%	−74%	99%/99%	−36%	97%	−39%
	94%	−46%	98%	−46%	99%	−61%	95%	−73%	96%	−82%	96%/97%	−57%	90%	−62%
	88%	−62%	94%	−71%	96%	−78%	91%	−83%	91%	−89%	90%/90%	−75%	76%	−80%

Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
 - Labeled Data Scarcity → Data-Efficient Fine-tuning
 - PLMs Are Gigantic → Reducing the Number of Parameters
 - Summary
- Closing Remarks

Reducing the Number of Parameters: Summary

- Parameter-efficient fine-tuning: Reduce the task-specific parameters in downstream task
- Early exit: Reduce the models that are involved during inference

Outline

- Background knowledge
- The Problems of PLMs
- The Solutions of Those Problems
- Closing Remarks

Closing Remarks

- What we address in this lecture
 - Making PLM smaller, faster, and more parameter-efficient
 - Deploying PLMs when the labeled data in the downstream task is scarce
- The problems are not completely solved yet
- The problems we discuss are just a small part of problems of PLMs
 - Why does self-supervised pre-training work
 - Interpretability of the model's prediction
 - Domain adaptation
 - Continual learning/lifelong learning
 - Security and privacy

To Learn More

- **AACL-IJCNLP 2022 Tutorial (11.24.2022)**
 - **Recent Advances in Pre-trained Language Models: Why Do They Work and How Do They Work.**

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